

Copy Protecting Cryptographic Functionalities over Entropic Inputs

Fuyuki Kitagawa^{1,2} and Takashi Yamakawa^{1,2}

¹NTT Social Informatics Laboratories, Tokyo, Japan
{fuyuki.kitagawa,takashi.yamakawa}@ntt.com

²NTT Research Center for Theoretical Quantum Information, Atsugi, Japan

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Abstract

We present improved definitions and constructions for copy-protected digital signatures and pseudorandom functions (PRFs). Our new security definitions support challenge messages or inputs chosen from arbitrary high min-entropy distributions and allow signing or evaluation queries. This extends prior definitions, which assumed uniformly random challenges and did not consider oracle access. We construct schemes that satisfy these stronger definitions using only polynomially secure indistinguishability obfuscation (iO) and one-way functions (OWFs), avoiding the subexponential assumptions and the Learning with Errors (LWE) assumption used in previous constructions, even in the uniform-challenge and query-free setting. Moreover, our constructions and security proofs are arguably simpler than existing ones.

We also propose a new security notion for unclonable puncturable obfuscation (UPO), which primarily extends prior definitions to support challenge inputs drawn from arbitrary high min-entropy distributions, along with some additional refinements. We construct a UPO satisfying this notion from polynomially secure iO and the LWE assumption, thereby avoiding the subexponential assumptions and unproven conjectures required in previous constructions, even in the uniform-challenge setting. In fact, in the uniform-challenge case, we show that iO and OWFs alone suffice, further removing the need for LWE. Again, our constructions and security proofs are arguably simpler than existing ones. As applications, we show that a UPO satisfying this notion is sufficient to copy-protect a variety of puncturable functionalities beyond those studied in the prior work.

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1 Introduction

Quantum information processing enables novel functionalities that are impossible in the classical setting. A particularly striking example is the *no-cloning theorem*, which states that an arbitrary quantum state cannot be copied. This fundamental principle opens up new paradigms in cryptography—collectively referred to as *unclonable cryptography*—encompassing applications such as quantum money [Wie83], quantum copy-protection [Aar09], and beyond.

Copy-protected signatures and PRFs. Among the various unclonable cryptographic primitives, a particularly compelling class involves the copy-protection of cryptographic functionalities. These include, for instance, the copy protection of decryption keys in public key encryption (PKE) schemes (known as *single-decryptor encryption* or SDE) [GZ20, CLLZ21], signing keys in digital signature schemes [LLQZ22], and secret keys in pseudorandom functions (PRFs) [CLLZ21].

This work primarily focuses on the latter two: copy-protection of digital signatures and PRFs. Previous works [CLLZ21, LLQZ22] have provided constructions of these primitives. However, the security definitions used in these works exhibit significant limitations. In the following, we begin by focusing on copy-protected signatures and explaining the issues. Similar concerns also arise in the context of copy-protected PRFs as well. To illustrate these issues, we begin by reviewing the existing security definition for copy-protected signatures [LLQZ22].

For a tuple of adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$, consider the following game:

- $\mathcal{A}$ receives a quantum signing key sk and a classical verification key vk .
- $\mathcal{A}$ attempts to “copy” sk by generating a bipartite quantum state on two registers, which are then sent to $\mathcal{B}$ and $\mathcal{C}$ respectively.
- $\mathcal{B}$ and $\mathcal{C}$ are each given uniformly random challenge messages m_B and m_C , and output signatures σ_B and σ_C .

The adversary wins if both signatures are valid, i.e., σ_B is valid for m_B and σ_C for m_C under vk . The security guarantees assert that any quantum polynomial-time (QPT) adversary can win the game only with negligible probability.

We identify the following critical limitations in the above security definition.

1. **Uniform challenge distribution.** The challenge messages m_B and m_C are sampled uniformly at random. However, this assumption is unrealistic: in practical scenarios, messages often originate from structured or application-specific distributions rather than the uniform distribution. As a result, the current definition fails to provide meaningful security guarantees in real-world settings, where the messages targeted by an adversary are typically non-uniform.
2. **Absence of signing queries.** The definition does not allow the adversary to make signing queries. This is problematic, since access to a signature on a message that is correlated with the challenge (e.g., with a single bit flipped) could potentially enable forgery. Ideally, the security definition should remain robust as long as the adversary does not query the exact challenge messages—analogue to the classical notion of existential unforgeability under chosen-message attacks (EUF-CMA).¹

We note that the first issue—the assumption of a uniform challenge distribution—can be circumvented by allowing stateful signing. However, no known solution exists in the standard stateless signing setting; see Section 1.2 for details.

In addition to the above issues, existing constructions of copy-protected signatures and PRFs rely on the combination of strong assumptions—namely, subexponentially secure iO and OWFs, along with polynomial hardness of LWE. While this reliance is partially justified by the fact that such primitives imply public-key quantum money—which itself relies on iO and OWFs [Zha19], except for a few constructions based on non-standard assumptions [FGH⁺12, KSS21, LMZ23, Zha23, BNZ24]—there remains a gap between the assumptions used for public-key quantum money and those used for copy-protected signatures and PRFs. Recent work by Kitagawa and Yamakawa [KY25] has closed this gap for SDE by constructing it from *polynomially* secure iO and OWFs, without relying on LWE. However, no such result is known for copy-protected signatures and PRFs, even under the existing security definitions. Thus, the following question remains open:

¹ However, unlike in the classical EUF-CMA setting, we should not allow the adversary to specify the challenge messages, since in our case, $\mathcal{A}$ could generate signatures on the challenge messages and forward them to $\mathcal{B}$ and $\mathcal{C}$.

Can we construct copy-protected signatures and PRFs, secure against non-uniform challenge distributions and robust to signing/evaluation queries, using only polynomially secure iO and OWFs?

Unclonable puncturable obfuscation. Some works explore copy-protection for general classes of functionalities, rather than targeting specific primitives. In this direction, Ananth and Behera [AB24] introduced the notion of *unclonable puncturable obfuscation* (UPO) as a foundational building block for copy-protecting general *puncturable* functionalities and beyond.

While UPO is a powerful abstraction, their construction and security definition suffer from several drawbacks. First, their security proof relies on a certain unproven conjecture, in addition to assuming subexponentially secure iO, subexponentially secure OWFs, and the polynomial hardness of LWE. Second, as in the case of copy-protected signatures and PRFs, their security definition only considers uniformly random challenge inputs. That is, the guarantee is limited to ensuring that one quantum obfuscated program cannot be cloned into two separate instances that successfully evaluate on uniformly random inputs. As discussed above, this assumption is unrealistic in practical settings, where inputs typically follow structured or application-dependent distributions. Security under more general distributions is therefore preferable. This motivates the following question:

Can we construct UPO without relying on conjectures that is secure against non-uniform challenge distributions?

1.1 Our Results

We answer the above questions affirmatively.

Copy-protected signatures and PRFs over non-uniform challenges. We generalize the security definitions of copy-protected signatures and PRFs from [CLLZ21, LLQZ22] to handle challenge messages and inputs drawn from arbitrary high min-entropy distributions and to support signing and evaluation queries. We then present a construction that satisfies these extended definitions, based on polynomially secure iO and OWFs. That is, our scheme improves upon the constructions in [CLLZ21, LLQZ22] along three dimensions simultaneously:

- **Distribution.** We handle arbitrary high min-entropy challenge distributions rather than only uniform ones;
- **Oracle.** Security hold in the presence of signing/evaluation oracles;
- **Assumption.** We rely only on polynomially secure iO and OWFs, avoiding additional subexponential or algebraic assumptions used in prior work.

Our constructions rely on coset states, similar to previous works [CLLZ21, LLQZ22], but are significantly simpler: the quantum component of the key consists of just a single coset state [CLLZ21], whereas prior works require multiple such states.

The key technical ingredient is a generalization of the computational strong monogamy property of coset states [CLLZ21] to a setting where the adversary receives auxiliary inputs that may leak partial information about the hidden subset. Once this generalization is established, our security proofs become relatively simple. Nonetheless, the proofs require certain cryptographic techniques reminiscent of those used in the security proof of SDE in [KY25], as well as a new cryptographic abstraction that we introduce, which we call key-robust non-committing encryption.

New security definition of UPO. [AB24] considered two variants of UPO: one in the independent challenge input setting, where two parties receive independently and uniformly random inputs, and the other in the identical challenge input setting, where both parties receive the same uniformly random input.² In this work, we focus on generalizing UPO in the independent challenge input setting, since UPO in the identical challenge input setting implies unclonable encryption [BL20], a primitive for which no construction is currently known in the plain model without relying on unproven conjectures.

²Note that in both variants, the challenge bits are identical for both parties. Here, the *challenge bits* refer to the bits that the adversaries are supposed to guess, while the *challenge inputs* are the inputs they receive in the second stage. Concretely, in the actual definitions (Definitions 6.9 and B.1), the challenge bits correspond to coin_B and coin_C (or coin in the latter), and the challenge inputs correspond to x_B and x_C .

As a natural extension of security in the independent challenge input setting, we consider the case where challenge inputs are sampled not necessarily from the uniform distribution, but from arbitrary independent distributions. We refer to this as $\mathcal{D}$ -UPO security, where $\mathcal{D}$ denotes the challenge distribution.³

We go further and introduce a stronger notion, which we call $\mathcal{D}$ -UPO⁺ security, inspired by a similar strengthening recently proposed for SDE [KY25]. We show that $\mathcal{D}$ -UPO⁺ security implies $\mathcal{D}$ -UPO security, assuming UPO satisfies security as iO. This assumption is relatively benign, as all known constructions of UPO—including ours—are based on iO and thus satisfy iO security.⁴

We also show new applications of UPO beyond those given in [AB24]. For instance, it enables copy-protection of the secret key of any puncturable PRF under an indistinguishability-based security notion. In contrast, such a result was not given in [AB24], where only search-based security was achieved for copy-protected PRFs. As another application, we demonstrate how to copy-protect the secret key of any puncturable PKE scheme, under a mild assumption on the min-entropy of the ciphertexts. We emphasize that this application is enabled by our extension to accommodate non-uniform challenge distributions.

More generally, UPO with $\mathcal{D}$ -UPO⁺ security appears to support a generic compiler that lifts any puncturable functionality to its unclonable version, as long as the functionality can be punctured at two distinct inputs. We do not attempt to formalize this compiler formally, as defining a general notion of puncturable functionalities would involve subtleties that lie beyond the scope of this work.

Construction of UPO. We construct a UPO satisfying $\mathcal{D}$ -UPO⁺ security for any distribution $\mathcal{D}$ with sufficiently large min-entropy, assuming polynomially secure iO and the polynomial hardness of LWE. This improves upon the construction in [AB24] in three key aspects:⁵ it supports general high-min-entropy challenge distributions (rather than only uniform), removes reliance on a conjecture, and weakens the underlying assumptions from subexponential to polynomial security. Moreover, in the special case where $\mathcal{D}$ is uniformly random—as considered in [AB24]—our construction can be based solely on polynomially secure iO and polynomially secure OWFs, eliminating the need for LWE altogether.

The construction and proof techniques are similar to those used for our copy-protected digital signatures and PRFs. Interestingly, our construction closely resembles the general copy-protection scheme relative to a classical oracle by [ALL⁺21], with the difference being that we use coset states instead of subset states. Despite this similarity, our security proof departs from theirs and requires novel techniques.

1.2 Related Work

Dealing with non-uniform challenge distributions via stateful signing. Amos et al. [AGKZ20] introduced the notion of single-signer signatures as an application of the primitive called one-shot signatures. These are closely related to the copy-protected signatures studied in this work, with a key distinction that they allow stateful signing. That is, the signer must maintain state information from each signing to use in subsequent signings. In particular, their construction requires the signer to store all previously generated signatures as part of its internal state and append them to each new signature, resulting in a signature size that grows linearly with the number of signed messages.⁶ Furthermore, the only known (and very recent) instantiations of one-shot signatures rely on subexponentially secure iO and subexponential LWE—significantly stronger assumptions than polynomially secure iO and OWFs [SZ25].

Copy-protection relative to a classical oracle. Aaronson et al. [ALL⁺21] constructed a copy-protection scheme for general unlearnable functionalities relative to a classical oracle. On the other hand, such general copy-protection is known to be impossible in the plain model [AL21].

³Precisely, $\mathcal{D}$ specifies two challenge distributions—one for each party—since we allow the challenge distributions to differ between parties.

⁴Even if a UPO does not satisfy iO security by default, it can be generically added by first applying iO to the circuit, followed by UPO.

⁵We emphasize that our improvements apply to their UPO construction in the independent challenge setting, and not to the identical challenge setting.

⁶Although the authors outline a potential approach to prevent the signature size from growing—by employing a strong variant of succinct non-interactive arguments of knowledge (SNARKs)—the details are omitted. Moreover, such an approach would likely rely on non-standard cryptographic assumptions or the random oracle model. Also, it would still not eliminate the need for stateful signing.

Collusion-resistant copy-protection Several works [LLQZ22, CG24] have studied *collusion-resistance* in the context of copy-protecting cryptographic functionalities such as SDE, copy-protected digital signatures, and PRFs. This notion ensures that an adversary given k copies of the protected state cannot generate $k + 1$ valid copies. In contrast, this paper focuses solely on the basic $1 \rightarrow 2$ unclonability setting and does not address collusion resistance.

Copy-protection with classical communication. Several works [RS22, Shm22, GMP23, CHV23] study copy-protection schemes with classical communication (often referred to as *semi-quantum cryptography*), where the copy-protected quantum object can be generated through classical interaction with a classical party. In contrast, this paper focuses on the setting where the copy-protected keys are transmitted directly over quantum channels.

Secure key leasing. Secure key leasing (also known as *key-revocable cryptography*) [KN22a, AKN⁺23, APV23] is a weaker variant of copy-protection for cryptographic keys. In this setting, the adversary is not prevented from copying the key, but is guaranteed to lose the ability to use it once the key is returned. While the security guarantees are weaker, a key advantage is that secure key leasing can be achieved from weaker assumptions—particularly, without relying on iO.

2 Technical Overview

In the first part of this overview (up to the end of Section 2.2), we focus on *copy-protected unpredictable functions* (UPFs) *without evaluation queries*. This setting is the most elementary one that still suffices to convey our key ideas. We later briefly explain how it naturally extends to copy-protected signatures and PRFs with signing and evaluation queries.

A copy-protected UPF is specified by three algorithms:

- $\mathcal{KG}$: On input the security parameter 1^λ , it outputs a classical *master secret key* msk and a quantum *secret key* $s\mathcal{K}$.
- Eval : Given (msk, x) , it deterministically outputs $y \in \{0, 1\}^\ell$.
- $Q\text{Eval}$: Given $(s\mathcal{K}, x)$, it outputs y , which must equal $\text{Eval}(\text{msk}, x)$.

As security, we require the following: Fix a pair of challenge distributions $\mathcal{D} = (\mathcal{D}_B, \mathcal{D}_C)$ over the input space.⁷ The following game is played between a challenger and an adversary $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$:

1. The challenger runs $(\text{msk}, s\mathcal{K}) \leftarrow \mathcal{KG}(1^\lambda)$ and sends $s\mathcal{K}$ to $\mathcal{A}$.
2. $\mathcal{A}$ prepares a bipartite state on registers R_B, R_C and sends them to $\mathcal{B}$ and $\mathcal{C}$, respectively.
3. Independently sample $x_B \leftarrow \mathcal{D}_B$ and $x_C \leftarrow \mathcal{D}_C$ and send them to $\mathcal{B}$ and $\mathcal{C}$, respectively.
4. $\mathcal{B}$ outputs y_B and $\mathcal{C}$ outputs y_C .

The adversary *wins* if $y_B = \text{Eval}(\text{msk}, x_B) \wedge y_C = \text{Eval}(\text{msk}, x_C)$. A copy-protected UPF is secure for $\mathcal{D}$ if every QPT adversary wins with at most negligible probability.

When $\mathcal{D}_B$ and $\mathcal{D}_C$ are uniform, such security was proven in [CLLZ21]. Below we explain why their proof does not extend to general high-min-entropy distributions and introduce our new technique that does.

Limitations of the hidden-trigger technique At a high level, the proof in [CLLZ21] relies on the *hidden-trigger* technique [SW21]: in the reduction, the challenger replaces truly random challenges with pseudorandom “triggers” that secretly encode a hidden coset. For this technique to work, it is essential that (at least some part of) the challenges look uniform. When the challenge distribution has only high min-entropy, the technique breaks down, as embedding information about a hidden coset into such a general source becomes difficult. A similar obstacle arises in the copy-protected signature scheme of [LLQZ22], highlighting the need for new techniques.

⁷In the main body we also allow auxiliary inputs correlated with the challenge inputs, which we omit here for clarity.

A naive Hash-and-Eval attempt (and why it fails). A natural idea is to hash the high-min-entropy input into an almost-uniform string using universal hashing, and then apply a UPF that is already secure for uniform inputs. While this may seem to work at first glance, there is a fundamental issue in the reduction: the reduction cannot efficiently *invert* the hash on its own challenge (x_B, x_C) to sample pre-images according to the correct conditional distribution. As a result, such a black-box reduction does not appear to work.⁸

Nevertheless, the high-level idea of the Hash-and-Eval approach does appear in our final construction—though our analysis is specifically tailored to our construction.

Coset states and computational strong monogamy property. Our construction relies on coset states and the *computational strong monogamy of entanglement property*. We introduce the relevant notations following [CLLZ21, AKL⁺22]. For a subspace $A \subseteq \mathbb{F}_2^n$, let

- $A^\perp$ denote its dual,
- $\text{Can}_A(s)$ be the lexicographically least element of $A + s$ (computable in $\text{poly}(n)$ time),
- $\text{CS}(A) := \{\text{Can}_A(s) : s \in \mathbb{F}_2^n\}$.

For canonical representatives $(s, t) \in \text{CS}(A) \times \text{CS}(A^\perp)$, define the *coset state*

$$|A_{s,t}\rangle = \frac{1}{\sqrt{|A|}} \sum_{a \in A} (-1)^{\langle a, t \rangle} |a + s\rangle.$$

Applying $H^{\otimes n}$ maps $|A_{s,t}\rangle$ to $|A_{t,s}^\perp\rangle$.

We often write $i\mathcal{O}(A + s)$ and $i\mathcal{O}(A^\perp + t)$ to denote obfuscated programs by $i\mathcal{O}$ that verify membership in $A + s$ and $A^\perp + t$, respectively.

In this work, we use the following *computational strong monogamy property*, proven by [CLLZ21, CV22]. Consider the following experiment played by a tuple of adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$.

1. $\mathcal{A}$ is given a coset state $|A_{s,t}\rangle$, where $A \subseteq \mathbb{F}_2^n$ is a uniformly random subspace of dimension $n/2$, and (s, t) are uniformly drawn from $\text{CS}(A) \times \text{CS}(A^\perp)$. Additionally, $\mathcal{A}$ receives two obfuscated membership checking programs $i\mathcal{O}(A + s)$ and $i\mathcal{O}(A^\perp + t)$. $\mathcal{A}$ outputs a bipartite state q over the registers R_B and R_C .
2. $\mathcal{B}$ (resp. $\mathcal{C}$) is given the register R_B (resp. R_C) and the description of A , and outputs s_B (resp. t_C).

The property guarantees that assuming that $i\mathcal{O}$ is secure iO and there exist OWFs, for any tuple of QPT adversaries, the probability that $s_B = s$ and $t_C = t$ is negligible.⁹

2.1 Construction for Uniform Challenges

While copy-protected UPF for uniform challenges is already known, we first present our construction in this setting, because it already contains many of the key ideas that we will later generalize to arbitrary challenge distributions.

Let F be a puncturable PRF (PPRF).

- $\mathcal{KG}(1^\lambda)$
 1. Sample two independent PPRF keys $K, K^\perp$.
 2. Sample a random coset state $|A_{s,t}\rangle$ where $(s, t) \leftarrow \text{CS}(A) \times \text{CS}(A^\perp)$ and set

$$C_{\text{mc}} \leftarrow i\mathcal{O}(A + s), \quad C_{\text{mc}}^\perp \leftarrow i\mathcal{O}(A^\perp + t).$$

⁸The reduction would work in the quantum random oracle model.

⁹In the original statement in [CLLZ21] (stated as Theorem 5.5), the pair (s, t) is sampled uniformly from the full space $\mathbb{F}_2^n \times \mathbb{F}_2^n$, and the winning condition is that $s_B \in A + s$ and $t_C \in A^\perp + t$. In contrast, in the formulation here, (s, t) is sampled from $\text{CS}(A) \times \text{CS}(A^\perp)$, and the winning condition is $s_B = s$ and $t_C = t$. One can easily verify that these two formulations are equivalent, since for any $s_B \in A + s$ and $t_C \in A^\perp + t$, we have $\text{Can}_A(s_B) = s$ and $\text{Can}_{A^\perp}(t_C) = t$ if $(s, t) \in \text{CS}(A) \times \text{CS}(A^\perp)$. We remark that a similar formulation to ours is used in [AKL⁺22].

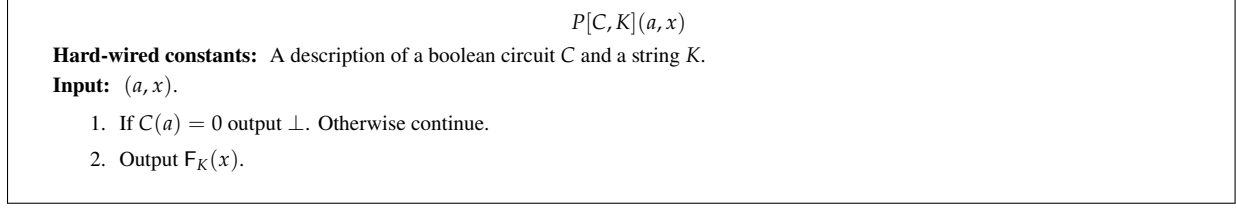


Figure 1: Description of the circuit P .

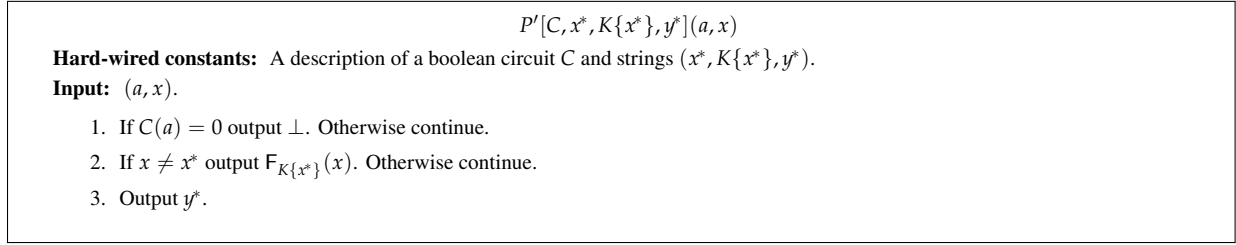


Figure 2: Description of the circuit P' .

3. Compute the obfuscations

$$\tilde{P} \leftarrow i\mathcal{O}(P[C_{\text{mc}}, K]), \quad \tilde{P}^\perp \leftarrow i\mathcal{O}(P[C_{\text{mc}}^\perp, K^\perp])$$

where P is the circuit described in Figure 1.

4. Output

$$\text{msk} \leftarrow (K, K^\perp), \quad s\mathcal{K} \leftarrow (|A_{s,t}\rangle, \tilde{P}, \tilde{P}^\perp).$$

- $\text{Eval}(\text{msk} = (K, K^\perp), x)$: Output $F_K(x) \parallel F_{K^\perp}(x)$.
- $Q\text{Eval}(|A_{s,t}\rangle, \tilde{P}, \tilde{P}^\perp)$: Starting with $|A_{s,t}\rangle$, coherently evaluate $\tilde{P}$ to obtain $F_K(x)$, uncompute to recover $|A_{s,t}\rangle$, apply $H^{\otimes n}$ to get $|A_{t,s}^\perp\rangle$, coherently evaluate $\tilde{P}^\perp$ to obtain $F_{K^\perp}(x)$, and output $F_K(x) \parallel F_{K^\perp}(x)$.

Correctness is immediate. Below we outline the ideas used in the security proof.

By a standard puncturing argument we can replace the obfuscations

$$\tilde{P} = i\mathcal{O}(P[C_{\text{mc}}, K]) \quad \text{and} \quad \tilde{P}^\perp = i\mathcal{O}(P[C_{\text{mc}}^\perp, K^\perp])$$

with

$$i\mathcal{O}(P'[C_{\text{mc}}, x_{\mathcal{B}}, K\{x_{\mathcal{B}}\}, y_{\mathcal{B}}]), \quad i\mathcal{O}(P'[C_{\text{mc}}^\perp, x_{\mathcal{C}}, K\{x_{\mathcal{C}}\}, y_{\mathcal{C}}]),$$

where P' is the circuit described in Figure 2, $x_{\mathcal{B}}, x_{\mathcal{C}}$ are the uniformly random challenge inputs of $\mathcal{B}, \mathcal{C}$, and $y_{\mathcal{B}}, y_{\mathcal{C}}$ are uniformly random outputs. Because $y_{\mathcal{B}}, y_{\mathcal{C}}$ also appear in the winning condition, for the adversary to win, $\mathcal{B}$ and $\mathcal{C}$ must output $y_{\mathcal{B}}$ in the first half and $y_{\mathcal{C}}$ in the second half, respectively.

First idea: shifting the outputs. Suppose we *shift* the reprogrammed outputs, replacing $y_{\mathcal{B}}, y_{\mathcal{C}}$ with $y_{\mathcal{B}} \oplus s, y_{\mathcal{C}} \oplus t$. Because $y_{\mathcal{B}}, y_{\mathcal{C}}$ are uniform, the shifted values remain uniform. Hence the distribution is unchanged, and any successful $\mathcal{B}$ and $\mathcal{C}$ must now output $y_{\mathcal{B}} \oplus s$ in the first half and $y_{\mathcal{C}} \oplus t$ in the second half, respectively. The hope is that, if we can simulate the security experiment while knowing only $(y_{\mathcal{B}}, y_{\mathcal{C}})$ —but *not* the hidden shifts (s, t) —then successful $\mathcal{B}$ and $\mathcal{C}$ would necessarily reveal s and t , respectively, yielding a reduction to the computational strong monogamy property. Unfortunately, the reduction algorithm still needs the *shifted* values $(y_{\mathcal{B}} \oplus s, y_{\mathcal{C}} \oplus t)$, because the obfuscated programs embedded in $s\mathcal{K}$ are now

$$i\mathcal{O}(P'[C_{\text{mc}}, x_{\mathcal{B}}, K\{x_{\mathcal{B}}\}, y_{\mathcal{B}} \oplus s]) \quad \text{and} \quad i\mathcal{O}(P'[C_{\text{mc}}^\perp, x_{\mathcal{C}}, K\{x_{\mathcal{C}}\}, y_{\mathcal{C}} \oplus t]),$$

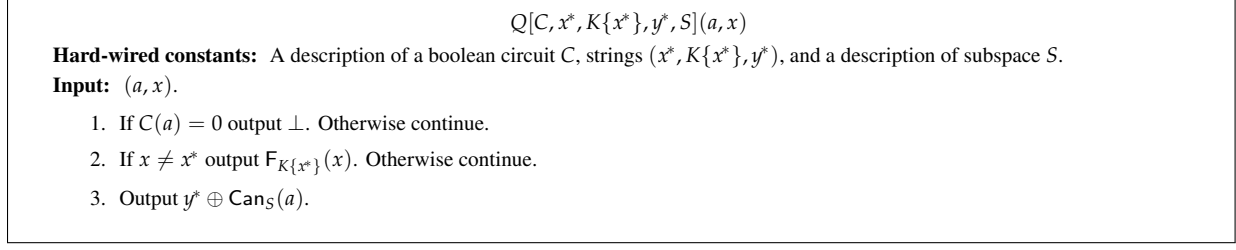


Figure 3: Description of the circuit Q .

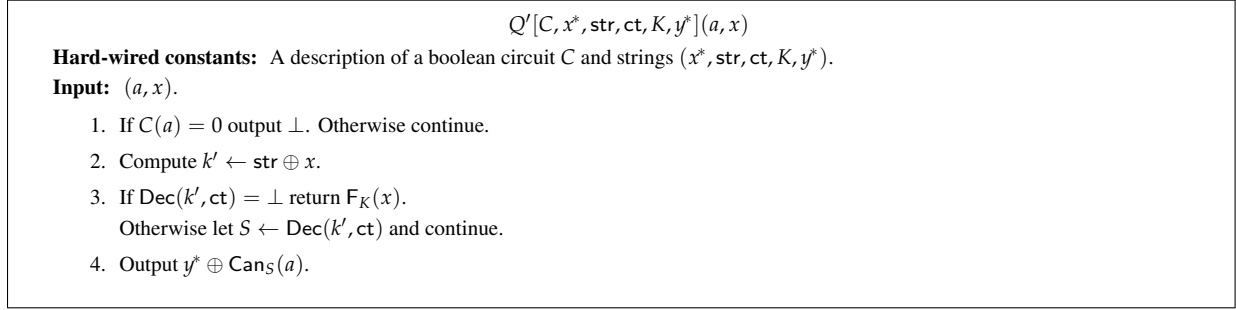


Figure 4: Description of the circuit Q' .

which evidently contain information about $(y_B \oplus s, y_C \oplus t)$.

To overcome this problem, we adopt the idea of [KY25]. Let Q be the circuit defined in Figure 3. Then the circuits

$$P'[C_{\text{mc}}, x_B, K\{x_B\}, y_B \oplus s] \quad \text{and} \quad P'[C_{\text{mc}}^\perp, x_C, K\{x_C\}, y_C \oplus t]$$

are functionally equivalent to

$$Q[C_{\text{mc}}, x_B, K\{x_B\}, y_B, A] \quad \text{and} \quad Q[C_{\text{mc}}^\perp, x_C, K\{x_C\}, y_C, A^\perp],$$

because $\text{Can}_A(a) = s$ for $a \in A + s$ and, analogously, $\text{Can}_{A^\perp}(a) = t$ for $a \in A^\perp + t$, where we used $(s, t) \in \text{CS}(A) \times \text{CS}(A^\perp)$. Replacing the obfuscations accordingly (using the security of iO) removes the dependence on $(y_B \oplus s, y_C \oplus t)$.

Nevertheless, this is still insufficient: the obfuscated programs must be given to the adversary in the *first* stage as part of s_K , whereas in the computational strong monogamy game, the description of A is revealed only in the *second* stage. This mismatch makes the approach incompatible.¹⁰ Consequently, we require an additional idea to remove the hardwired descriptions of A and $A^\perp$ from the obfuscated programs.

Second idea: key-robust non-committing encryption. The rough idea is as follows. We replace the hardwired descriptions of A and $A^\perp$ inside the obfuscated programs by their “encryption” whose decryption keys will later be derived from the challenge inputs x_B and x_C . Because we eventually need to remove information about A , the ciphertext must be simulatable *without* knowing A . At the same time, to preserve functionality, the simulated ciphertext must remain *openable* to an arbitrary value—either A or $A^\perp$ —once the appropriate key is provided.

To realize the above idea, we formalize a simple primitive called *key-robust (one-time) non-committing encryption* specified by the following algorithms:

- $\text{Gen}(1^\lambda)$: Output a key k .
- $\text{Enc}(k, m)$: On key k and message m , output a ciphertext ct .

¹⁰For SDE, this issue does not arise because the corresponding obfuscated circuit represents ciphertexts supplied in the second stage; see [KY25].

- $\text{Dec}(k, \text{ct})$: On key k and ciphertext ct , output m' or $\perp$.
- $\text{Fake}(1^\lambda)$: Output a fake ciphertext ct^* and state information st .
- $\text{Open}(\text{st}, m^*)$: On st and target message m^* , output a simulated key k^* .

Besides standard correctness of decryption, the primitive satisfies:¹¹

Key-robustness For any distinct keys $k \neq k'$, $\text{Dec}(k', \text{Enc}(k, m)) = \perp$ for all m .

Non-committing For every m , the following distributions are computationally indistinguishable:

$$(\text{ct}, k) \stackrel{c}{\approx} (\text{ct}^*, k^*),$$

where $k \leftarrow \text{Gen}(1^\lambda)$, $\text{ct} \leftarrow \text{Enc}(k, m)$, $(\text{ct}^*, \text{st}) \leftarrow \text{Fake}(1^\lambda)$, $k^* \leftarrow \text{Open}(\text{st}, m)$.

Key-robust non-committing encryption can be easily constructed from any injective one-way function, which in turn can be built from iO and OWFs; see Section 4.

We now explain how to employ this primitive in the security proof. Generate

$$k_B, k_C \leftarrow \text{Gen}(1^\lambda), \quad \text{ct}_B \leftarrow \text{Enc}(k_B, A), \quad \text{ct}_C \leftarrow \text{Enc}(k_C, A^\perp),$$

and set $\text{str}_B = k_B \oplus x_B$, $\text{str}_C = k_C \oplus x_C$. By key-robustness and decryption correctness, the circuits

$$Q[C_{\text{mc}}, x_B, K\{x_B\}, y_B, A] \quad \text{and} \quad Q[C_{\text{mc}}^\perp, x_C, K\{x_C\}, y_C, A^\perp]$$

are functionally equivalent to

$$Q'[C_{\text{mc}}, x_B, \text{str}_B, \text{ct}_B, K\{x_B\}, y_B] \quad \text{and} \quad Q'[C_{\text{mc}}^\perp, x_C, \text{str}_C, \text{ct}_C, K\{x_C\}, y_C],$$

where Q' is defined in Figure 4. Hence, by invoking the security of iO, we can replace the obfuscations accordingly.

Next we invoke the non-committing property to replace ct_B and ct_C by *fake* ciphertexts and, at a later stage, open those ciphertexts to A and $A^\perp$ via Open , thereby obtaining simulated keys k_B and k_C . This modification is computationally indistinguishable by non-committing security.

A subtlety remains: the obfuscated programs must embed $\text{str}_B = k_B \oplus x_B$ and $\text{str}_C = k_C \oplus x_C$ which seem to require knowledge of k_B and k_C —and thus of A and $A^\perp$ —*before* the programs are generated. However, because the challenge inputs x_B, x_C are uniform, the strings $\text{str}_B, \text{str}_C$ are themselves uniform. Hence we may simply choose $\text{str}_B, \text{str}_C$ uniformly at random during the first stage. After the description of A becomes available, we open ct_B, ct_C to A and $A^\perp$ to obtain the simulated keys k_B, k_C , and then set $m_B := \text{str}_B \oplus k_B$ and $m_C := \text{str}_C \oplus k_C$ accordingly. The resulting distribution is identical to the previous one, while the generation of the obfuscated programs no longer depends on A or $A^\perp$.

At this point the first stage requires no knowledge of A , s , or t beyond the coset state $|A_{s,t}\rangle$ and the obfuscated programs $C_{\text{mc}} = i\mathcal{O}(A + s)$, $C_{\text{mc}}^\perp = i\mathcal{O}(A^\perp + t)$, which are precisely what are available in the computational strong monogamy game. As previously explained, we can now extract (s, t) from the outputs of B, C whenever they win the security game of the copy-protected UPF, thereby completing the reduction.

2.2 Extending to Non-Uniform Challenge Distributions

We now explain how to adapt the above construction and security proof to challenge distributions that are *not* uniform. Observe that uniformity was used only at the very end of the proof, when we swapped the sampling order of (x_B, x_C) and $(\text{str}_B, \text{str}_C)$. In the hybrid before the swap, we first sample x_B, x_C and then set $\text{str}_B = k_B \oplus x_B$ and $\text{str}_C = k_C \oplus x_C$. Uniformity allowed us to *reverse* this order—i.e. sample $\text{str}_B, \text{str}_C$ uniformly and then define $x_B = k_B \oplus \text{str}_B$ and $x_C = k_C \oplus \text{str}_C$. For non-uniform distributions such a swap is no longer justified.

¹¹Negligible errors are omitted for simplicity.

Hash-and-Eval remedy. To remedy this, we apply the Hash-and-Eval technique already discussed: first pass the challenge input through a universal hash function h , then run the uniform-case construction on the hash output. After incorporating this modification, in the corresponding hybrid, we have

$$\text{str}_{\mathcal{B}} = k_{\mathcal{B}} \oplus h(x_{\mathcal{B}}), \quad \text{str}_{\mathcal{C}} = k_{\mathcal{C}} \oplus h(x_{\mathcal{C}}).$$

By the leftover hash lemma, if $x_{\mathcal{B}}$ and $x_{\mathcal{C}}$ have sufficient min-entropy, $h(x_{\mathcal{B}})$ and $h(x_{\mathcal{C}})$ are statistically close to uniform, and so are $\text{str}_{\mathcal{B}}, \text{str}_{\mathcal{C}}$. One might therefore hope to swap the sampling order as before: first choose $\text{str}_{\mathcal{B}}, \text{str}_{\mathcal{C}}$ uniformly, then—after $k_{\mathcal{B}}, k_{\mathcal{C}}$ become known—sample $x_{\mathcal{B}}, x_{\mathcal{C}}$ conditioned on $\text{str}_{\mathcal{B}} = k_{\mathcal{B}} \oplus h(x_{\mathcal{B}})$ and $\text{str}_{\mathcal{C}} = k_{\mathcal{C}} \oplus h(x_{\mathcal{C}})$. Unfortunately, efficiently sampling from this conditional distribution is infeasible in general. In fact, this is exactly the obstacle encountered in the generic Hash-and-Eval approach.

Third idea: computational strong monogamy with simulatable auxiliary input. The crux is that, although the strings $\text{str}_{\mathcal{B}}$ and $\text{str}_{\mathcal{C}}$ do depend on the hidden subspace A , their distributions are *near-uniform*, so on its own it should not benefit the adversary. The complication is that $x_{\mathcal{B}}$ and $x_{\mathcal{C}}$ are revealed later, potentially granting the adversary extra leverage.

To overcome this obstacle, we strengthen the computational strong monogamy property so that it admits *simulatable auxiliary inputs*. Concretely, each party $\mathcal{A}, \mathcal{B}, \mathcal{C}$ may receive an efficiently generated auxiliary input that depends on the hidden subspace A , but the auxiliary input of $\mathcal{A}$ must remain *statistically simulatable* without using A . We formalize and prove this refined notion, which we call *computational strong monogamy with simulatable auxiliary inputs*. Once this is established, the security proof of our scheme for non-uniform challenge distributions can be completed by a straightforward reduction to the property.

Below, we sketch the idea for the proof of the above property. Recall the proof of computational strong monogamy in [CLLZ21]: start with the *information-theoretic* version (no obfuscated programs), then apply the subspace-hiding technique of [Zha19] to obtain the computational statement. Our key observation is that, in the information-theoretic setting, the simulatable auxiliary input is useless—the adversary can simply perform inefficient reverse sampling on its own. After showing this, we replay the transformation of [CLLZ21] to get the computational version, noting that the auxiliary inputs are generated efficiently, so the argument goes through unchanged.¹²

2.3 Handling Evaluation Queries.

So far, we have not considered evaluation queries. We now allow the adversary to make an arbitrary number of evaluation queries, with the restriction that adversaries $\mathcal{B}$ and $\mathcal{C}$ are not allowed to query on $x_{\mathcal{B}}$ and $x_{\mathcal{C}}$, respectively, in order to prevent trivial attacks.

The security proof for the uniform challenge setting extends naturally to the case with evaluation queries. This is because, throughout the proof, the reduction has access to either the full keys $K, K^{\perp}$ or the punctured keys $K\{x_{\mathcal{B}}\}, K^{\perp}\{x_{\mathcal{C}}\}$. These are sufficient to simulate the evaluation oracle, as the adversary is prohibited from querying the challenge inputs themselves.¹³ Thus, security in the presence of evaluation queries follows with no additional complications in the uniform challenge setting.

However, the situation becomes more subtle in the case of a general high min-entropy challenge distribution. In this setting, we first apply a universal hash function h to the input, and then apply the construction designed for uniform challenge inputs. Consequently, in the security proof, we must now puncture the keys at $h(x_{\mathcal{B}})$ and $h(x_{\mathcal{C}})$, rather than at $x_{\mathcal{B}}$ and $x_{\mathcal{C}}$ themselves. This introduces a potential vulnerability: if the adversary can find inputs $x'_{\mathcal{B}}$ or $x'_{\mathcal{C}}$ such that $h(x'_{\mathcal{B}}) = h(x_{\mathcal{B}})$ or $h(x'_{\mathcal{C}}) = h(x_{\mathcal{C}})$, then they can query these to the evaluation oracle. Since the UPF values depend only on the hashed inputs, such queries would return the same values as on $x_{\mathcal{B}}$ and $x_{\mathcal{C}}$, effectively bypassing the challenge restriction and enabling an attack. As h is a universal hash function without any assumed cryptographic hardness, finding such collisions may be feasible.

¹²In the technical section, we prove the computational strong monogamy property with simulatable auxiliary input (Corollary 5.4) as a corollary of a stronger, indistinguishability-based variant (Theorem 5.2). However, the direct proof presented here would also be possible.

¹³Strictly speaking, $\mathcal{A}$ may query $x_{\mathcal{B}}$ and $x_{\mathcal{C}}$, and $\mathcal{B}$ and $\mathcal{C}$ may query $x_{\mathcal{C}}$ and $x_{\mathcal{B}}$, respectively, but this happens with negligible probability since $x_{\mathcal{B}}$ and $x_{\mathcal{C}}$ are sampled uniformly at random.

A simple fix to this issue would be to assume that h is also target collision-resistant. In that case, the above attack would no longer be possible, and the proof would go through.¹⁴ Unfortunately, we do not know how to construct a hash function that is both universal and target collision-resistant. Therefore, we instead assume two separate components: a universal hash function h and a target collision-resistant hash function H_t .

Using them, consider the following construction: for any input x , we compute both $h(x)$ and $H_t(x)$, concatenate them as $h(x)||H_t(x)$, and apply the construction for the uniform setting to this combined value. This approach mitigates the attack: while $h(x)$ alone may admit collisions, the use of H_t ensures that it is hard to find any x' such that $H_t(x') = H_t(x_B)$ or $H_t(x') = H_t(x_C)$, due to the target collision resistance of H_t .

One might worry that the concatenated value $h(x)||H_t(x)$ is no longer close to uniform. However, this is not an issue. As long as some component of the combined input—namely, the output of the universal hash $h(x)$ —remains close to uniform, the security proof essentially carries over with appropriate adaptations.

2.4 Copy-Protected Signatures

A copy-protected UPF already can be regarded as a copy-protected message authentication code (MAC)—that is, a privately verifiable version of a signature scheme. To obtain full-fledged copy-protected signatures, we simply need to add a *public* verification algorithm. This can be achieved using the standard puncturing-and-programming technique: we prepare two additional obfuscated programs that allow verification of (x, Y) , specifically to check whether Y equals the UPF value on x .

Introducing these obfuscated verification programs does not interfere with any of the game hops. All transitions in the proof still go through, relying on similar puncturing arguments as before. The only subtle point arises in how the obfuscated verification programs check the reprogrammed outputs on the challenge inputs x_B and x_C . Recall that, after the “shifting” step, the first half of the UPF value at x_B and the second half at x_C are replaced with $y_B \oplus s$ and $y_C \oplus t$, respectively.

At first glance, it may seem that the obfuscated verification programs would need to *know* the shift values s and t , which would compromise the reduction. However, this issue is resolved using the following simple observation: for the obfuscated circuits

$$C_{mc} = i\mathcal{O}(A + s) \quad \text{and} \quad C_{mc}^\perp = i\mathcal{O}(A^\perp + t),$$

since $(s, t) \in \text{CS}(A) \times \text{CS}(A^\perp)$, the following equivalences hold:

$$s' = s \iff (C_{mc}(s') = 1 \wedge \text{Can}_A(s') = s'), \quad t' = t \iff (C_{mc}^\perp(t') = 1 \wedge \text{Can}_{A^\perp}(t') = t').$$

Thus, equality with s or t can be checked without revealing s or t themselves, using only the obfuscated circuits C_{mc} , $C_{mc}^\perp$, and the description of A . The dependence on A is subsequently removed exactly as in the original UPF security proof, via key-robust non-committing encryption.

With these techniques, we obtain copy-protected signatures that inherit both support for non-uniform challenge distributions and security in the presence of signing queries.

2.5 Copy-Protected PRFs

We briefly discuss how to extend our construction to copy-protected PRFs under an indistinguishability-based security notion. Very roughly, the security guarantees that a single quantum secret key cannot be used to generate two (possibly entangled) quantum states that enable distinguishing the PRF values on challenge inputs x_B and x_C —sampled from high min-entropy distributions—from uniformly random values, simultaneously.¹⁵

A crucial difference here is that this is a decision game: the adversary’s outputs are just bits. Thus, the proof strategy used for copy-protected UPFs and signatures—where we extract the values s and t from the adversary’s output—no longer applies. Instead, we take a different approach. Rather than reducing security to the computational strong monogamy property with simulatable auxiliary inputs (as before), we consider its indistinguishability-based version and reduce to that.

¹⁴Note that target collision resistance—rather than full collision resistance—suffices here, since x_B and x_C are sampled independently of h .

¹⁵Our actual definition adopts the “+”-style security introduced by [KY25] in the context of SDE. See Section 6.2 for details.

The indistinguishability-based version is similar in structure, except that the second stage adversaries are each given a random string r_B and r_C , respectively. Their goal is then to guess $\langle r_B, s \rangle$ and $\langle r_C, t \rangle$, rather than to recover the full s and t .¹⁶ We prove this indistinguishability-based variant of the computational strong monogamy property with simulatable auxiliary inputs via the simultaneous quantum Goldreich-Levin lemma [AKY25], together with the techniques previously explained for the search-based version.

Our strategy is to embed $\langle r_B, s \rangle$ and $\langle r_C, t \rangle$ into the challenge bits of the copy-protected PRF security game. In this way, if the adversary correctly guesses the challenge bits, it enables the reduction algorithm to correctly compute $\langle r_B, s \rangle$ and $\langle r_C, t \rangle$.¹⁷ While we omit the details, this embedding is achieved using the same key-robust non-committing encryption technique described earlier. Similarly, the techniques for handling non-uniform challenge distributions and evaluation queries carry over directly from the UPF case, as previously discussed.

2.6 Construction of UPO

Finally, we briefly explain how to construct a UPO. A UPO is a type of obfuscator that encodes a classical circuit into a quantum form in such a way that it becomes copy-protected. While we omit the details here (see Section 6.3 for formal definitions), its security is formulated as a decision-based game. As a result, we apply a similar strategy to that used for copy-protected PRFs.

In fact, when the challenge distribution is uniform, essentially the same proof applies, allowing us to construct UPOs from polynomially secure iO and OWFs. Note that this already significantly improves upon the result of [AB24] in the independent challenge input setting, as it removes the reliance on a conjecture and weakens the underlying assumptions. A subtlety here is that the security definition of UPO in [AB24] considers identical challenge *bits* even in the independent challenge *input* setting.¹⁸ In contrast, our proof thus far focuses on the setting where both the challenge *bits* and *inputs* are independent. The key insight that resolves this discrepancy stems from a simple observation made in [KY25], which was itself inspired by [AKY25]. In that work, within the context of SDE, the authors argue that security in the setting where the challenge *inputs* are independent but the challenge *bits* are identical can be reduced to security in the setting where both the challenge *bits* and *inputs* are independent—under a slightly strengthened “+”-style security notion, where the adversary is considered successful not only when both parties’ guesses are *correct*, but also when both are *incorrect*. Our proof naturally achieves this “+”-style security notion, as it builds on the simultaneous quantum Goldreich-Levin theorem from [AKY25], which already provides it. Therefore, our result also implies security in the setting with identical challenge *bits* and independent challenge *inputs*.

On the other hand, the situation is less clear when the challenge distribution is non-uniform. In a nutshell, the issue is that we might want to apply a universal hash function h and then use the construction for the uniform distribution case. However, doing so alters the functionality: if we intend to obfuscate a circuit C , the resulting quantum circuit instead evaluates $C \circ h$, which does not preserve the original functionality.

To address this, we introduce an alternative technique that avoids applying the hash function h in the construction itself and instead uses it only within the security proof. While we omit the details here, the approach relies on the use of lossy functions and therefore requires an additional algebraic assumption, such as the LWE assumption. This technique eliminates the need for key-robust non-committing encryption, but still relies on the computational strong monogamy-of-entanglement property with simulatable auxiliary input.

3 Preliminaries

Notations and conventions. In this paper, standard math or sans serif font stands for classical algorithms (e.g., C or Gen) and classical variables (e.g., x or pk). Calligraphic font stands for quantum algorithms (e.g., $\mathcal{G}$ or $\mathcal{Gen}$) and calligraphic font and/or the bracket notation for (mixed) quantum states (e.g., q or $|\psi\rangle$).

Let $[\ell]$ denote the set of integers $\{1, \dots, \ell\}$, λ denote a security parameter, and $y := z$ denote that y is set, defined, or substituted by z . For a finite set X and a distribution D , $x \leftarrow X$ denotes selecting an element from X uniformly at random, $x \leftarrow D$ denotes sampling an element x according to D . Let $y \leftarrow A(x)$ and $y \leftarrow \mathcal{A}(x)$ denote assigning to y

¹⁶Due to a technical reason, we must also allow for auxiliary inputs that depend on r_B and r_C , as formalized in Theorem 5.2.

¹⁷The idea of embedding $\langle r_B, s \rangle$ and $\langle r_C, t \rangle$ into challenge bits also appears in the work of [KY25] in the context of SDE.

¹⁸See Footnote 2 for the meaning of *challenge bits* and *challenge inputs*.

the output of a probabilistic or deterministic algorithm A and a quantum algorithm $\mathcal{A}$ on an input x and χ , respectively. When we explicitly show that A uses randomness r , we write $y \leftarrow A(x; r)$. PPT and QPT algorithms stand for probabilistic polynomial-time algorithms and polynomial-time quantum algorithms, respectively. Let negl denote a negligible function, non-negl denote a non-negligible function, and poly denote a positive polynomial. For a string $x \in \{0, 1\}^n$, $x[i]$ is its i -th bit. For strings $x, y \in \{0, 1\}^n$, $\langle x, y \rangle$ denotes $\bigoplus_{i \in [n]} x[i] \cdot y[i]$. We often use the similar notation for vectors $x, y \in \mathbb{F}_2^n$ by identifying elements of $\mathbb{F}_2$ and $\{0, 1\}$ in the natural manner.

We use gray sans serif font (e.g., $\mathbb{X}$) to stand for a quantum register. For a quantum state q over registers R_1 and R_2 , we write $q[R_1]$ to denote the portion of the state on register R_1 . Similarly, for a pure quantum state $|\psi\rangle$ over registers R_1 and R_2 , we write $|\psi\rangle_{R_1}$ to denote the portion of the state on register R_1 . For quantum states q and q' , $\|q - q'\|_{tr}$ denotes their trace distance.

3.1 Min-Entropy, Universal Hash Functions, and Leftover Hash Lemma

Definition 3.1 (Min-Entropy of Distributions). Let D be a distribution over output space $\mathcal{X}$ and auxiliary information space $\{0, 1\}^{\ell_{\text{aux}}}$. We say that D has conditional min-entropy at least p if for any unbounded $\mathcal{A}$, we have

$$\Pr[\mathcal{A}(z) = x | (x, z) \leftarrow D] \leq 2^{-p}.$$

Definition 3.2 (Universal Hash Functions). A family of function $\mathcal{H} : \mathcal{X} \rightarrow \mathcal{Y}$ is said to be universal, if it holds that for all $x, x' \in \mathcal{X}$ such that $x \neq x'$, we have

$$\Pr_{h \leftarrow \mathcal{H}}[h(x) = h(x')] \leq \frac{1}{|\mathcal{Y}|}.$$

Lemma 3.3 (Leftover Hash Lemma). Let $\mathcal{H} : \mathcal{X} \rightarrow \mathcal{Y}$ be a universal hash function family. Let D be a distribution over $\mathcal{X} \times \{0, 1\}^*$ with conditional min-entropy p . If $\log |\mathcal{Y}| + 2 \log \left(\frac{1}{\epsilon}\right) \leq p$, then the statistical distance between

$$\{h, h(x), z : h \leftarrow \mathcal{H}, (x, z) \leftarrow D\} \quad \text{and} \quad \{h, y, z : h \leftarrow \mathcal{H}, (x, z) \leftarrow D, y \leftarrow \mathcal{Y}\}$$

is bounded by ϵ .

3.2 Classical Cryptographic Primitives

Indistinguishability obfuscation

Definition 3.4 (Indistinguishability Obfuscator [BGI⁺12]). A PPT algorithm $i\mathcal{O}$ is a secure iO for a classical circuit class $\{\mathcal{C}_\lambda\}_{\lambda \in \mathbb{N}}$ if it satisfies the following two conditions.

Functionality: For any security parameter $\lambda \in \mathbb{N}$, circuit $C \in \mathcal{C}_\lambda$, and input x , we have that

$$\Pr[C'(x) = C(x) \mid C' \leftarrow i\mathcal{O}(C)] = 1.$$

Indistinguishability: For any PPT Sampler and QPT distinguisher $\mathcal{A}$, the following holds:

If $\Pr[\forall x \ C_0(x) = C_1(x) \wedge |C_0| = |C_1| \mid (C_0, C_1, \text{aux}) \leftarrow \text{Sampler}(1^\lambda)] > 1 - \text{negl}(\lambda)$, then we have

$$\begin{aligned} \text{Adv}_{i\mathcal{O}, \mathcal{D}}^{\text{io}}(\lambda) := & \left| \Pr \left[\mathcal{A}(i\mathcal{O}(C_0), \text{aux}) = 1 \mid (C_0, C_1, \text{aux}) \leftarrow \text{Sampler}(1^\lambda) \right] \right. \\ & \left. - \Pr \left[\mathcal{A}(i\mathcal{O}(C_1), \text{aux}) = 1 \mid (C_0, C_1, \text{aux}) \leftarrow \text{Sampler}(1^\lambda) \right] \right| \leq \text{negl}(\lambda). \end{aligned}$$

There are several candidates of secure iO for polynomial-size classical circuits against quantum adversaries [BGMZ18, CHVW19, AP20, BDGM20, WW21, GP21, DQV⁺21, BDGM22, BDJ⁺24, HJL25, CLW25].

Pseudorandom functions.

Definition 3.5 (Puncturable PRF). A puncturable PRF (PPRF) is a tuple of algorithms $\text{PPRF} = (\text{PRF.Gen}, \text{F}, \text{Puncture})$ where $\{F_K : \{0, 1\}^{\ell_{\text{inp}}} \rightarrow \{0, 1\}^{\ell_{\text{out}}} \mid K \in \{0, 1\}^\lambda\}$ is a PRF family and satisfies the following two conditions.

Punctured correctness: For any polynomial-size set $S \subseteq \{0, 1\}^{\ell_{\text{inp}}}$ and any $x \in \{0, 1\}^{\ell_{\text{inp}}} \setminus S$, it holds that

$$\Pr[F_K(x) = F_{K\{S\}}(x) \mid K \leftarrow \text{PRF.Gen}(1^\lambda), K\{S\} \leftarrow \text{Puncture}(K, S)] = 1.$$

Pseudorandom at punctured point: For any polynomial-size set $S \subseteq \{0, 1\}^{\ell_{\text{inp}}}$ and any QPT distinguisher $\mathcal{A}$, it holds that

$$\left| \Pr[\mathcal{A}(F_{K\{S\}}, \{F_K(x_i)\}_{x_i \in S}) \rightarrow 1] - \Pr[\mathcal{A}(F_{K\{S\}}, (\mathcal{U}_{\ell_{\text{out}}})^{|S|}) \rightarrow 1] \right| \leq \text{negl}(\lambda),$$

where $K \leftarrow \text{PRF.Gen}(1^\lambda)$, $K\{S\} \leftarrow \text{Puncture}(K, S)$ and $\mathcal{U}_{\ell_{\text{out}}}$ denotes the uniform distribution over $\{0, 1\}^{\ell_{\text{out}}}$.

If $S = \{x_1^*, x_2^*, \dots, x_n^*\}$ for some $n \in \mathbb{N}$, we simply denote the punctured key by $K\{x_1^*, x_2^*, \dots, x_n^*\}$ instead of $K\{\{x_1^*, x_2^*, \dots, x_n^*\}\}$.

Goldwasser-Goldreich-Micali tree-based construction of PRFs (GGM PRF) [GGM86] from OWF yield puncturable PRFs where the size of the punctured key grows polynomially with the size of the set S being punctured [BW13, BG14, KPTZ13].

Target collision resistant (TCR) hash functions.

Definition 3.6 (Target Collision Resistance Hash Functions). A target collision resistant hash function TCR is given by a PPT algorithm Gen that takes 1^λ as input, and outputs a description of a classical-polynomial-time-computable hash function $H_t : \{0, 1\}^{\ell_{\text{inp}}} \rightarrow \{0, 1\}^{\ell_{\text{out}}}$. It satisfies that for all QPT algorithms $\mathcal{A}$ and $x \in \{0, 1\}^{\ell_{\text{inp}}}$, we have

$$\Pr[H_t(x) = H_t(x') \mid H_t \leftarrow \text{Gen}(1^\lambda), x' \leftarrow \mathcal{A}(H_t, x)] = \text{negl}(\lambda).$$

A target collision resistant hash function can be constructed from any OWF [NY89, Rom90].

Keyed injective one-way functions.

Definition 3.7 (Keyed Injective One-Way Functions). A keyed injective one-way function IOWF is given by a PPT algorithm Gen that takes 1^λ as input, and outputs a description of a classical-polynomial-time-computable function $F : \{0, 1\}^{\ell_{\text{inp}}} \rightarrow \{0, 1\}^{\ell_{\text{out}}}$. It satisfies the following properties.

Injectivity: With overwhelming probability over the choice of $F \leftarrow \text{Gen}(1^\lambda)$, F is injective.

One-Wayness: For all QPT algorithms $\mathcal{A}$,

$$\Pr[F(x') = F(x) \mid F \leftarrow \text{Gen}(1^\lambda), x \leftarrow \{0, 1\}^{\ell_{\text{inp}}}, x' \leftarrow \mathcal{A}(1^\lambda, F, F(x))] = \text{negl}(\lambda).$$

Keyed injective one-way functions exist assuming the existence of iO and OWFs [BPW16].

Lossy functions.

Definition 3.8 (Lossy Functions). A collection of $(\ell_{\text{inp}}, \ell_{\text{loss}})$ -lossy functions LF is given by a pair of two PPT algorithms $(\text{Gen}_{\text{inj}}, \text{Gen}_{\text{loss}})$.

- The injective function generation algorithm Gen_{inj} takes the security parameter 1^λ as an input, and outputs a description of a classical-polynomial-time-computable injective function F_{inj} over the domain $\{0, 1\}^{\ell_{\text{inp}}}$.

- The lossy function generation algorithm Gen_{loss} takes the security parameter 1^λ as an input, and outputs a description of a classical-polynomial-time-computable function F_{loss} over the domain $\{0, 1\}^{\ell_{\text{inp}}}$ whose image size is at most $2^{\ell_{\text{inp}} - \ell_{\text{loss}}}$.

LF satisfies the following mode indistinguishability.

Mode Indistinguishability: We have $F_{\text{inj}} \stackrel{c}{\approx} F_{\text{loss}}$, where $F_{\text{inj}} \leftarrow \text{Gen}_{\text{inj}}(1^\lambda)$ and $F_{\text{loss}} \leftarrow \text{Gen}_{\text{loss}}(1^\lambda)$.

It is known that lossy functions can be constructed assuming the hardness of LWE [PW11]. The following formulation is adapted from [CLLZ21].

Theorem 3.9 ([PW11]). Assuming the polynomial hardness of LWE , for any constant $c > 0$, there exists a collection of $(\ell_{\text{inp}}, \ell_{\text{loss}})$ -lossy functions such that $\ell_{\text{inp}} - \ell_{\text{loss}} \leq \ell_{\text{inp}}^c$.

3.3 Quantum Goldreich-Levin

Ananth et al. [AKY25] extended the quantum Goldreich-Levin lemma [AC02, CLLZ21] to the non-local setting, where two non-communicating algorithms with potentially entangled quantum inputs are involved.¹⁹

Lemma 3.10 (Simultaneous Quantum Goldreich-Levin [AKY25]). There exists a QPT oracle algorithm $\mathcal{Ext}$ that satisfies the following. Let $n \in \mathbb{N}$, $x_{\mathcal{B}}, x_{\mathcal{C}} \in \{0, 1\}^n$, $\epsilon \in [0, 1/2]$, and q be a quantum state over registers $\mathcal{R}_{\mathcal{B}}$ and $\mathcal{R}_{\mathcal{C}}$. Let $\mathcal{B}$ (resp. $\mathcal{C}$) be a quantum algorithm that takes $\mathcal{R}_{\mathcal{B}}$ (resp. $\mathcal{R}_{\mathcal{C}}$) and an n -bit string $r_{\mathcal{B}}$ (resp. $r_{\mathcal{C}}$) as input, and outputs a bit. If we have

$$\Pr \left[b_{\mathcal{B}} \oplus b_{\mathcal{C}} = \langle r_{\mathcal{B}}, x_{\mathcal{B}} \rangle \oplus \langle r_{\mathcal{C}}, x_{\mathcal{C}} \rangle : \begin{array}{l} r_{\mathcal{B}}, r_{\mathcal{C}} \leftarrow \{0, 1\}^n \\ b_{\mathcal{B}} \leftarrow \mathcal{B}(q[\mathcal{R}_{\mathcal{B}}], r_{\mathcal{B}}) \\ b_{\mathcal{C}} \leftarrow \mathcal{C}(q[\mathcal{R}_{\mathcal{C}}], r_{\mathcal{C}}) \end{array} \right] \geq \frac{1}{2} + \epsilon,$$

then, we have

$$\Pr [\mathcal{Ext}(\mathcal{B}, q[\mathcal{R}_{\mathcal{B}}]) \rightarrow x_{\mathcal{B}} \wedge \mathcal{Ext}(\mathcal{C}, q[\mathcal{R}_{\mathcal{C}}]) \rightarrow x_{\mathcal{C}}] \geq 4\epsilon^2.$$

where $\mathcal{B}$ and $\mathcal{C}$ in the input of $\mathcal{Ext}$ mean their descriptions.

Lemma 3.10 was originally proven in [AKY25, Lemma 5] for the special case of $x_{\mathcal{B}} = x_{\mathcal{C}}$. However, their proof does not rely on this assumption, and the same argument holds for the more general case where $x_{\mathcal{B}} \neq x_{\mathcal{C}}$, as observed in [KY25].

4 Key-Robust Non-committing Encryption.

We introduce the notion of key-robust (one-time) non-committing encryption, which serves as one of the building blocks in our copy-protection schemes. We construct such key-robust non-committing encryption from any keyed injective one-way function, which exist assuming the existence of iO and one-way functions [BPW16].

Definition 4.1 (Key-Robust Non-committing Encryption). A key-robust non-committing encryption scheme NCE with the message space $\mathcal{M}$ is a tuple of five PPT algorithms $(\text{Gen}, \text{Enc}, \text{Dec}, \text{Fake}, \text{Open})$.

$\text{Gen}(1^\lambda)$: The key generation algorithm takes as input a security parameter 1^λ and outputs a key k .

$\text{Enc}(k, m) \rightarrow \text{ct}$ The encryption algorithm takes as input a key k and a message $m \in \mathcal{M}$, and outputs a ciphertext ct .

$\text{Dec}(k, \text{ct}) \rightarrow m'$ The decryption algorithm is a deterministic algorithm that takes as input a key k and a ciphertext ct , and outputs $m' \in \mathcal{M} \cup \{\perp\}$.

¹⁹A slightly weaker version is also proven in [AKL23]

$\text{Fake}(1^\lambda) \rightarrow (\text{ct}^*, \text{st})$ The fake algorithm takes as input a security parameter 1^λ , and outputs a fake ciphertext ct^* and a state information st .

$\text{Open}(\text{st}, m) \rightarrow k^*$ The open algorithm takes as input a state information st and a message $m \in \mathcal{M}$, and outputs a key k^* .

We require it to satisfy the following three properties.

Correctness: With overwhelming probability over the choice of $k \leftarrow \text{Gen}(1^\lambda)$, for any $m \in \mathcal{M}$,

$$\Pr[\text{Dec}(k, \text{Enc}(k, m)) = m] = 1.$$

Key-Robustness: With overwhelming probability over the choice of $k \leftarrow \text{Gen}(1^\lambda)$, for any $m \in \mathcal{M}$ and any $k' \neq k$,

$$\Pr[\text{Dec}(k', \text{Enc}(k, m)) = \perp] = 1.$$

Non-committing: For any message $m \in \mathcal{M}$, we have

$$(\text{ct}, k) \stackrel{\mathcal{C}}{\approx} (\text{ct}^*, k^*),$$

where $k \leftarrow \text{Gen}(1^\lambda)$, $\text{ct} \leftarrow \text{Enc}(k, m)$, $(\text{ct}^*, \text{st}) \leftarrow \text{Fake}(1^\lambda)$, and $k^* \leftarrow \text{Open}(\text{st}, \text{ct}^*)$.

We say that a key k is good if both $\Pr[\text{Dec}(k, \text{Enc}(k, m)) = m] = 1$ and $\Pr[\text{Dec}(k', \text{Enc}(k, m)) = \perp] = 1$ hold for all m and $k' \neq k$. By the union bound, a key generated by $\text{Gen}(1^\lambda)$ is good with an overwhelming probability.

Theorem 4.2. Assuming the existence of keyed injective OWFs, for any polynomial ℓ_m , there exists a key-robust non-committing encryption scheme with the message space $\{0, 1\}^{\ell_m}$, where the key length is linear in ℓ_m .

Proof. It suffices to construct a scheme for single-bit messages since we can expand the message space by a simple bit-by-bit encryption (under independent keys for each bit). Let FGen be a generation algorithm for a keyed injective one-way function. Then we construct a key-robust non-committing encryption scheme with the message space $\{0, 1\}$ and key space $\{0, 1\}^{\ell_{\text{inp}}}$ as follows.

$\text{Gen}(1^\lambda)$: On input 1^λ , generate $F \leftarrow \text{FGen}(1^\lambda)$ and $x \leftarrow \{0, 1\}^{\ell_{\text{inp}}}$, and output $k := (F, x)$.

$\text{Enc}(k, m)$: On input (F, x) and $m \in \{0, 1\}$, sample $x' \leftarrow \{0, 1\}^{\ell_{\text{inp}}}$ and $r_0, r_1 \leftarrow \{0, 1\}^{\ell_{\text{inp}}}$, sets

$$\text{ct}_m := (F(x), r_m, \langle r_m, x \rangle)$$

and

$$\text{ct}_{m \oplus 1} := (F(x'), r_{m \oplus 1}, \langle r_{m \oplus 1}, x' \rangle \oplus 1),$$

and output $\text{ct} := (F, \text{ct}_0, \text{ct}_1)$.

$\text{Dec}(k, \text{ct})$: On input k and $\text{ct} = (F, \text{ct}_0, \text{ct}_1)$, if the first component of k disagrees with F , output $\perp$. Otherwise, parse $k = (F, x)$ and $\text{ct}_b = (y_b, r_b, \beta_b)$ for $b \in \{0, 1\}$. If there is unique $b \in \{0, 1\}$ such that $y_b = F(x)$ and $\beta_b = \langle r_b, x \rangle$, then output b , and output $\perp$ otherwise.

$\text{Fake}(1^\lambda)$: On input 1^λ , generate $F \leftarrow \text{FGen}(1^\lambda)$, sample $x_0, x_1 \leftarrow \{0, 1\}^{\ell_{\text{inp}}}$ and $r_0, r_1 \leftarrow \{0, 1\}^{\ell_{\text{inp}}}$, sets

$$\text{ct}_b^* := (F(x_b), r_b, \langle r_b, x_b \rangle)$$

for $b \in \{0, 1\}$, and output $\text{ct}^* := (F, \text{ct}_0^*, \text{ct}_1^*)$ and $\text{st} := (F, x_0, x_1)$.

$\text{Open}(\text{st}, m)$: On input st and $m \in \{0, 1\}$, parse $\text{st} = (F, x_0, x_1)$ and output $k^* := (F, x_m)$.

Correctness and key-robustness are clear from the injectivity of FGen . Below, we show the non-committing property. Fix $m \in \{0, 1\}$ and consider the following hybrids of distributions.

D_0 : This corresponds to the distribution in the LHS of the non-committing property. That is, it samples as follows: Generate $F \leftarrow \text{FGen}(1^\lambda)$, sample $x, x' \leftarrow \{0, 1\}^{\ell_{\text{inp}}}$, and $r_0, r_1 \leftarrow \{0, 1\}^{\ell_{\text{inp}}}$, set $\text{ct}_m := (F(x), r_m, \langle r_m, x \rangle)$ and $\text{ct}_{m \oplus 1} := (F(x'), r_{m \oplus 1}, \langle r_{m \oplus 1}, x' \rangle \oplus 1)$, and output $(\text{ct} = (F, \text{ct}_0, \text{ct}_1), k = (F, x))$.

D_1 : This is identical to D_0 except that $\text{ct}_{m \oplus 1}$ is generated as $\text{ct}_{m \oplus 1} := (F(x'), r_{m \oplus 1}, \langle r_{m \oplus 1}, x' \rangle)$. By the one-wayness of FGen and the Goldreich-Levin theorem [GL89], $D_1 \stackrel{\epsilon}{\approx} D_2$.

D_2 : This corresponds to the distribution in the LHS of the non-committing property. That is, it samples as follows: Generate $F \leftarrow \text{FGen}(1^\lambda)$, sample $x_0, x_1 \leftarrow \{0, 1\}^{\ell_{\text{inp}}}$, and $r_0, r_1 \leftarrow \{0, 1\}^{\ell_{\text{inp}}}$, set $\text{ct}_b := (F(x_b), r_b, \langle r_b, x_b \rangle)$ for $b \in \{0, 1\}$, and output $(\text{ct} = (\text{ct}_0, \text{ct}_1), k = (F, x_m))$.

It is easy to verify that D_1 and D_2 are identical: $D_1 \equiv D_2$.

Combining above, $D_0 \stackrel{\epsilon}{\approx} D_2$, and thus the non-committing property is proven. $\square$

5 Strong Monogamy Property of Coset States with Auxiliary Inputs

In this section, we introduce a new variant of the strong monogamy property of coset states that considers certain auxiliary inputs, and prove it.

Notations. For a subspace $A \subseteq \mathbb{F}_2^n$, we use the following notations introduced by [CLLZ21, AKL⁺22].

- $A^\perp := \{b \in \mathbb{F}_2^n : \forall a \in A, \langle a, b \rangle = 0\}$.
- Can_A is a function such that for any $s \in \mathbb{F}_2^n$, $\text{Can}_A(s)$ is the lexicographically smallest vector contained in $A + s$, which we call the canonical representative of coset $A + s$. For any $s' \in A + s$, we have $\text{Can}_A(s') = \text{Can}_A(s)$. Can_A is polynomial-time computable given the description of A .
- $\text{CS}(A) = \{\text{Can}_A(s) : s \in \mathbb{F}_2^n\}$.

For a subspace $A \subseteq \mathbb{F}_2^n$ and $(s, t) \in \mathbb{F}_2^n \times \mathbb{F}_2^n$, we define the *coset state* $|A_{s,t}\rangle$ as

$$|A_{s,t}\rangle := \frac{1}{\sqrt{|A|}} \sum_{a \in A} (-1)^{\langle a, t \rangle} |a + s\rangle.$$

By applying $H^{\otimes n}$ to $|A_{s,t}\rangle$, we obtain $|A_{t,s}^\perp\rangle$, where H is the Hadamard gate (thus $H^{\otimes n}$ is the Quantum Fourier Transformation over $\mathbb{F}_2^n$.)

When the context is clear, we often abuse notation by writing A to refer to its description, represented as a tuple of basis vectors. We often write $i\mathcal{O}(A + s)$ to denote an obfuscation by $i\mathcal{O}$ of the program that verifies membership in $A + s$.

We remark the following easy lemma, which is obvious from the definitions.

Lemma 5.1. *For a subspace $A \subseteq \mathbb{F}_2^n$, $s \in \text{CS}(A)$, and $s' \in \mathbb{F}_2^n$, the following two conditions are equivalent:*

1. $s = s'$
2. $s' \in A + s$ and $\text{Can}_A(s') = s'$.

Strong monogamy property. An information-theoretic property known as the strong monogamy property of coset states (Theorem 5.5) was conjectured by Coladangelo et al. [CLLZ21] and later proven by Culf and Vidick [CV22]. Coladangelo et al. [CLLZ21] also extended this property to a computational version using $i\mathcal{O}$. In this paper, we introduce a further variant of the property that considers an indistinguishability game instead of a search game, and, more importantly, allows the adversary to receive certain auxiliary inputs.²⁰ The formal statement is given below:

²⁰An indistinguishability version without auxiliary inputs was already introduced in [KY25].

Theorem 5.2 (Computational Strong Indistinguishability Monogamy Property of Coset States with Simulatable Auxiliary Inputs). Let $i\mathcal{O}$ be $i\mathcal{O}$. Let $n \in \mathbb{N}$ be a polynomial in λ . Let $\mathcal{Aux} = (\mathcal{AuxSetup}, \mathcal{AuxGen}_{\mathcal{B}}, \mathcal{AuxGen}_{\mathcal{C}})$ be a tuple of QPT algorithms with the following syntax.

$\mathcal{AuxSetup}(1^\lambda)$: This algorithm takes a security parameter 1^λ , and outputs a classical public parameter pp .

$\mathcal{AuxGen}_X(\text{pp}, A, r_X)$ where $X \in \{\mathcal{B}, \mathcal{C}\}$: This algorithm takes a public parameter pp , a description of a subspace $A \subseteq \mathbb{F}_2^n$, and $r_X \in \{0, 1\}^n$, and outputs a pair of classical strings $(z_{1,X}, z_{2,X})$.

We assume that, the first coordinate of the output of $\mathcal{AuxGen}_X(\text{pp}, A, r_X)$ can be statistically simulated from pp alone, without using (A, r_X) . That is, for $X \in \{\mathcal{B}, \mathcal{C}\}$, there exists a (not necessarily polynomial-time) quantum algorithm Sim_X such that for any subspace $A \subseteq \mathbb{F}_2^n$ and any $r_X \in \{0, 1\}^n$, the following distributions are statistically indistinguishable:

$$\left\{ (\text{pp}, z_{1,X}) : \begin{array}{l} \text{pp} \leftarrow \mathcal{AuxSetup}(1^\lambda) \\ (z_{1,X}, z_{2,X}) \leftarrow \mathcal{AuxGen}_X(\text{pp}, A, r_X) \end{array} \right\} \approx \left\{ (\text{pp}, z_{1,X}) : \begin{array}{l} \text{pp} \leftarrow \mathcal{AuxSetup}(1^\lambda) \\ z_{1,X} \leftarrow \text{Sim}_X(\text{pp}) \end{array} \right\}.$$

Consider the following experiment $\text{Exp}_{i\mathcal{O}, \mathcal{Aux}, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-ind-moe-aux}}(\lambda)$ between a challenger and an adversary $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$.

1. The challenger picks a uniformly random subspace $A \subseteq \mathbb{F}_2^n$ of dimension $n/2$, and two uniformly random canonical representatives $(s, t) \in \text{CS}(A) \times \text{CS}(A^\perp)$. The challenger picks $r_{\mathcal{B}}, r_{\mathcal{C}} \leftarrow \{0, 1\}^n$, and generates $\text{pp} \leftarrow \mathcal{AuxSetup}(1^\lambda)$, $(z_{1,\mathcal{B}}, z_{2,\mathcal{B}}) \leftarrow \mathcal{AuxGen}_{\mathcal{B}}(\text{pp}, A, r_{\mathcal{B}})$, and $(z_{1,\mathcal{C}}, z_{2,\mathcal{C}}) \leftarrow \mathcal{AuxGen}_{\mathcal{C}}(\text{pp}, A, r_{\mathcal{C}})$. It sends $|A_{s,t}\rangle$, $i\mathcal{O}(A + s)$, $i\mathcal{O}(A^\perp + t)$, pp , $z_{1,\mathcal{B}}$, and $z_{1,\mathcal{C}}$ to $\mathcal{A}$.
2. $\mathcal{A}$ creates a bipartite state q over registers $R_{\mathcal{B}}$ and $R_{\mathcal{C}}$. Then, $\mathcal{A}$ sends register $R_{\mathcal{B}}$ to $\mathcal{B}$ and register $R_{\mathcal{C}}$ to $\mathcal{C}$.
3. The challenger sends $(A, r_{\mathcal{B}}, z_{2,\mathcal{B}})$ to $\mathcal{B}$ and $(A, r_{\mathcal{C}}, z_{2,\mathcal{C}})$ to $\mathcal{C}$.
4. $\mathcal{B}$ and $\mathcal{C}$ respectively output $b_{\mathcal{B}}$ and $b_{\mathcal{C}}$. The challenger outputs 1 if $b_{\mathcal{B}} \oplus b_{\mathcal{C}} = \langle r_{\mathcal{B}}, s \rangle \oplus \langle r_{\mathcal{C}}, t \rangle$ and outputs 0 otherwise.

Assuming $i\mathcal{O}$ is a secure $i\mathcal{O}$ and there exist OWFs, for any QPT adversary $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$, it holds that

$$\Pr \left[\text{Exp}_{i\mathcal{O}, \mathcal{Aux}, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-ind-moe-aux}}(\lambda) = 1 \right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

Remark 5.3. We do not need to explicitly provide $(A, r_{\mathcal{B}})$ to $\mathcal{B}$ and $(A, r_{\mathcal{C}})$ to $\mathcal{C}$, as these can be included in $z_{2,\mathcal{B}}$ and $z_{2,\mathcal{C}}$ without loss of generality. However, we present them explicitly here for clarity.

We defer the proof of Theorem 5.2 to Section 5.1.

Theorem 5.2 immediately implies the following corollary, which can be seen as a search (as opposed to indistinguishability) version of Theorem 5.2.

Corollary 5.4 (Strong Monogamy Property of Coset States with Simulatable Auxiliary Inputs). Let $i\mathcal{O}$ be $i\mathcal{O}$. Let $n \in \mathbb{N}$ be a polynomial in λ . Let $\mathcal{Aux} = (\mathcal{AuxSetup}, \mathcal{AuxGen}_{\mathcal{B}}, \mathcal{AuxGen}_{\mathcal{C}})$ be a tuple of QPT algorithms with syntax similar to those in Theorem 5.2, except that $\mathcal{AuxGen}_X$, where $X \in \{\mathcal{B}, \mathcal{C}\}$, takes input (pp, A) and does not take r_X . We assume that $\mathcal{Aux}$ satisfies a simulatability condition analogous to that in Theorem 5.2. That is, for $X \in \{\mathcal{B}, \mathcal{C}\}$, there exists a (not necessarily polynomial-time) quantum algorithm Sim_X such that for any subspace $A \subseteq \mathbb{F}_2^n$, the following distributions are statistically indistinguishable:

$$\left\{ (\text{pp}, z_{1,X}) : \begin{array}{l} \text{pp} \leftarrow \mathcal{AuxSetup}(1^\lambda) \\ (z_{1,X}, z_{2,X}) \leftarrow \mathcal{AuxGen}_X(\text{pp}, A) \end{array} \right\} \approx \left\{ (\text{pp}, z_{1,X}) : \begin{array}{l} \text{pp} \leftarrow \mathcal{AuxSetup}(1^\lambda) \\ z_{1,X} \leftarrow \text{Sim}_X(\text{pp}) \end{array} \right\}.$$

Consider the following experiment $\text{Exp}_{i\mathcal{O}, \mathcal{Aux}, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-moe-aux}}(\lambda)$ between a challenger and an adversary $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$.

1. The challenger picks a uniformly random subspace $A \subseteq \mathbb{F}_2^n$ of dimension $n/2$, and two uniformly random canonical representatives $(s, t) \in \text{CS}(A) \times \text{CS}(A^\perp)$. The challenger generates $\text{pp} \leftarrow \mathcal{AuxSetup}(1^\lambda)$, $(z_{1,\mathcal{B}}, z_{2,\mathcal{B}}) \leftarrow \mathcal{AuxGen}_{\mathcal{B}}(\text{pp}, A, r_{\mathcal{B}})$, and $(z_{1,\mathcal{C}}, z_{2,\mathcal{C}}) \leftarrow \mathcal{AuxGen}_{\mathcal{C}}(\text{pp}, A, r_{\mathcal{C}})$. It sends $|A_{s,t}\rangle$, $i\mathcal{O}(A + s)$, $i\mathcal{O}(A^\perp + t)$, pp , $z_{1,\mathcal{B}}$, and $z_{1,\mathcal{C}}$ to $\mathcal{A}$.
2. $\mathcal{A}$ creates a bipartite state q over registers $R_{\mathcal{B}}$ and $R_{\mathcal{C}}$. Then, $\mathcal{A}$ sends register $R_{\mathcal{B}}$ to $\mathcal{B}$ and register $R_{\mathcal{C}}$ to $\mathcal{C}$.
3. The challenger sends $(A, z_{2,\mathcal{B}})$ to $\mathcal{B}$ and the description of $(A, z_{2,\mathcal{C}})$ to $\mathcal{C}$.
4. $\mathcal{B}$ and $\mathcal{C}$ respectively output $s_{\mathcal{B}}$ and $t_{\mathcal{C}}$. The challenger outputs 1 if $s_{\mathcal{B}} = s$ and $t_{\mathcal{C}} = t$, and outputs 0 otherwise.

Assuming $i\mathcal{O}$ is a secure $i\mathcal{O}$ and there exist OWFs, for any QPT adversary $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$, it holds that

$$\Pr \left[\text{Exp}_{i\mathcal{O}, \mathcal{Aux}, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-moe-aux}}(\lambda) = 1 \right] \leq \text{negl}(\lambda).$$

Proof. Let $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ and $\mathcal{Aux} = (\mathcal{AuxSetup}, \mathcal{AuxGen}_{\mathcal{B}}, \mathcal{AuxGen}_{\mathcal{C}})$ be QPT algorithms as in Corollary 5.4. Then we construct QPT algorithms $\mathcal{A}'_{\text{moe}} = (\mathcal{A}', \mathcal{B}', \mathcal{C}')$ and $\mathcal{Aux}' = (\mathcal{AuxSetup}', \mathcal{AuxGen}'_{\mathcal{B}}, \mathcal{AuxGen}'_{\mathcal{C}})$ as in Theorem 5.2 as follows:

$\mathcal{AuxSetup}'(1^\lambda)$: This is identical to $\mathcal{AuxSetup}(1^\lambda)$.

$\mathcal{AuxGen}'_{\mathcal{X}}(\text{pp}, A, r_{\mathcal{X}})$ where $\mathcal{X} \in \{\mathcal{B}, \mathcal{C}\}$: This behaves identically to $\mathcal{AuxGen}_{\mathcal{X}}(1^\lambda, A)$, disregarding the input $r_{\mathcal{X}}$.

$\mathcal{A}'$: This is identical to $\mathcal{A}$.

$\mathcal{B}'$: Upon receiving $q[R_{\mathcal{B}}]$ from $\mathcal{A}'$ and $(A, r_{\mathcal{B}}, z_{2,\mathcal{B}})$ from the challenger, run $\mathcal{B}$ on $(q[R_{\mathcal{B}}], A, z_{2,\mathcal{B}})$ to obtain $s_{\mathcal{B}}$, and output $b_{\mathcal{B}} = \langle r_{\mathcal{B}}, s_{\mathcal{B}} \rangle$.

$\mathcal{C}'$: Upon receiving $q[R_{\mathcal{C}}]$ from $\mathcal{A}'$ and $(A, r_{\mathcal{C}}, z_{2,\mathcal{C}})$ from the challenger, run $\mathcal{C}$ on $(q[R_{\mathcal{C}}], A, z_{2,\mathcal{C}})$ to obtain $t_{\mathcal{C}}$, and output $b_{\mathcal{C}} = \langle r_{\mathcal{C}}, t_{\mathcal{C}} \rangle$.

It is clear that $\mathcal{Aux}'$ satisfies the simulatability condition in Theorem 5.2 if we assume that $\mathcal{Aux}$ satisfies the simulatability condition in Corollary 5.4.

In the execution of $\text{Exp}_{i\mathcal{O}, \mathcal{Aux}', \mathcal{A}'_{\text{moe}}}^{\text{comp-strong-ind-moe-aux}}(\lambda)$, $\mathcal{A}'_{\text{moe}}$ perfectly simulates $\text{Exp}_{i\mathcal{O}, \mathcal{Aux}, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-moe-aux}}(\lambda)$ for $\mathcal{A}_{\text{moe}}$. Thus, with probability $\Pr \left[\text{Exp}_{i\mathcal{O}, \mathcal{Aux}, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-moe-aux}}(\lambda) = 1 \right]$, we have $s_{\mathcal{B}} = s$ and $t_{\mathcal{C}} = t$. When this occurs, it clearly holds that

$$\langle r_{\mathcal{B}}, s_{\mathcal{B}} \rangle \oplus \langle r_{\mathcal{C}}, t_{\mathcal{C}} \rangle = \langle r_{\mathcal{B}}, s \rangle \oplus \langle r_{\mathcal{C}}, t \rangle.$$

On the other hand, if either $s_{\mathcal{B}} \neq s$ or $t_{\mathcal{C}} \neq t$, then the probability that

$$\langle r_{\mathcal{B}}, s_{\mathcal{B}} \rangle \oplus \langle r_{\mathcal{C}}, t_{\mathcal{C}} \rangle = \langle r_{\mathcal{B}}, s \rangle \oplus \langle r_{\mathcal{C}}, t \rangle$$

is $1/2$, since $r_{\mathcal{B}}$ and $r_{\mathcal{C}}$ are chosen uniformly at random, independently of s , t , $s_{\mathcal{B}}$, and $t_{\mathcal{C}}$.

Thus, in total, we have:

$$\begin{aligned} & \Pr \left[\text{Exp}_{i\mathcal{O}, \mathcal{Aux}', \mathcal{A}'_{\text{moe}}}^{\text{comp-strong-ind-moe-aux}}(\lambda) = 1 \right] \\ &= \Pr \left[\text{Exp}_{i\mathcal{O}, \mathcal{Aux}, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-moe-aux}}(\lambda) = 1 \right] \cdot 1 + \left(1 - \Pr \left[\text{Exp}_{i\mathcal{O}, \mathcal{Aux}, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-moe-aux}}(\lambda) = 1 \right] \right) \cdot \frac{1}{2} \\ &= \frac{1}{2} + \frac{1}{2} \cdot \Pr \left[\text{Exp}_{i\mathcal{O}, \mathcal{Aux}, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-moe-aux}}(\lambda) = 1 \right]. \end{aligned}$$

By Theorem 5.2, the above expression is upper bounded by $1/2 + \text{negl}(\lambda)$. Therefore, we conclude that

$$\Pr \left[\text{Exp}_{i\mathcal{O}, \mathcal{Aux}, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-moe-aux}}(\lambda) = 1 \right] \leq \text{negl}(\lambda).$$

This completes the proof of Corollary 5.4. □

Looking ahead, we rely on Theorem 5.2 in the security proofs of our copy-protected PRF and UPO in Sections 8 and 9 whereas we rely on Corollary 5.4 in the security proof of our copy-protected signatures in Section 7.

5.1 Proof of Theorem 5.2

In this subsection, we prove Theorem 5.2. Our proof builds on the strong monogamy property of coset states [CLLZ21, CV22], which we gradually extend to eventually establish the desired result.

We begin by recalling the strong monogamy property of coset states.

Theorem 5.5 (Strong Monogamy Property of Coset States [CLLZ21, CV22]). *Let $n \in \mathbb{N}$ be a polynomial in λ . Consider the following experiment $\text{Exp}_{\mathcal{A}_{\text{moe}}}^{\text{strong-moe}}(\lambda)$ between a challenger and an adversary $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$.*

1. *The challenger picks a uniformly random subspace $A \subseteq \mathbb{F}_2^n$ of dimension $n/2$, and vectors $(s, t) \leftarrow \mathbb{F}_2^n \times \mathbb{F}_2^n$. It sends $|A_{s,t}\rangle$ to $\mathcal{A}$.*
2. *$\mathcal{A}$ creates a bipartite state q over registers R_B and R_C . Then, $\mathcal{A}$ sends register R_B to $\mathcal{B}$ and register R_C to $\mathcal{C}$.*
3. *The challenger sends A to both $\mathcal{B}$ and $\mathcal{C}$.*
4. *$\mathcal{B}$ and $\mathcal{C}$ respectively output s_B and t_C . The challenger outputs 1 if $s_B \in (A + s)$ and $t_C \in (A^\perp + t)$, and outputs 0 otherwise*

For any (not necessarily polynomial-time) quantum adversary $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$, it holds that

$$\Pr \left[\text{Exp}_{\mathcal{A}_{\text{moe}}}^{\text{strong-moe}}(\lambda) = 1 \right] \leq \text{negl}(\lambda).$$

Remark 5.6. In Theorem 5.5, the pair (s, t) is sampled uniformly from the entire space $\mathbb{F}_2^n \times \mathbb{F}_2^n$, rather than from $\text{CS}(A) \times \text{CS}(A^\perp)$ as in Theorem 5.2 and Corollary 5.4. Although both versions are equivalent at this point, this choice is motivated by the fact that the proof of the extended version of the property, presented in Corollary 5.7, appears to rely on (s, t) being uniformly distributed over $\mathbb{F}_2^n \times \mathbb{F}_2^n$. The discrepancy in the distribution of (s, t) is resolved later in the proof, specifically during the game hops from Hyb_0 to Hyb_2 in the proof of Theorem 5.2, given at the end of this subsection.

[CLLZ21] showed that Theorem 5.5 implies the following corollary, which we refer to as the *extended strong monogamy property* of coset states. In this variant, the adversary $\mathcal{A}$ additionally receives the descriptions of a uniformly random subspace $B \subseteq \mathbb{F}_2^n$ of dimension $3n/4$ that contains A , and a uniformly random subspace $C \subseteq A$ of dimension $n/4$, along with the vectors $s + v$ and $t + w$, where $v \leftarrow B$ and $w \leftarrow C^\perp$. The differences from Theorem 5.5 are highlighted in bold red text.

Corollary 5.7 (Extended Strong Monogamy Property of Coset States [CLLZ21]²¹). *Let $n \in \mathbb{N}$ be a polynomial in λ . Consider the following experiment $\text{Exp}_{\mathcal{A}_{\text{moe}}}^{\text{ex-strong-moe}}(\lambda)$ between a challenger and an adversary $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$.*

1. *The challenger picks a uniformly random subspace $A \subseteq \mathbb{F}_2^n$ of dimension $n/2$, **a uniformly random subspace B of dimension $3n/4$ that contains A , and a uniformly random subspace $C \subseteq A$ of dimension $n/4$** . The challenger picks vectors $(s, t) \leftarrow \mathbb{F}_2^n \times \mathbb{F}_2^n$ **and $(v, w) \leftarrow B \times C^\perp$** . It sends $|A_{s,t}\rangle$, **the descriptions of B and C , and vectors $s + v$ and $t + w$** to $\mathcal{A}$.*
2. *$\mathcal{A}$ creates a bipartite state q over registers R_B and R_C . Then, $\mathcal{A}$ sends register R_B to $\mathcal{B}$ and register R_C to $\mathcal{C}$.*
3. *The challenger sends A to both $\mathcal{B}$ and $\mathcal{C}$.*
4. *$\mathcal{B}$ and $\mathcal{C}$ respectively output s_B and t_C . The challenger outputs 1 if $s_B \in (A + s)$ and $t_C \in (A^\perp + t)$, and outputs 0 otherwise*

For any (not necessarily polynomial-time) quantum adversary $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$, it holds that

$$\Pr \left[\text{Exp}_{\mathcal{A}_{\text{moe}}}^{\text{ex-strong-moe}}(\lambda) = 1 \right] \leq \text{negl}(\lambda).$$

²¹The reduction from Corollary 5.7 to Theorem 5.5 is provided in Lemma C.16 of the arXiv version of [CLLZ21]: <https://arxiv.org/pdf/2107.05692>.

In Corollary 5.7, note that the statement remains equivalent even if the goals of $\mathcal{B}$ and $\mathcal{C}$ are modified to output $\text{Can}_A(s)$ and $\text{Can}_{A^\perp}(t)$, respectively.²² With this observation, the following corollary follows immediately by applying the simultaneous quantum Goldreich-Levin lemma (Lemma 3.10) to Corollary 5.7. The differences from Corollary 5.7 are highlighted in bold red text.

Corollary 5.8 (Extended Strong Indistinguishability Monogamy Property of Coset States). *Let $n \in \mathbb{N}$ be a polynomial in λ . Consider the following experiment $\text{Exp}_{\mathcal{A}_{\text{moe}}}^{\text{ex-strong-ind-moe}}(\lambda)$ between a challenger and an adversary $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$.*

1. *The challenger picks a uniformly random subspace $A \subseteq \mathbb{F}_2^n$ of dimension $n/2$, a uniformly random subspace B of dimension $3n/4$ that contains A , and a uniformly random subspace $C \subseteq A$ of dimension $n/4$. The challenger picks vectors $(s, t) \in \mathbb{F}_2^n \times \mathbb{F}_2^n$ and $(v, w) \leftarrow B \times C^\perp$.
It sends $|A_{s,t}\rangle$, the descriptions of B and C , and vectors $s + v$ and $t + w$ to $\mathcal{A}$.*
2. *$\mathcal{A}$ creates a bipartite state q over registers R_B and R_C . Then, $\mathcal{A}$ sends register R_B to $\mathcal{B}$ and register R_C to $\mathcal{C}$.*
3. **The challenger chooses $r_B, r_C \leftarrow \{0, 1\}^n$. The challenger sends (A, r_B) to $\mathcal{B}$, and (A, r_C) to $\mathcal{C}$.**
4. *$\mathcal{B}$ and $\mathcal{C}$ respectively output b_B and b_C . The challenger outputs 1 if $b_B \oplus b_C = \langle r_B, \text{Can}_A(s) \rangle \oplus \langle r_C, \text{Can}_{A^\perp}(t) \rangle$ and outputs 0 otherwise.*

For any (not necessarily polynomial-time) quantum adversary $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$, it holds that

$$\Pr \left[\text{Exp}_{\mathcal{A}_{\text{moe}}}^{\text{ex-strong-ind-moe}}(\lambda) = 1 \right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

Next, we consider a variant of the above property in which $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ receive auxiliary inputs as in Theorem 5.2, but do not receive the obfuscated programs. The differences from Corollary 5.8 are highlighted in bold red text.

Corollary 5.9 (Extended Strong Indistinguishability Monogamy Property of Coset States with Simulatable Auxiliary Inputs). *Let $n \in \mathbb{N}$ be a polynomial in λ . Let $\text{Aux} = (\text{AuxSetup}, \text{AuxGen}_B, \text{AuxGen}_C)$ and $(\text{Sim}_B, \text{Sim}_C)$ be (not necessarily polynomial-time) quantum algorithms satisfying the conditions in Theorem 5.2. Consider the following experiment $\text{Exp}_{\text{Aux}, \mathcal{A}_{\text{moe}}}^{\text{ex-strong-ind-moe-aux}}(\lambda)$ between a challenger and an adversary $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$.*

1. *The challenger picks a uniformly random subspace $A \subseteq \mathbb{F}_2^n$ of dimension $n/2$, a uniformly random subspace B of dimension $3n/4$ that contains A , and a uniformly random subspace $C \subseteq A$ of dimension $n/4$. The challenger picks vectors $(s, t) \in \mathbb{F}_2^n \times \mathbb{F}_2^n$ and $(v, w) \leftarrow B \times C^\perp$. **The challenger picks $r_B, r_C \leftarrow \{0, 1\}^n$, and generates $\text{pp} \leftarrow \text{AuxSetup}(1^\lambda)$, $(z_{1,B}, z_{2,B}) \leftarrow \text{AuxGen}_B(\text{pp}, A, r_B)$, and $(z_{1,C}, z_{2,C}) \leftarrow \text{AuxGen}_C(\text{pp}, A, r_C)$.**
It sends $|A_{s,t}\rangle$, the descriptions of B and C , vectors $s + v$ and $t + w$, **pp, $z_{1,B}$, and $z_{1,C}$** to $\mathcal{A}$.*
2. *$\mathcal{A}$ creates a bipartite state q over registers R_B and R_C . Then, $\mathcal{A}$ sends register R_B to $\mathcal{B}$ and register R_C to $\mathcal{C}$.*
3. *The challenger sends $(A, r_B, z_{2,B})$ to $\mathcal{B}$, and $(A, r_C, z_{2,C})$ to $\mathcal{C}$.*
4. *$\mathcal{B}$ and $\mathcal{C}$ respectively output b_B and b_C . The challenger outputs 1 if $b_B \oplus b_C = \langle r_B, \text{Can}_A(s) \rangle \oplus \langle r_C, \text{Can}_{A^\perp}(t) \rangle$ and outputs 0 otherwise.*

For any (not necessarily polynomial-time) quantum adversary $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$, it holds that

$$\Pr \left[\text{Exp}_{\text{Aux}, \mathcal{A}_{\text{moe}}}^{\text{ex-strong-ind-moe-aux}}(\lambda) = 1 \right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

Proof of Corollary 5.9. Let $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ be a (not necessarily polynomial-time) quantum adversary as in Corollary 5.9. Then we construct a (not necessarily polynomial-time) quantum adversary $\mathcal{A}'_{\text{moe}} = (\mathcal{A}', \mathcal{B}', \mathcal{C}')$ as in Corollary 5.8 as follows:

²²This is because, given the description of A , one can efficiently compute $\text{Can}_A(s_B) = \text{Can}_A(s)$ and $\text{Can}_{A^\perp}(t_C) = \text{Can}_{A^\perp}(t)$ from $s_B \in (A + s)$ and $t_C \in (A^\perp + t)$, respectively.

$\mathcal{A}'$: Upon receiving $(|A_{s,t}\rangle, B, C, s + v, t + w)$, it generates $\text{pp} \leftarrow \mathcal{AuxSetup}(1^\lambda)$, $z_{1,B} \leftarrow \text{Sim}_B(\text{pp})$, and $z_{1,C} \leftarrow \text{Sim}_C(\text{pp})$. Then it runs $\mathcal{A}$ on input $(|A_{s,t}\rangle, B, C, s + v, t + w, z_{1,B}, z_{1,C})$ to obtain a bipartite state q over registers R_B and R_C . It sends $z_{1,B}$ and $q[R_B]$ to $\mathcal{B}'$, and $z_{1,C}$ and $q[R_C]$ to $\mathcal{C}'$.

$\mathcal{B}'$: Upon receiving $(z_{1,B}, q[R_B])$ from $\mathcal{A}'$ and (A, r_B) from the challenger, it samples $(z_1, z_2) \leftarrow \mathcal{AuxGen}_B(\text{pp}, A, r_B)$ conditioned on $z_1 = z_{1,B}$ (using unbounded computational power), and sets $z_{2,B} = z_2$.²³ Finally, it runs $\mathcal{B}$ on input $(q[R_B], A, r_B, z_{2,B})$ to obtain b_B , and outputs b_B .

$\mathcal{C}'$: Upon receiving $(z_{1,C}, q[R_C])$ from $\mathcal{A}'$ and (A, r_C) from the challenger, it samples $(z_1, z_2) \leftarrow \mathcal{AuxGen}_C(\text{pp}, A, r_C)$ conditioned on $z_1 = z_{1,C}$ (using unbounded computational power), and sets $z_{2,C} = z_2$.²⁴ Finally, it runs $\mathcal{C}$ on input $(q[R_C], A, r_C, z_{2,C})$ to obtain b_C , and outputs b_C .

By the simulatability condition, for any A and r , with overwhelming probability over the choice of $\text{pp} \leftarrow \mathcal{AuxSetup}(1^\lambda)$, $\text{Sim}_X(\text{pp})$ statistically simulates the first coordinate of $\mathcal{AuxGen}_X(\text{pp}, A, r_X)$ for $X \in \{B, C\}$. Therefore, the joint distribution of $(z_{1,B}, z_{2,B}, z_{1,C}, z_{2,C})$ generated by $\mathcal{A}'_{\text{moe}}$ is statistically close to that in $\text{Exp}_{\mathcal{AuxGen}, \mathcal{A}_{\text{moe}}}^{\text{ex-strong-ind-moe-aux}}(\lambda)$. Apart from this, $\mathcal{A}'_{\text{moe}}$ perfectly simulates the execution of $\text{Exp}_{\mathcal{AuxGen}, \mathcal{A}_{\text{moe}}}^{\text{ex-strong-ind-moe-aux}}(\lambda)$ for $\mathcal{A}_{\text{moe}}$. Therefore, the difference between $\Pr[\text{Exp}_{\mathcal{AuxGen}, \mathcal{A}_{\text{moe}}}^{\text{ex-strong-ind-moe-aux}}(\lambda) = 1]$ and $\Pr[\text{Exp}_{\mathcal{A}'_{\text{moe}}}^{\text{ex-strong-ind-moe}}(\lambda) = 1]$ is negligible. Hence, Corollary 5.9 follows directly from Corollary 5.8. $\square$

We reduce Theorem 5.2 to Corollary 5.9 using a technique similar to that of [CLLZ21], which upgrades the strong monogamy property of coset states (Theorem 5.5) to its computational analogue. This technique relies on a result by Zhandry [Zha19], which shows that, assuming the existence of OWFs, any iO also serves as a subspace-hiding obfuscator.

Below, we state the definition of a subspace-hiding obfuscator.

Definition 5.10 ([Zha19]). A subspace hiding obfuscator (shO) for a field $\mathbb{F}$ and dimensions d_0, d_1 is a PPT algorithm shO satisfying the following:

Syntax. shO takes as input the description of a linear subspace $S \subseteq \mathbb{F}^n$ of dimension $d \in \{d_0, d_1\}$ and outputs a classical circuit $\hat{S}$.

Correctness. For any linear subspace $S \subseteq \mathbb{F}^n$ of dimension $d \in \{d_0, d_1\}$, it holds that

$$\Pr[\forall x \in \mathbb{F}^n, S(x) = \hat{S}(x) : \hat{S} \leftarrow \text{shO}(S)] \geq 1 - \text{negl}(\lambda)$$

where $S(x)$ is the function that decides membership in S .

Security. Consider the following game between an adversary and a challenger, indexed by a bit b .

- The adversary submits to the challenger a subspace S_0 of dimension d_0
- The challenger chooses a random subspace $S_1 \subseteq \mathbb{F}^n$ of dimension d_1 such that $S_0 \subseteq S_1$. It then runs $\hat{S} \leftarrow \text{shO}(S_b)$, and gives $\hat{S}$ to the adversary
- The adversary makes a guess b' for b .

For all QPT adversaries, it holds that $|\Pr[b' = b] - 1/2| \leq \text{negl}(\lambda)$ in the above game.

Lemma 5.11 ([Zha19]). If OWFs exist, then any iO, appropriately padded, is also a subspace hiding obfuscator for field $\mathbb{F}$ and dimensions d_0, d_1 , as long as $|\mathbb{F}|^{n-d_1}$ is exponential.

Remark 5.12. The original statement in [Zha19] assumes the existence of injective OWFs, but it is easy to see that keyed injective OWFs (Definition 3.7) suffice, which are known to exist if iO and OWFs exist [BPW16].

²³If $z_{1,B}$ is not in the support of the first coordinate of the output of $\mathcal{AuxGen}(\text{pp}, A, r_B)$, the algorithm simply aborts.

²⁴If $z_{1,C}$ is not in the support of the first coordinate of the output of $\mathcal{AuxGen}(\text{pp}, A, r_C)$, the algorithm simply aborts.

Then we prove Theorem 5.2.

Proof of Theorem 5.2. Let shO be a subspace hiding obfuscator for field $\mathbb{F}$ and dimensions $d_0 = n/2$ and $d_1 = 3n/4$. Such a subspace hiding obfuscator exists assuming the existence of iO and OWFs by Lemma 5.11.

For a QPT adversary $\mathcal{A}_{\text{moe}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$, we consider the following sequence of hybrids.

Hyb₀: This is the original security experiment $\text{Exp}_{i\mathcal{O}, \mathcal{A}_{\text{moe}}, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-ind-moe-aux}}(\lambda)$ described in Theorem 5.2.

Hyb₁: This is identical to Hyb₀ except that the hybrid outputs 1 if $b_{\mathcal{B}} \oplus b_{\mathcal{C}} = \langle r_{\mathcal{B}}, \text{Can}_A(s) \rangle \oplus \langle r_{\mathcal{C}}, \text{Can}_{A^\perp}(t) \rangle$ and outputs 0 otherwise.

Since $s \in \text{CS}(A)$ and $t \in \text{CS}(A^\perp)$, we have $\text{Can}_A(s) = s$ and $\text{Can}_{A^\perp}(t) = t$. Thus, we clearly have

$$\Pr[\text{Hyb}_1 = 1] = \Pr[\text{Hyb}_0 = 1].$$

Hyb₂: This is identical to Hyb₁ except that s and t are sampled uniformly from $\mathbb{F}_2^n$ instead of $\text{CS}(A)$ and $\text{CS}(A^\perp)$.

Note that the only information of (s, t) given to the adversary is $|A_{s,t}\rangle$, $i\mathcal{O}(A + s)$, and $i\mathcal{O}(A^\perp + t)$. For any $(s, t) \in \mathbb{F}_2^n \times \mathbb{F}_2^n$, $|A_{s,t}\rangle = |A_{\text{Can}_A(s), \text{Can}_{A^\perp}(t)}\rangle$, and $i\mathcal{O}(A + s)$ and $i\mathcal{O}(A^\perp + t)$ are computationally indistinguishable from $i\mathcal{O}(A + \text{Can}_A(s))$, and $i\mathcal{O}(A^\perp + \text{Can}_{A^\perp}(t))$, respectively, by the security of iO. Moreover, we have $\text{Can}_A(s) = \text{Can}_A(\text{Can}_A(s))$ and $\text{Can}_{A^\perp}(t) = \text{Can}_{A^\perp}(\text{Can}_{A^\perp}(t))$. This means that the probability to output 1 only negligibly changes even if we modify Hyb₂ by replacing s and t with $\text{Can}_A(s)$ and $\text{Can}_{A^\perp}(t)$, respectively. For uniformly random $(s, t) \in \mathbb{F}_2^n \times \mathbb{F}_2^n$, $(\text{Can}_A(s), \text{Can}_{A^\perp}(t))$ is uniformly random over $\text{CS}(A) \times \text{CS}(A^\perp)$. Thus the game obtained by replacing s and t in Hyb₂ with $\text{Can}_A(s)$ and $\text{Can}_{A^\perp}(t)$ is identical to Hyb₁. Thus, we have

$$|\Pr[\text{Hyb}_2 = 1] - \Pr[\text{Hyb}_1 = 1]| \leq \text{negl}(\lambda).$$

Hyb₃: This is identical to Hyb₂ except that the first obfuscated program $i\mathcal{O}(A + s)$ given to $\mathcal{A}$ is replaced with $i\mathcal{O}(\text{shO}(A)(\cdot - s))$ where $\text{shO}(A)(\cdot - s)$ is a program that takes $x \in \mathbb{F}^n$ as input and outputs $\text{shO}(A)(x - s)$.

By the correctness of shO, $\text{shO}(A)(\cdot - s)$ is functionally equivalent to the membership testing program of $A + s$ with an overwhelming probability. Thus, by the security of $i\mathcal{O}$, we have

$$|\Pr[\text{Hyb}_3 = 1] - \Pr[\text{Hyb}_2 = 1]| \leq \text{negl}(\lambda).$$

Hyb₄: This is identical to Hyb₃ except that the first obfuscated program $i\mathcal{O}(\text{shO}(A)(\cdot - s))$ given to $\mathcal{A}$ is replaced with $i\mathcal{O}(\text{shO}(B)(\cdot - s))$ where $B \subseteq \mathbb{F}_2^n$ is a uniformly random subspace of dimension $3n/4$ that contains A .

By the security of shO, we have

$$|\Pr[\text{Hyb}_4 = 1] - \Pr[\text{Hyb}_3 = 1]| \leq \text{negl}(\lambda).$$

Hyb₅: This is identical to Hyb₄ except that the first obfuscated program $i\mathcal{O}(\text{shO}(B)(\cdot - s))$ given to $\mathcal{A}$ is replaced with $i\mathcal{O}(\text{shO}(B)(\cdot - (s + v)))$ for a uniformly random element $v \in B$.

By the correctness of shO, $\text{shO}(B)(\cdot - s)$ and $\text{shO}(B)(\cdot - (s + v))$ are functionally equivalent with an overwhelming probability. Thus, by the security of $i\mathcal{O}$, we have

$$|\Pr[\text{Hyb}_5 = 1] - \Pr[\text{Hyb}_4 = 1]| \leq \text{negl}(\lambda).$$

Hyb₆: This is identical to Hyb₅ except that the second obfuscated program $i\mathcal{O}(A^\perp + t)$ given to $\mathcal{A}$ is replaced with $i\mathcal{O}(\text{shO}(C^\perp)(\cdot - (t + w)))$ where C is a uniformly random subspace of A of dimension $n/4$ and $w \in C^\perp$ is a uniformly random element of $C^\perp$.

By a similar argument to the hops from Hyb_2 to Hyb_5 , we have

$$|\Pr[\text{Hyb}_6 = 1] - \Pr[\text{Hyb}_5 = 1]| \leq \text{negl}(\lambda).$$

One can see that Hyb_6 is identical to $\text{Exp}_{\mathcal{A}_{\text{aux}}, \mathcal{A}_{\text{moe}}}^{\text{ex-strong-ind-moe-aux}}(\lambda)$ except that $\mathcal{A}$ is given $i\mathcal{O}(\text{shO}(B)(\cdot - (s + v)))$ and $i\mathcal{O}(\text{shO}(C^\perp)(\cdot - (t + w)))$ instead of $(B, C, s + v, t + w)$. Since $i\mathcal{O}(\text{shO}(B)(\cdot - (s + v)))$ and $i\mathcal{O}(\text{shO}(C^\perp)(\cdot - (t + w)))$ can be efficiently generated from $(B, C, s + v, t + w)$, a straightforward reduction to Corollary 5.9 gives us the following:

$$\Pr[\text{Hyb}_6 = 1] \leq 1/2 + \text{negl}(\lambda).$$

Combining the above, we can conclude that for any QPT adversary $\mathcal{A}_{\text{moe}}$,

$$\Pr[\text{Hyb}_0 = 1] \leq 1/2 + \text{negl}(\lambda),$$

completing the proof of Theorem 5.2. $\square$

6 Definitions for Copy-Protection

We introduce copy-protected signature and PRF, and unclonable puncturable obfuscation (UPO) [AB24].

6.1 Copy-Protected Signature

Definition 6.1 (Copy-Protected Signature). A copy-protected signature scheme CP-SIG for the message space $\mathcal{M}$ is a tuple of four algorithms $(\mathcal{KG}, \text{Sign}, \text{QSign}, \text{Vrfy})$.

$\mathcal{KG}(1^\lambda) \rightarrow (\text{vk}, \text{msk}, s\kappa)$: The key generation algorithm is a QPT algorithm that takes a security parameter 1^λ , and outputs a verification key vk , a master signing key msk , and a quantum secret key $s\kappa$.

$\text{Sign}(\text{msk}, m) \rightarrow \sigma$: The signing algorithm is a classical polynomial-time algorithm that takes a master secret key msk and a message $m \in \mathcal{M}$, and outputs a signature σ .

$\text{QSign}(s\kappa, m) \rightarrow \sigma$: The quantum signing algorithm is a QPT algorithm that takes a quantum secret key $s\kappa$ and a message $m \in \mathcal{M}$, and outputs a signature σ .

$\text{Vrfy}(\text{vk}, m, \sigma)$: The verification algorithm is a deterministic classical algorithm that takes a verification key vk , a message m , and a signature σ , and outputs $\top$ or $\perp$.

Verification correctness: For every $m \in \mathcal{M}$, we have

$$\Pr \left[\text{Vrfy}(\text{vk}, \sigma) = \top \mid \begin{array}{l} (\text{vk}, \text{msk}, s\kappa) \leftarrow \mathcal{KG}(1^\lambda) \\ \sigma \leftarrow \text{Sign}(\text{msk}, m) \end{array} \right] = 1 - \text{negl}(\lambda)$$

and

$$\Pr \left[\text{Vrfy}(\text{vk}, \sigma) = \top \mid \begin{array}{l} (\text{vk}, \text{msk}, s\kappa) \leftarrow \mathcal{KG}(1^\lambda) \\ \sigma \leftarrow \text{QSign}(s\kappa, m) \end{array} \right] = 1 - \text{negl}(\lambda).$$

The above verification correctness does not imply the reusability of the quantum secret key. We introduce the following strong correctness requirement that implies the reusability.

Definition 6.2 (Perfect Correctness with Deterministic Signing). CP-SIG = $(\mathcal{KG}, \text{Sign}, \text{QSign}, \text{Vrfy})$ is said to be perfect correct if it satisfies the followings.

- Sign is a classical deterministic polynomial time algorithm.

- For any $m \in \mathcal{M}$, we have

$$\Pr [\text{Vrfy}(\text{vk}, \text{Sign}(\text{msk}, m)) = \top \mid (\text{vk}, \text{msk}, s\kappa) \leftarrow \mathcal{KG}(1^\lambda)] = 1.$$

and

$$\Pr [\text{QSign}(s\kappa, m) = \text{Sign}(\text{msk}, m) \mid (\text{vk}, \text{msk}, s\kappa) \leftarrow \mathcal{KG}(1^\lambda)] = 1.$$

Perfect correctness with deterministic signing implies the reusability of the quantum secret key. This is because the output of the quantum signing algorithm is unique, and thus such an operation can be done without disturbing the input state by the gentle measurement lemma [Win99].

We define the following security notion for copy-protected signatures, which generalizes the definition given by [LLQZ22] by allowing non-uniform challenge distributions.

Definition 6.3 ($\mathcal{D}$ -UF Anti-Piracy Security). Let $\text{CP-SIG} = (\mathcal{KG}, \text{Sign}, \text{QSign})$ be a copy-protected signature scheme. Let $\mathcal{D} = (\mathcal{D}_\mathcal{B}, \mathcal{D}_\mathcal{C})$ be a pair of distributions over $\mathcal{M} \times \{0, 1\}^{\ell_{\text{aux}}}$. We consider the $\mathcal{D}$ -UF anti-piracy game $\text{Exp}_{\text{CP-SIG}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{uf-anti-piracy}}(\lambda)$ between the challenger and an adversary $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ below.

1. The challenger generates $(\text{vk}, \text{msk}, s\kappa) \leftarrow \mathcal{KG}(1^\lambda)$, $(m_\mathcal{B}, \text{aux}_\mathcal{B}) \leftarrow \mathcal{D}_\mathcal{B}(1^\lambda)$, and $(m_\mathcal{C}, \text{aux}_\mathcal{C}) \leftarrow \mathcal{D}_\mathcal{C}(1^\lambda)$. The challenger sends vk , $s\kappa$, $\text{aux}_\mathcal{B}$, and $\text{aux}_\mathcal{C}$ to $\mathcal{A}$.
2. The challenge prepares a list L_{query} that is initialized to empty. $\mathcal{A}$ may make an arbitrary number of queries to the signing oracle, which on input $m \in \mathcal{M}$, returns $\sigma \leftarrow \text{Sign}(\text{msk}, m)$, and adds (m, σ) to L_{query} .
3. $\mathcal{A}$ creates a bipartite state q over registers $R_\mathcal{B}$ and $R_\mathcal{C}$. Then, $\mathcal{A}$ sends register $R_\mathcal{B}$ to $\mathcal{B}$ and register $R_\mathcal{C}$ to $\mathcal{C}$.
4. The challenger sends $m_\mathcal{B}$ and $m_\mathcal{C}$ to $\mathcal{B}$ and $\mathcal{C}$, respectively.
5. $\mathcal{B}$ and $\mathcal{C}$ may make an arbitrary number of queries to the signing oracle described in the second item.
6. $\mathcal{B}$ and $\mathcal{C}$ respectively output $\sigma'_\mathcal{B}$ and $\sigma'_\mathcal{C}$. The challenger outputs 1 if $\text{Vrfy}(\text{vk}, m_\mathcal{B}, \sigma'_\mathcal{B}) = \top$, $\text{Vrfy}(\text{vk}, m_\mathcal{C}, \sigma'_\mathcal{C}) = \top$, and neither $(m_\mathcal{B}, \sigma'_\mathcal{B})$ nor $(m_\mathcal{C}, \sigma'_\mathcal{C})$ is included in L_{query} . Otherwise, the challenger outputs 0.

We say that CP-SIG is $\mathcal{D}$ -UF anti-piracy secure if for any QPT adversary $\mathcal{A}_{\text{cp}}$, it holds that

$$\Pr [\text{Exp}_{\text{CP-SIG}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{uf-anti-piracy}}(\lambda) = 1] \leq \text{negl}(\lambda).$$

6.2 Copy-Protected PRF

Definition 6.4 (Copy-Protected PRF). A copy-protected PRF scheme CP-PRF for the domain D_{prf} and the range R_{prf} is a tuple of three algorithms $(\mathcal{KG}, \text{Eval}, \text{QEval})$.

$\mathcal{KG}(1^\lambda) \rightarrow (\text{msk}, s\kappa)$: The key generation algorithm is a QPT algorithm that takes a security parameter 1^λ , and outputs a master secret key msk and a quantum secret key $s\kappa$.

$\text{Eval}(\text{msk}, x) \rightarrow y$: The evaluation algorithm is a deterministic classical polynomial-time algorithm that takes a master secret key msk and an input $x \in D_{\text{prf}}$, and outputs a value y .

$\text{QEval}(s\kappa, x) \rightarrow y$: The quantum evaluation algorithm is a QPT algorithm that takes a quantum secret key $s\kappa$ and an input $x \in D_{\text{prf}}$, and outputs a value y .

Evaluation correctness: For every $x \in D_{\text{prf}}$, we have

$$\Pr [\text{QEval}(s\kappa, x) = \text{Eval}(\text{msk}, x) \mid (\text{msk}, s\kappa) \leftarrow \mathcal{KG}(1^\lambda)] = 1 - \text{negl}(\lambda).$$

We say that it satisfies the perfect correctness if the above probability is 1.

Evaluation correctness implies the usability of the quantum secret key thanks to the gentle measurement lemma.

We define the security notion for copy-protected PRF. Not only does it consider non-uniform challenge samplers and the evaluation oracle, but it also adopts the “+”-style definition introduced by [KY25], where the adversary wins if the guesses of $\mathcal{B}$ and $\mathcal{C}$ are either both correct or both incorrect. In Appendix A, we also define both search-style and standard indistinguishability-style security notions, which are direct analogues of existing definitions [CLLZ21] adapted to non-uniform challenge distributions, and show that the following definition implies both.

Definition 6.5 ($\mathcal{D}$ -IND⁺Anti-Piracy Security). Let $\text{CP-PRF} = (\mathcal{KG}, \text{Eval}, Q\text{Eval})$ be a copy protected PRF scheme. Let $\mathcal{D} = (\mathcal{D}_B, \mathcal{D}_C)$ be a pair of distributions over $D_{\text{prf}} \times \{0, 1\}^{\ell_{\text{aux}}}$. We consider the $\mathcal{D}$ -IND⁺anti-piracy game $\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{indplus-anti-piracy}}(\lambda)$ between the challenger and an adversary $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ below.

1. The challenger generates $(\text{msk}, s_K) \leftarrow \mathcal{KG}(1^\lambda)$, $(x_B, \text{aux}_B) \leftarrow \mathcal{D}_B(1^\lambda)$, and $(x_C, \text{aux}_C) \leftarrow \mathcal{D}_C(1^\lambda)$. The challenger sends s_K , aux_B , and aux_C to $\mathcal{A}$.
2. $\mathcal{A}$ may make an arbitrary number of queries to the evaluation oracle, which, on input $x \in D_{\text{prf}}$, returns $\text{Eval}(\text{msk}, x)$.
3. $\mathcal{A}$ creates a bipartite state q over registers R_B and R_C . Then, $\mathcal{A}$ sends register R_B to $\mathcal{B}$ and register R_C to $\mathcal{C}$.
4. The challenger does the following.
 - Choose $\text{coin}_B \leftarrow \{0, 1\}$, and generate $y_{B,0} \leftarrow \text{Eval}(\text{msk}, x_B)$ and $y_{B,1} \leftarrow R_{\text{prf}}$.
 - Choose $\text{coin}_C \leftarrow \{0, 1\}$, and generate $y_{C,0} \leftarrow \text{Eval}(\text{msk}, x_C)$, and $y_{C,1} \leftarrow R_{\text{prf}}$.

The challenger sends $(x_B, y_{B, \text{coin}_B})$ and $(x_C, y_{C, \text{coin}_C})$ to $\mathcal{B}$ and $\mathcal{C}$, respectively.

5. $\mathcal{B}$ and $\mathcal{C}$ may make an arbitrary number of queries to the evaluation oracle described in the second item, with the restriction that $\mathcal{B}$ and $\mathcal{C}$ are not allowed to query x_B and x_C , respectively.
6. $\mathcal{B}$ and $\mathcal{C}$ respectively output coin'_B and coin'_C . The challenger outputs 1 if $\text{coin}'_B \oplus \text{coin}'_C = \text{coin}_B \oplus \text{coin}_C$ otherwise outputs 0.

We say that CP-PRF is $\mathcal{D}$ -IND⁺anti-piracy secure if for any QPT adversary $\mathcal{A}_{\text{cp}}$, it holds that

$$\Pr \left[\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{indplus-anti-piracy}}(\lambda) = 1 \right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

Remark 6.6. The definition is equivalent even if we replace the required inequality with

$$\left| \Pr \left[\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{indplus-anti-piracy}}(\lambda) = 1 \right] - \frac{1}{2} \right| \leq \text{negl}(\lambda).$$

6.3 Unclonable Puncturable Obfuscation

Definition 6.7 (Unclonable Puncturable Obfuscation (UPO)). A UPO scheme UPO for the circuit class $\text{Circ} = \{C : \{0, 1\}^{\ell_{\text{cin}}} \rightarrow \{0, 1\}^{\ell_{\text{cout}}}\}$ is a pair of two algorithms $(\text{Obf}, \text{Eval})$.

$\text{Obf}(1^\lambda, C) \rightarrow \rho_C$: The obfuscation algorithm is a QPT algorithm that takes a security parameter 1^λ and a circuit $C \in \text{Circ}$, and outputs a quantum state ρ_C .

$\text{Eval}(\rho_C, x) \rightarrow y$: The evaluation algorithm is a QPT algorithm that takes a quantum state ρ_C and an input $x \in \{0, 1\}^{\ell_{\text{cin}}}$, and output a value y .

Evaluation correctness: For every $C \in \text{Circ}$ and $x \in \{0, 1\}^{\ell_{\text{cin}}}$, we have

$$\Pr [\text{Eval}(\rho_C, x) = C(x) \mid \rho_C \leftarrow \text{Obf}(1^\lambda, C)] = 1 - \text{negl}(\lambda).$$

We say that it satisfies perfect correctness if the above probability is 1.

Evaluation correctness implies the reusability of the quantum obfuscated state ρ_C thanks to the gentle measurement lemma.

To define the security notion of UPO, we first introduce notation for punctured circuits.

Definition 6.8 (Punctured Circuit). Let $C : \{0,1\}^{\ell_{\text{cin}}} \rightarrow \{0,1\}^{\ell_{\text{cout}}}$ be a circuit. Let $x_B, x_C \in \{0,1\}^{\ell_{\text{cin}}}$ and $y_B, y_C \in \{0,1\}^{\ell_{\text{out}}}$. We define the punctured circuit $C^*[x_B, x_C, y_B, y_C]$ as²⁵

$$C^*[x_B, x_C, y_B, y_C](x) = \begin{cases} C(x), & x \in \{0,1\}^{\ell_{\text{cin}}} \setminus \{x_B, x_C\} \\ y_B, & x = x_B \\ y_C, & x = x_C \end{cases}.$$

We then define a security notion of UPO, which we call $\mathcal{D}$ -UPO⁺ anti-piracy security. This extends the original definition from [AB24] by adopting the “+”-style formulation introduced by [KY25], similarly to the $\mathcal{D}$ -IND⁺ anti-piracy security definition for copy-protected PRFs given in Section 6.2. We show in Appendix B that $\mathcal{D}$ -UPO⁺ anti-piracy security implies a non-uniform challenge variant of the original definition (which we refer to $\mathcal{D}$ -UPO security in Definition B.1), assuming additionally that the UPO satisfies security as iO. Note that this additional requirement is arguably mild in the context of UPO, as all existing constructions of UPO (including ours) already rely on iO and satisfy iO security by default. Moreover, if this is not the case, iO security can be easily added by first applying iO, followed by UPO.

Definition 6.9 ($\mathcal{D}$ -UPO⁺ Security). Let $\text{UPO} = (\text{Obf}, \text{Eval})$ be a UPO scheme for the circuit class $\text{Circ} = \{C : \{0,1\}^{\ell_{\text{cin}}} \rightarrow \{0,1\}^{\ell_{\text{cout}}}\}$. Let $\mathcal{D} = (\mathcal{D}_B, \mathcal{D}_C)$ be a pair of distributions over $\{0,1\}^{\ell_{\text{cin}}} \times \{0,1\}^{\ell_{\text{aux}}}$. We consider the $\mathcal{D}$ -UPO⁺ anti-piracy game $\text{Expt}_{\text{UPO}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{upoplus}}(\lambda)$ between the challenger and an adversary $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ below.

1. On input 1^λ , $\mathcal{A}$ sends $C \in \text{Circ}$ together with two circuits $\mu_B : \{0,1\}^{\ell_{\text{cin}}} \rightarrow \{0,1\}^{\ell_{\text{cout}}}$ and $\mu_C : \{0,1\}^{\ell_{\text{cin}}} \rightarrow \{0,1\}^{\ell_{\text{cout}}}$.
2. The challenger does the following.
 - Choose $\text{coin}_B \leftarrow \{0,1\}$, generate $(x_B, \text{aux}_B) \leftarrow \mathcal{D}_B(1^\lambda)$, $y_{B,0} \leftarrow C(x_B)$, and $y_{B,1} \leftarrow \mu_B(x_B)$.
 - Choose $\text{coin}_C \leftarrow \{0,1\}$, generate $(x_C, \text{aux}_C) \leftarrow \mathcal{D}_C(1^\lambda)$, $y_{C,0} \leftarrow C(x_C)$, and $y_{C,1} \leftarrow \mu_C(x_C)$.

The challenger generates $\rho_C \leftarrow \text{Obf}(1^\lambda, C^*[x_B, x_C, y_{B, \text{coin}_B}, y_{C, \text{coin}_C}])$ and sends $(\rho_C, \text{aux}_B, \text{aux}_C)$ to $\mathcal{A}$.

3. $\mathcal{A}$ creates a bipartite state q over registers R_B and R_C . Then, $\mathcal{A}$ sends register R_B to $\mathcal{B}$ and register R_C to $\mathcal{C}$.
4. The challenger sends x_B and x_C to $\mathcal{B}$ and $\mathcal{C}$, respectively.
5. $\mathcal{B}$ and $\mathcal{C}$ respectively output coin'_B and coin'_C . The challenger outputs 1 if $\text{coin}'_B \oplus \text{coin}'_C = \text{coin}_B \oplus \text{coin}_C$ otherwise outputs 0.

We say that UPO is $\mathcal{D}$ -UPO⁺ secure if for any QPT adversary $\mathcal{A}_{\text{cp}}$, it holds that

$$\Pr[\text{Expt}_{\text{UPO}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{upoplus}}(\lambda) = 1] \leq \frac{1}{2} + \text{negl}(\lambda).$$

Remark 6.10. The definition is equivalent even if we replace the required inequality with

$$\left| \Pr[\text{Expt}_{\text{UPO}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{upoplus}}(\lambda) = 1] - \frac{1}{2} \right| \leq \text{negl}(\lambda).$$

As explained above, a UPO that satisfies both $\mathcal{D}$ -UPO⁺ anti-piracy security and iO security also satisfies $\mathcal{D}$ -UPO anti-piracy security. Therefore, such a UPO can be used to copy-protect any puncturable functionality under search-based notions, such as the copy-protection of puncturable signatures, as demonstrated in [AB24]. In Appendix C, we show that the same UPO can also be applied to decision-type applications—specifically, to the copy-protected PRFs and SDE schemes.

²⁵If $x_B = x_C$, we define the output on $x_B = x_C$ to be y_B , ignoring y_C . This case occurs only with negligible probability throughout this paper, as both x_B and x_C have high min-entropy. We include this definition for completeness.

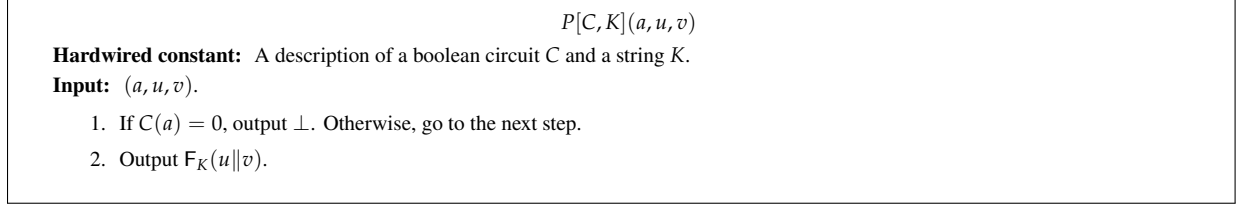


Figure 5: The description of the circuit P .

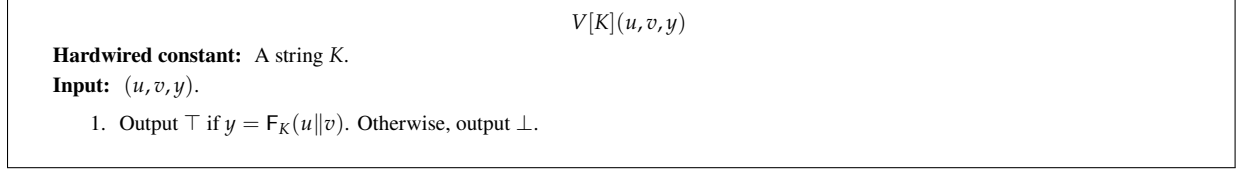


Figure 6: The description of the circuit V .

7 Construction of Copy-Protected Signature

In this section, we prove the following theorem.

Theorem 7.1. *Assuming the existence of polynomially secure iO and OWFs, for any constant $c > 0$ and any polynomial ℓ_{inp} , there exists a copy-protected signature scheme with the message space $\{0, 1\}^{\ell_{\text{inp}}}$ that satisfies $\mathcal{D}$ -UF anti-piracy security for any QPT-samplable distributions $\mathcal{D} = (\mathcal{D}_B, \mathcal{D}_C)$, where $\mathcal{D}_B$ and $\mathcal{D}_C$ have conditional min-entropy at least λ^c .*

7.1 Construction

Let $n, \ell_k, \ell_{\text{tcr}}, \ell_{\text{out}}$, and ℓ_{inp} be polynomials in the security parameter λ . Below, we assume $\ell_{\text{out}} = n$ for simplicity though a similar construction and security proof work as long as $\ell_{\text{out}} \geq n$.

We use the following building blocks.

- iO for polynomial size classical circuits $i\mathcal{O}$.
- PPRF $\text{PPRF} = (F, \text{Puncture})$ with the domain $\{0, 1\}^{\ell_k + \ell_{\text{tcr}}}$ and the range $\{0, 1\}^{\ell_{\text{out}}}$.
- Universal hash function family $\mathcal{H} : \{0, 1\}^{\ell_{\text{inp}}} \rightarrow \{0, 1\}^{\ell_k}$.
- TCR hash function TCR.Gen with the domain $\{0, 1\}^{\ell_{\text{inp}}}$ and the range $\{0, 1\}^{\ell_{\text{tcr}}}$.

We below construct a copy-protected signature scheme $\text{CP-SIG} = \text{CP-SIG}(\mathcal{KG}, \text{Sign}, \text{QSign})$ with the message space $\{0, 1\}^{\ell_{\text{inp}}}$.

$\text{CP-SIG}.\mathcal{KG}(1^\lambda)$:

- Generate a uniformly random subspace $A \subseteq \mathbb{F}_2^n$ of dimension $\frac{n}{2}$, and two uniformly random elements $(s, t) \in \text{CS}(A) \times \text{CS}(A^\perp)$.
- Define circuits $C_{\text{mc}} := i\mathcal{O}(A + s)$ and $C_{\text{mc}}^\perp := i\mathcal{O}(A^\perp + t)$.
- Generate keys $K, K^\perp \leftarrow \{0, 1\}^\lambda$ for PPRF.
- Generate $h \leftarrow \mathcal{H}$ and $H_t \leftarrow \text{TCR.Gen}(1^\lambda)$.
- Generate two obfuscated circuits $\tilde{P} \leftarrow i\mathcal{O}(P[C_{\text{mc}}, K])$ and $\tilde{P}^\perp \leftarrow i\mathcal{O}(P[C_{\text{mc}}^\perp, K^\perp])$, where P is the circuit described in Figure 5.

- Generate an obfuscated circuits $\tilde{V} \leftarrow i\mathcal{O}(V[K])$ and $\tilde{V}^\perp \leftarrow i\mathcal{O}(V[K^\perp])$, where V is the circuit described in Figure 6.
- Output $\text{vk} := (\tilde{V}, \tilde{V}^\perp, h, H_t)$, $\text{msk} := (K, K^\perp, h, H_t)$, and $s\mathcal{K} := (|A_{s,t}\rangle, \tilde{P}, \tilde{P}^\perp, h, H_t)$.

CP-SIG.Sign(msk, m):

- Parse $\text{msk} = (K, K^\perp, h, H_t)$.
- Compute $u \leftarrow h(m)$ and $v \leftarrow H_t(m)$.
- Output $\sigma := (F_K(u\|v), F_{K^\perp}(u\|v))$.

CP-SIG.QSign($s\mathcal{K}, m$):

- Parse $s\mathcal{K} = (\rho, \tilde{P}, \tilde{P}^\perp, h, H_t)$, where ρ is a quantum state, and $\tilde{P}$ and $\tilde{P}^\perp$ are interpreted as descriptions of circuits.
- Compute $u \leftarrow h(m)$ and $v \leftarrow H_t(m)$.
- Let E and $E^\perp$ be unitaries that respectively work as follows:

$$|z_1\rangle |z_2\rangle \mapsto |z_1\rangle |z_2 \oplus \tilde{P}(z_1, u, v)\rangle \quad \text{and} \quad |z_1\rangle |z_2\rangle \mapsto |z_1\rangle |z_2 \oplus \tilde{P}^\perp(z_1, u, v)\rangle$$

for any $z_1 \in \{0, 1\}^n$ and $z_2 \in \{0, 1\}^{\ell_{\text{out}}}$.

- Apply E to $\rho \otimes |0^{\ell_{\text{out}}}\rangle$ and measure the second register to obtain an outcome y . Then, trace out the second register and obtain the state ρ' .
- Apply $E^\perp$ to $H^{\otimes n} \rho' \otimes |0^{\ell_{\text{out}}}\rangle$ and measure the second register to obtain an outcome $y^\perp$.
- Output $\sigma := (y, y^\perp)$.

CP-SIG.Vrfy(vk, m, σ):

- Parse $\text{vk} = (\tilde{V}, \tilde{V}^\perp, h, H_t)$ and $\sigma = (y, y^\perp)$.
- Compute $u \leftarrow h(m)$ and $v \leftarrow H_t(m)$.
- Output $\top$ if $\tilde{V}(u, v, y) = \tilde{V}^\perp(u, v, y^\perp) = \top$. Otherwise, output $\perp$.

Perfect correctness with deterministic signing. The classical signing algorithm CP-SIG.Sign is deterministic as required in Definition 6.2. Also, from the correctness of $i\mathcal{O}$, for any $m \in \{0, 1\}^{\ell_{\text{inp}}}$, we can immediately obtain

$$\Pr [\text{CP-SIG.Vrfy}(\text{vk}, \text{CP-SIG.Sign}(\text{msk}, m)) = \top \mid (\text{vk}, \text{msk}, s\mathcal{K}) \leftarrow \text{CP-SIG.KG}(1^\lambda)] = 1.$$

We then prove

$$\Pr [\text{CP-SIG.QSign}(s\mathcal{K}, m) = \text{CP-SIG.Sign}(\text{msk}, m) \mid (\text{vk}, \text{msk}, s\mathcal{K}) \leftarrow \text{CP-SIG.KG}(1^\lambda)] = 1$$

for any $m \in \{0, 1\}^{\ell_{\text{inp}}}$. Suppose we generate $\text{vk} := (\tilde{V}, h, H_t)$, $\text{msk} := (K, K^\perp, h, H_t)$, and $s\mathcal{K} := (|A_{s,t}\rangle, \tilde{P}, \tilde{P}^\perp, h, H_t)$ as prescribed in the description of CP-SIG.KG. Fix $m \in \{0, 1\}^{\ell_{\text{inp}}}$ and let $u \leftarrow h(m)$ and $v \leftarrow H_t(m)$. We have $\text{CP-SIG.Sign}(\text{msk}, m) = (F_K(u\|v), F_{K^\perp}(u\|v))$. Letting $(y, y^\perp) \leftarrow \text{CP-SIG.QSign}(s\mathcal{K}, m)$, we show that $y = F_K(u\|v)$ and $y^\perp = F_{K^\perp}(u\|v)$ holds with probability 1.

- We prove $y = F_K(u\|v)$. For any $a \in A + s$, we have $\tilde{P}(a, u, v) = F_K(u\|v)$ from the correctness of $i\mathcal{O}$. Thus, given that $|A_{s,t}\rangle = \frac{1}{\sqrt{|A|}} \sum_{a \in A} (-1)^{\langle a, t \rangle} |a + s\rangle$, if we apply the unitary E to $|A_{s,t}\rangle \otimes |0^{\ell_{\text{out}}}\rangle$, we obtain the state $|A_{s,t}\rangle \otimes |F_K(u\|v)\rangle$. This means $y = F_K(u\|v)$ holds, and the state obtained by tracing out the second register after measuring y is $|A_{s,t}\rangle$.

- We then prove $y^\perp = F_{K^\perp}(u\|v)$. For any $a \in A^\perp + t$, we have $\tilde{P}_2(a, u, v) = F_{K^\perp}(u\|v)$ from the correctness of $i\mathcal{O}$. Thus, given that $H^{\otimes n} |A_{s,t}\rangle = \frac{1}{\sqrt{|A^\perp|}} \sum_{a \in A^\perp} (-1)^{\langle a, s \rangle} |a + t\rangle$, if we apply the unitary $E^\perp$ to $H^{\otimes n} |A_{s,t}\rangle \otimes |0^{\ell_{\text{out}}}\rangle$, we obtain the state $H^{\otimes n} |A_{s,t}\rangle \otimes |F_{K^\perp}(u\|v)\rangle$. This means $y^\perp = F_{K^\perp}(u\|v)$ holds.

The above shows the perfect correctness of CP-SIG.

7.2 Proof of Security

In the security proof, we additionally use the following primitive, where $\ell_A := n^2/2$ is the number of bits needed to represent a subspace $A \subseteq \mathbb{F}_2^n$ of dimension $n/2$.

- A key-robust non-committing encryption scheme $\text{NCE} = (\text{NCE.Enc}, \text{NCE.Dec}, \text{NCE.Fake}, \text{NCE.Open})$ with the message space $\{0, 1\}^{\ell_A + n}$ and the key space $\{0, 1\}^{\ell_k}$.

We prove the following theorem.

Theorem 7.2. *Let $\mathcal{D} = (\mathcal{D}_B, \mathcal{D}_C)$ be a pair of QPT-samplable distributions over $\{0, 1\}^{\ell_{\text{inp}}} \times \{0, 1\}^{\ell_{\text{aux}}}$ such that both $\mathcal{D}_B$ and $\mathcal{D}_C$ have conditional min-entropy at least $\ell_k + \ell_{\text{tcr}} + \omega(\log \lambda)$. Then, CP-SIG satisfies $\mathcal{D}$ -UF anti-piracy security.*

We may choose ℓ_k and ℓ_{tcr} to be arbitrarily small polynomials, while allowing ℓ_{inp} to be any polynomial. Therefore, Theorem 7.1 follows.

Proof of Theorem 7.2. Let $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ be a QPT adversary against $\mathcal{D}$ -UF anti-piracy security of CP-SIG. We consider the following sequence of hybrids.

Hyb₀: This is the original security experiment $\text{Exp}_{\text{CP-SIG}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{uf-anti-piracy}}(\lambda)$ in Definition 6.3. More specifically, it works as follows.

1. The challenger generates $\text{vk} := (\tilde{V}, h, H_t)$, $\text{msk} := (K, K^\perp, h, H_t)$, and $s\mathcal{K} := (|A_{s,t}\rangle, \tilde{P}, \tilde{P}^\perp, h, H_t)$ as prescribed in the description of CP-SIG. $\mathcal{KG}$. The challenger also generates $(m_B, \text{aux}_B) \leftarrow \mathcal{D}_B(1^\lambda)$ and $(m_C, \text{aux}_C) \leftarrow \mathcal{D}_C(1^\lambda)$. The challenger sends vk , $s\mathcal{K}$, aux_B , and aux_C to $\mathcal{A}$.
2. The challenger prepares an empty list L_{query} . $\mathcal{A}$ can make arbitrarily many queries to the signing oracle, which on input $m \in \mathcal{M}$, returns $\sigma \leftarrow \text{Sign}(\text{msk}, m)$, and adds (m, σ) to L_{query} .
3. $\mathcal{A}$ creates a bipartite state q over registers R_B and R_C . Then, $\mathcal{A}$ sends R_B to $\mathcal{B}$, and R_C to $\mathcal{C}$.
4. The challenger sends m_B and m_C to $\mathcal{B}$ and $\mathcal{C}$, respectively.
5. $\mathcal{B}$ and $\mathcal{C}$ can make arbitrarily many queries to the signing oracle described in the second item.
6. $\mathcal{B}$ and $\mathcal{C}$ respectively output $\sigma_B = (w_B, w_B^\perp)$ and $\sigma_C = (w_C, w_C^\perp)$. The challenger outputs 1 if $\tilde{V}(u_B, v_B, w_B, w_B^\perp) = \top$, $\tilde{V}(u_C, v_C, w_C, w_C^\perp) = \top$, and neither (m_B, σ_B) nor (m_C, σ_C) is included in L_{query} , where $u_B \leftarrow h(m_B)$, $v_B \leftarrow H_t(m_B)$, $u_C \leftarrow h(m_C)$, and $v_C \leftarrow H_t(m_C)$. Otherwise, the challenger outputs 0.

By the definition, we clearly have

$$\Pr[\text{Hyb}_0 = 1] = \Pr[\text{Exp}_{\text{CP-SIG}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{uf-anti-piracy}}(\lambda) = 1].$$

Hyb₁: This is identical to Hyb₀ except for the following changes.

- It immediately outputs 0 if $v_B = v_C$.
- The signing oracle returns $\perp$ on input m if $H_t(m) = v_B$ or $H_t(m) = v_C$ holds.

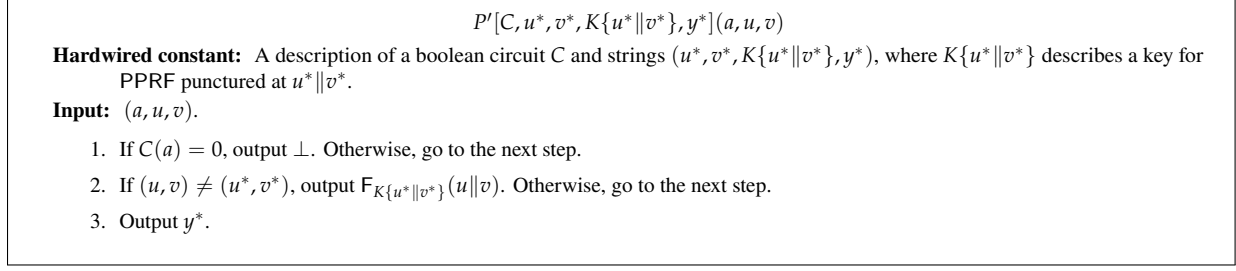


Figure 7: The description of the circuit P' .

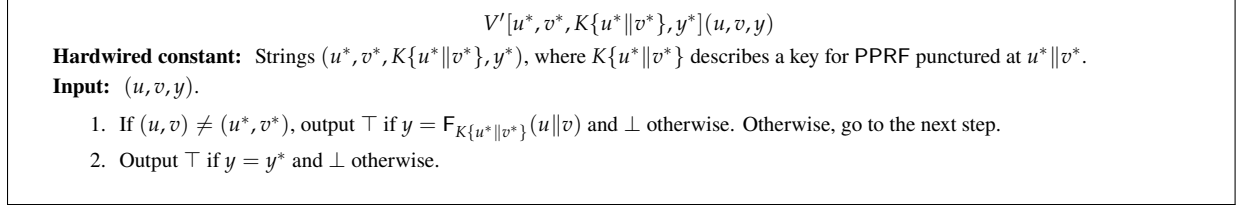


Figure 8: The description of the circuit V' .

By the min-entropy condition of $\mathcal{D}$, we have $m_B \neq m_C$ with overwhelming probability. Thus, by the security of TCR, $v_B = v_C$ occurs with negligible probability.

For the second item, note that if the adversary queries the signing oracle on either m_B or m_C , then both hybrids Hyb_0 and Hyb_1 automatically output 0. On the other hand, by the security of TCR, the probability that the adversary queries the signing oracle on m such that $(H_t(m) = v_B \wedge m \neq m_B)$ or $(H_t(m) = v_C \wedge m \neq m_C)$ is negligible. Therefore, we have

$$|\Pr[\text{Hyb}_0 = 1] - \Pr[\text{Hyb}_1 = 1]| = \text{negl}(\lambda).$$

We below use the circuits P' and V^* respectively described in Figure 7 and Figure 8.

Hyb₂: This is identical to Hyb_1 except for the following changes, where $K\{u_B \| v_B\} \leftarrow \text{Puncture}(K, u_B \| v_B)$, $K^\perp\{u_C \| v_C\} \leftarrow \text{Puncture}(K^\perp, u_C \| v_C)$, $y_B \leftarrow F_K(u_B \| v_B)$, and $y_C^\perp \leftarrow F_{K^\perp}(u_C \| v_C)$:

- $\tilde{P}$ is generated as $\tilde{P} \leftarrow i\mathcal{O}(P'[C_{mc}, u_B, v_B, K\{u_B \| v_B\}, y_B])$.
- $\tilde{P}^\perp$ is generated as $\tilde{P}^\perp \leftarrow i\mathcal{O}(P'[C_{mc}^\perp, u_C, v_C, K^\perp\{u_C \| v_C\}, y_C^\perp])$.
- $\tilde{V}$ is generated as $\tilde{V} \leftarrow i\mathcal{O}(V'[u_B, v_B, K\{u_B \| v_B\}, y_B])$.
- $\tilde{V}^\perp$ is generated as $\tilde{V}^\perp \leftarrow i\mathcal{O}(V'[u_C, v_C, K^\perp\{u_C \| v_C\}, y_C^\perp])$.

Moreover, the adversary's winning condition (i.e., the condition that the experiment returns 1) is modified to $\sigma_B = (w_B, w_B^\perp) = (y_B, F_{K^\perp}(u_B \| v_B))$ and $\sigma_C = (w_C, w_C^\perp) = (F_K(u_C \| v_C), y_C^\perp)$.

From the correctness of $i\mathcal{O}$, the adversary's winning condition is equivalent to that in the previous hybrid. Moreover, it is easy to verify that each replacement of the obfuscated circuits described above involves functionally equivalent circuits. Thus, from the security of $i\mathcal{O}$, we have

$$|\Pr[\text{Hyb}_1 = 1] - \Pr[\text{Hyb}_2 = 1]| = \text{negl}(\lambda).$$

Hyb₃: This is identical to Hyb_2 except that y_B and $y_C^\perp$ are generated as uniformly random strings instead of $F_K(u_B \| v_B)$ and $F_{K^\perp}(u_C \| v_C)$, respectively.

Note that simulation of the signing oracle and check of the adversary's winning condition can be done with $K\{u_B\|v_B\}$ and $K^\perp\{u_C\|v_C\}$ due to the modifications made in Hyb₁. Thus, from the security of PPRF, we have

$$|\Pr[\text{Hyb}_2 = 1] - \Pr[\text{Hyb}_3 = 1]| = \text{negl}(\lambda).$$

Hyb₄: This is identical to Hyb₃ except that y_B and $y_C^\perp$ are replaced with $y_B \oplus s$ and $y_C^\perp \oplus t$, respectively. More specifically, the following changes are applied.

- $\tilde{P}$ is generated as $\tilde{P} \leftarrow i\mathcal{O}(P'[C_{\text{mc}}, u_B, v_B, K\{u_B\|v_B\}, y_B \oplus s])$.
- $\tilde{P}^\perp$ is generated as $\tilde{P}^\perp \leftarrow i\mathcal{O}(P'[C_{\text{mc}}^\perp, u_C, v_C, K^\perp\{u_C\|v_C\}, y_C^\perp \oplus t])$.
- $\tilde{V}$ is generated as $\tilde{V} \leftarrow i\mathcal{O}(V'[u_B, v_B, K\{u_B\|v_B\}, y_B \oplus s])$.
- $\tilde{V}^\perp$ is generated as $\tilde{V}^\perp \leftarrow i\mathcal{O}(V'[u_C, v_C, K^\perp\{u_C\|v_C\}, y_C^\perp \oplus t])$.
- The adversary's winning condition is modified to the following: $\sigma_B = (w_B, w_B^\perp) = (y_B \oplus s, F_{K^\perp}(u_B\|v_B))$ and $\sigma_C = (w_C, w_C^\perp) = (F_K(u_C\|v_C), y_C^\perp \oplus t)$.

Since $(y_B, y_C^\perp)$ is distributed uniformly at random, so is $(y_B \oplus s, y_C^\perp \oplus t)$. Thus, we have

$$\Pr[\text{Hyb}_3 = 1] = \Pr[\text{Hyb}_4 = 1].$$

We below use the circuits P'' and V'' respectively described in Figure 9 and Figure 10.

Hyb₅: This is identical to Hyb₄ except for the following changes.

- $\tilde{P}$ is generated as $\tilde{P} \leftarrow i\mathcal{O}(P''[C_{\text{mc}}, v_B, \text{str}_B, \text{ct}_B, K, y_B])$, where $k_B \leftarrow \text{NCE.Gen}(1^\lambda)$, $\text{ct}_B \leftarrow \text{NCE.Enc}(k_B, A)$, and $\text{str}_B = k_B \oplus u_B$.
- $\tilde{P}^\perp$ is generated as $\tilde{P}^\perp \leftarrow i\mathcal{O}(P''[C_{\text{mc}}^\perp, v_C, \text{str}_C, \text{ct}_C, K^\perp, y_C^\perp])$, where $k_C \leftarrow \text{NCE.Gen}(1^\lambda)$, $\text{ct}_C \leftarrow \text{NCE.Enc}(k_C, A^\perp)$, and $\text{str}_C = k_C \oplus u_C$.

Below, we assume that both k_B and k_C are good as per Definition 4.1, i.e., the perfect correctness and key-robustness hold under those keys, since this holds with overwhelming probability.

We prove that $P'[C_{\text{mc}}, u_B, v_B, K\{u_B\|v_B\}, y_B \oplus s]$ and $P''[C_{\text{mc}}, v_B, \text{str}_B, \text{ct}_B, K, y_B]$ are functionally equivalent. For an input (a, u, v) , we consider the following cases.

- The case where $C_{\text{mc}}(a) = 0$ (i.e., $a \notin A + s$). In this case, both circuits clearly output $\perp$ according to the first items of Figures 7 and 9, respectively.
- The case where $C_{\text{mc}}(x) = 1$ (i.e., $a \in A + s$) and $(u \neq u_B \text{ or } v \neq v_B)$.
 - If $u \neq u_B$, $P'[C_{\text{mc}}, u_B, v_B, K\{u_B\|v_B\}, y_B \oplus s]$ outputs $F_{K\{u^*\|v^*\}}(u\|v) = F_K(u\|v)$ according to the second item of Figure 7. We show that $P''[C_{\text{mc}}, v_B, \text{str}_B, \text{ct}_B, K, y_B]$ also outputs $F_K(u\|v)$. Let $k' \leftarrow \text{str}_B \oplus u$. Then $k' \neq \text{str}_B \oplus u_B = k_B$ since $u \neq u_B$. Thus, by the key-robustness of NCE and the fact that ct_B is a ciphertext under the key k_B , $\text{NCE.Dec}(k', \text{ct}_B) = \perp$. Hence, it also outputs $F_K(u\|v)$ according to the third item of Figure 9.
 - If $v \neq v_B$, both circuits clearly output $F_K(u\|v)$ according to the second item of Figure 7 and the third item of Figure 9, respectively.

Hence, in either case, both circuits output $F_K(u\|v)$.

- The case where $C_{\text{mc}}(a) = 1$ (i.e., $a \in A + s$), $u = u_B$, and $v = v_B$. In this case, $P'[C_{\text{mc}}, u_B, v_B, K\{u_B\|v_B\}, y_B \oplus s]$ outputs the hardwired value $y_B \oplus s$ according to the third item of Figure 7. We show that $P''[C_{\text{mc}}, v_B, \text{str}_B, \text{ct}_B, K, y_B]$ also outputs $y_B \oplus s$ in this case. Let $k' \leftarrow \text{str}_B \oplus u$. Since $u = u_B$ and $\text{str}_B = k_B \oplus u_B$, we have $k' = k_B$. Thus, by the correctness of NCE, we have $\text{NCE.Dec}(k', \text{ct}_B) = A$, and combining it with the fact that $v = v_B$, the computation goes to the forth item of Figure 9. Since $a \in A + s$ holds, we have $\text{Can}_A(a) = s$. We then can conclude that it outputs $y_B \oplus s$.

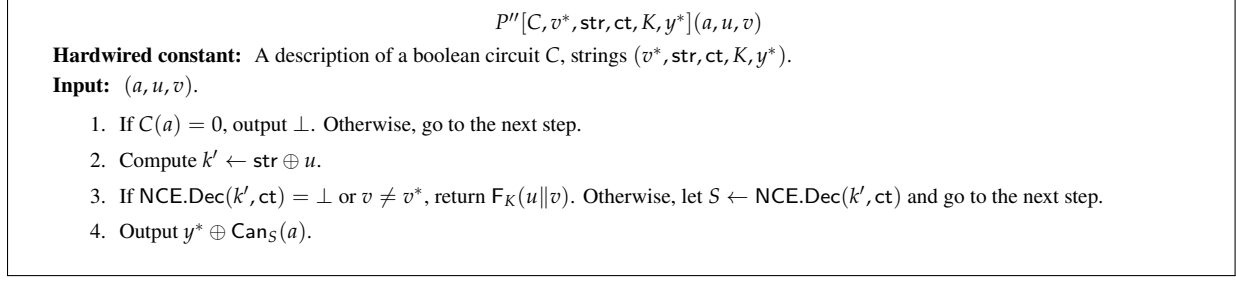


Figure 9: The description of the circuit P'' .

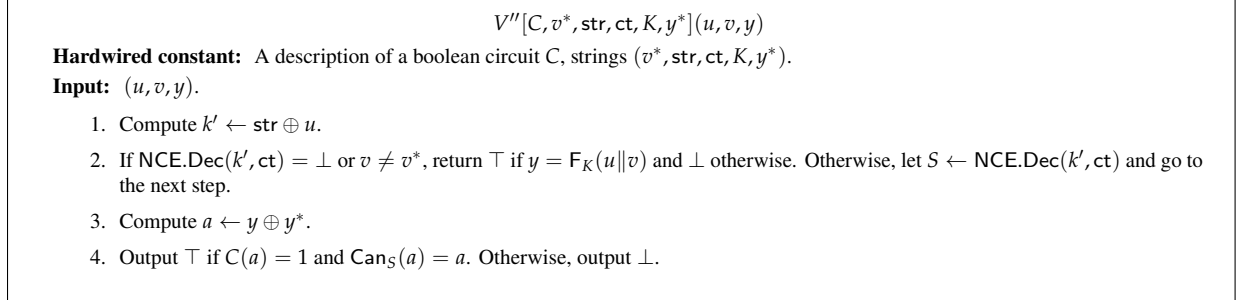


Figure 10: The description of the circuit V'' .

This shows that the two circuits are functionally equivalent.

We can similarly prove that $P'[C_{\text{mc}}^\perp, u_C, v_C, K^\perp\{u_C||v_C\}, y_C^\perp \oplus t]$ and $P''[C_{\text{mc}}^\perp, v_C, \text{str}_C, \text{ct}_C, K^\perp, y_C^\perp]$ are functionally equivalent. Then, from the above discussion and the security of $i\mathcal{O}$, we have

$$|\Pr[\text{Hyb}_4 = 1] - \Pr[\text{Hyb}_5 = 1]| = \text{negl}(\lambda).$$

Hyb₆: This is identical to Hyb₅ except for the following changes.

- $\tilde{V}$ is generated as $\tilde{V} \leftarrow i\mathcal{O}(V''[C_{\text{mc}}, v_B, \text{str}_B, \text{ct}_B, K, y_B])$, where ct_B is defined in Hyb₅.
- $\tilde{V}^\perp$ is generated as $\tilde{V}^\perp \leftarrow i\mathcal{O}(V''[C_{\text{mc}}^\perp, v_C, \text{str}_C, \text{ct}_C, K^\perp, y_C^\perp])$, where ct_C is defined in Hyb₅.

We prove that $V'[u_B, v_B, K\{u_B||v_B\}, y_B \oplus s]$ and $V''[C_{\text{mc}}, v_B, \text{str}_B, \text{ct}_B, K, y_B]$ are functionally equivalent. For an input (u, v, y) , we consider the following cases.

- The case where $u \neq u_B$ or $v \neq v_B$.
 - If $u \neq u_B$, $V'[u_B, v_B, K\{u_B||v_B\}, y_B \oplus s]$ outputs $\top$ if $y = F_{K\{u^*||v^*\}}(u||v) = F_K(u||v)$ and $\perp$ otherwise, according to the first item of Figure 8. We show that $V''[C_{\text{mc}}, v_B, \text{str}_B, \text{ct}_B, K, y_B]$ also works similarly. Let $k' \leftarrow \text{str}_B \oplus u$. Then $k' \neq \text{str}_B \oplus u_B = k_B$ since $u \neq u_B$. Thus, by the key-robustness of NCE and the fact that ct_B is a ciphertext under the key k_B , $\text{NCE.Dec}(k', \text{ct}_B) = \perp$. Hence, it also outputs $\top$ if $y = F_K(u||v)$ and $\perp$ otherwise according to the second item of Figure 10.
 - If $v \neq v_B$, both circuits clearly output $\top$ if $y = F_K(u||v)$ and $\perp$ otherwise, according to the first item of Figure 8 and the second item of Figure 10, respectively.

In either case, both circuits return the same outputs.

- The case where $u = u_B$ and $v = v_B$. In this case, $V'[u_B, v_B, K\{u_B||v_B\}, y_B \oplus s]$ outputs $\top$ if $y = y_B \oplus s$ and $\perp$ otherwise, according to the second item of Figure 8. We show that $V''[C_{\text{mc}}, v_B, \text{str}_B, \text{ct}_B, K, y_B]$ also works similarly. Let $k' \leftarrow \text{str}_B \oplus u$. Since $u = u_B$, we have $k' = k_B$. Thus, if we let $S \leftarrow \text{NCE.Dec}(k', \text{ct}_B)$,

$S = A$. Combining it with the fact that $v = v_B$, the computation goes to the forth item of Figure 10. By Lemma 5.1, $(C_{mc}(y \oplus y_B) = 1 \wedge \text{Can}_A(y \oplus y_B) = y \oplus y_B)$ is equivalent to $y \oplus y_B = s$, which in turn is equivalent to $y = y_B \oplus s$. Therefore, $V''[C_{mc}, v_B, \text{str}_B, \text{ct}_B, K, y_B]$ also outputs $\top$ if $y = y_B \oplus s$ and $\perp$ otherwise.

This shows that the two circuits are functionally equivalent.

We can similarly prove that $V'[u_C, v_C, K^\perp\{u_C\|v_C\}, y_C^\perp \oplus t]$ and $V''[C_{mc}^\perp, v_C, \text{str}_C, \text{ct}_C, K^\perp, y_C^\perp]$ are functionally equivalent. Then, from the above discussion and the security of $i\mathcal{O}$, we have

$$|\Pr[\text{Hyb}_5 = 1] - \Pr[\text{Hyb}_6 = 1]| = \text{negl}(\lambda).$$

Hyb₇: This is identical to Hyb₆ except for the following changes.

- (ct_B, k_B) are generated as $(\text{ct}_B, \text{st}_B) \leftarrow \text{NCE.Fake}(1^\lambda)$ and $k_B \leftarrow \text{NCE.Open}(\text{st}_B, A)$.
- (ct_C, k_C) are generated as $(\text{ct}_C, \text{st}_C) \leftarrow \text{NCE.Fake}(1^\lambda)$ and $k_C \leftarrow \text{NCE.Open}(\text{st}_C, A^\perp)$.

From the non-committing property of NCE, we have

$$|\Pr[\text{Hyb}_6 = 1] - \Pr[\text{Hyb}_7 = 1]| = \text{negl}(\lambda).$$

We below show that

$$\Pr[\text{Hyb}_7 = 1] \leq \text{negl}(\lambda)$$

by a reduction to the computational strong monogamy property of coset states with simulatable auxiliary input (Corollary 5.4) with respect to the following auxiliary input generator $\mathcal{Aux} = (\mathcal{AuxSetup}, \mathcal{AuxGen}_B, \mathcal{AuxGen}_C)$ and the corresponding simulator $\mathcal{Sim} = (\mathcal{Sim}_B, \mathcal{Sim}_C)$.

$\mathcal{AuxSetup}(1^\lambda)$:

- Generate $h \leftarrow \mathcal{H}$ and $H_t \leftarrow \text{TCR.KG}(1^\lambda)$.
- Output $\text{pp} = (h, H_t)$.

$\mathcal{AuxGen}_X(\text{pp}, A)$ where $X \in \{B, C\}$:

- Parse $\text{pp} = (h, H_t)$.
- Sample $(m_X, \text{aux}_X) \leftarrow \mathcal{D}_X(1^\lambda)$ and compute $u_X \leftarrow h(m_X)$ and $v_X \leftarrow H_t(m_X)$.
- Generate $(\text{ct}_X, \text{st}_X) \leftarrow \text{NCE.Fake}(1^\lambda)$ and

$$k_X \leftarrow \begin{cases} \text{NCE.Open}(\text{st}_B, A) & \text{if } X = B \\ \text{NCE.Open}(\text{st}_C, A^\perp) & \text{if } X = C \end{cases}$$

- Set $\text{str}_X := k_X \oplus u_X$.
- Output $z_{1,X} := (\text{aux}_X, \text{str}_X, v_X, \text{ct}_X)$ and $z_2 := m$.

$\mathcal{Sim}_X(\text{pp})$ where $X \in \{B, C\}$:

- Generate $(\text{aux}_X, v_X, \text{ct}_X)$ in the same way as in $\mathcal{AuxGen}_X$.
- Sample $\text{str}_X \leftarrow \{0, 1\}^{\ell_k}$.
- Output $z_{1,X} := (\text{aux}_X, \text{str}_X, v_X, \text{ct}_X)$.

First, we show that they satisfy the simulatability condition of Corollary 5.4.

Lemma 7.3. For $X \in \{\mathcal{B}, \mathcal{C}\}$ and any subspace $A \subseteq \mathbb{F}_2^n$,

$$\left\{ (\text{pp}, z_{1,X}) : \begin{array}{l} \text{pp} \leftarrow \text{AuxSetup}(1^\lambda) \\ (z_{1,X}, z_{2,X}) \leftarrow \text{AuxGen}_X(\text{pp}, A) \end{array} \right\} \approx \left\{ (\text{pp}, z_{1,X}) : \begin{array}{l} \text{pp} \leftarrow \text{AuxSetup}(1^\lambda) \\ z_{1,X} \leftarrow \text{Sim}_X(\text{pp}) \end{array} \right\}.$$

Proof of Lemma 7.3. In the execution of $\text{AuxGen}_X(\text{pp}, A)$, note that $\text{aux}_X, u_X = h(m_X)$, and $v_X = H_t(m_X)$ are the only variables that depend on m_X . By the assumption, m_X has conditional min-entropy at least $\ell_k + \ell_{\text{tcr}} + \omega(\log \lambda)$ given aux_X . Since v_X is an ℓ_{tcr} -bit string, by the chain rule, m_X has conditional min-entropy at least $\ell_k + \omega(\log \lambda)$ given both aux_X and v_X . Thus, by the leftover hash lemma (Lemma 3.3), $(h, H_t, \text{aux}_X, v_X, \text{ct}_X, k_X, h(m_X)) \approx (h, H_t, \text{aux}_X, v_X, \text{ct}_X, k_X, u'_X)$ where $u'_X \leftarrow \{0, 1\}^{\ell_k}$. This immediately implies Lemma 7.3. $\square$

We now construct the following adversary $\mathcal{A}_{\text{moe}} = (\hat{\mathcal{A}}, \hat{\mathcal{B}}, \hat{\mathcal{C}})$ for the experiment in Corollary 5.4 using $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$.

1. On input $|A_{s,t}\rangle$, $C_{\text{mc}} = i\mathcal{O}(A + s)$, $C_{\text{mc}}^\perp = i\mathcal{O}(A^\perp + t)$, $\text{pp} = (h, H_t)$, $z_{1,\mathcal{B}} = (\text{aux}_{\mathcal{B}}, \text{str}_{\mathcal{B}}, v_{\mathcal{B}}, \text{ct}_{\mathcal{B}})$, and $z_{1,\mathcal{C}} = (\text{aux}_{\mathcal{C}}, \text{str}_{\mathcal{C}}, v_{\mathcal{C}}, \text{ct}_{\mathcal{C}})$, $\hat{\mathcal{A}}$ generates $\text{vk} := (\tilde{V}, \tilde{V}^\perp, h, H_t)$, $\text{msk} := (K, K^\perp, h, H_t)$, and $s\kappa := (|A_{s,t}\rangle, \tilde{P}, \tilde{P}^\perp, h, H_t)$ as in Hyb₇. Note that $\hat{\mathcal{A}}$ can generate them with the given inputs and $(K, K^\perp, y_{\mathcal{B}}, y_{\mathcal{C}}^\perp)$ sampled by $\hat{\mathcal{A}}$ itself. $\hat{\mathcal{A}}$ then invokes $\mathcal{A}$ with vk , $s\kappa$, $\text{aux}_{\mathcal{B}}$, and $\text{aux}_{\mathcal{C}}$. $\mathcal{A}_{\text{moe}}$ below simulates the signing oracle for $\mathcal{A}_{\text{cp}}$ as in Hyb₇, which is possible with msk , $v_{\mathcal{B}}$, and $v_{\mathcal{C}}$.
2. When $\mathcal{A}$ outputs a bipartite state q over registers $R_{\mathcal{B}}$ and $R_{\mathcal{C}}$, $\hat{\mathcal{A}}$ sends $(\text{msk}, v_{\mathcal{B}}, v_{\mathcal{C}}, y_{\mathcal{B}}, q[R_{\mathcal{B}}])$ to $\hat{\mathcal{B}}$ and $(\text{msk}, v_{\mathcal{B}}, v_{\mathcal{C}}, y_{\mathcal{C}}^\perp, q[R_{\mathcal{C}}])$ to $\hat{\mathcal{C}}$.
3. $\hat{\mathcal{B}}$ and $\hat{\mathcal{C}}$ are respectively given $(A, r_{\mathcal{B}}, z_{2,\mathcal{B}} = m_{\mathcal{B}})$ and $(A, r_{\mathcal{C}}, z_{2,\mathcal{C}} = m_{\mathcal{C}})$, and behave as follows.
 - $\hat{\mathcal{B}}$ invokes $\mathcal{B}$ with $q[R_{\mathcal{B}}]$ and $m_{\mathcal{B}}$, and obtains $\sigma_{\mathcal{B}} = (w_{\mathcal{B}}, w_{\mathcal{B}}^\perp)$. $\hat{\mathcal{B}}$ outputs $s_{\mathcal{B}} := w_{\mathcal{B}} \oplus y_{\mathcal{B}}$.
 - $\hat{\mathcal{C}}$ invokes $\mathcal{C}$ with $q[R_{\mathcal{C}}]$ and $m_{\mathcal{C}}$, and obtains $\sigma_{\mathcal{C}} = (w_{\mathcal{C}}, w_{\mathcal{C}}^\perp)$. $\hat{\mathcal{C}}$ outputs $t_{\mathcal{C}} := w_{\mathcal{C}}^\perp \oplus y_{\mathcal{C}}^\perp$.

$\mathcal{A}_{\text{moe}}$ perfectly simulates Hyb₇ for $\mathcal{A}_{\text{cp}}$. If $\mathcal{A}_{\text{cp}}$ wins the simulated experiment (i.e., Hyb₇ outputs 1), we have $w_{\mathcal{B}} = y_{\mathcal{B}} \oplus s$ and $w_{\mathcal{C}}^\perp = y_{\mathcal{C}}^\perp \oplus t$, that is, $w_{\mathcal{B}} \oplus y_{\mathcal{B}} = s$ and $w_{\mathcal{C}}^\perp \oplus y_{\mathcal{C}}^\perp = t$. Thus, by Corollary 5.4 and Lemma 7.3, we have

$$\Pr[\text{Hyb}_7 = 1] = \Pr\left[\text{Exp}_{i\mathcal{O}, \text{Aux}_X, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-moe-aux}}(\lambda) = 1\right] \leq \text{negl}(\lambda).$$

Combining the above, we have

$$\Pr\left[\text{Exp}_{\text{CP-SIG}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{uf-anti-piracy}}(\lambda) = 1\right] \leq \text{negl}(\lambda).$$

This completes the proof of Theorem 7.2. $\square$

8 Construction of Copy-Protected PRF

We prove the following theorem.

Theorem 8.1. Assuming the existence of polynomially secure $i\mathcal{O}$ and OWFs, for any constant $c > 0$ and any polynomials ℓ_{inp} and ℓ_{out} , there exists a copy-protected PRF scheme with the domain $\{0, 1\}^{\ell_{\text{inp}}}$ and range $\{0, 1\}^{\ell_{\text{out}}}$ that satisfies $\mathcal{D}\text{-IND}^+$ anti-piracy security for any QPT-samplable distributions $\mathcal{D} = (\mathcal{D}_{\mathcal{B}}, \mathcal{D}_{\mathcal{C}})$, where $\mathcal{D}_{\mathcal{B}}$ and $\mathcal{D}_{\mathcal{C}}$ have conditional min-entropy at least λ^c .

8.1 Construction

Let $n, \ell_k, \ell_{\text{tcr}}, \ell_{\text{out}}$, and ℓ_{inp} be polynomials in the security parameter λ .

We use the following building blocks.

- iO for polynomial size classical circuits $i\mathcal{O}$.
- PPRF $\text{PPRF} = (\text{F}, \text{Puncture})$ with the domain $\{0, 1\}^{\ell_k + \ell_{\text{tcr}}}$ and the range $\{0, 1\}^{\ell_{\text{out}}}$.
- Universal hash function family $\mathcal{H} : \{0, 1\}^{\ell_{\text{inp}}} \rightarrow \{0, 1\}^{\ell_k}$.
- TCR hash function TCR.Gen with the domain $\{0, 1\}^{\ell_{\text{inp}}}$ and the range $\{0, 1\}^{\ell_{\text{tcr}}}$.

We below construct a copy-protected PRF scheme $\text{CP-PRF} = \text{CP-PRF}(\mathcal{KG}, \text{Eval}, \text{QEval})$ with the domain $\{0, 1\}^{\ell_{\text{inp}}}$ and the range $\{0, 1\}^{\ell_{\text{out}}}$.

$\text{CP-PRF}(\mathcal{KG}(1^\lambda))$:

- Generate a uniformly random subspace $A \subseteq \mathbb{F}_2^n$ of dimension $\frac{n}{2}$, and two uniformly random elements $(s, t) \in \text{CS}(A) \times \text{CS}(A^\perp)$.
- Define circuits $C_{\text{mc}} := i\mathcal{O}(A + s)$ and $C_{\text{mc}}^\perp := i\mathcal{O}(A^\perp + t)$.
- Generate keys $K, K^\perp \leftarrow \{0, 1\}^\lambda$ for PPRF.
- Generate $h \leftarrow \mathcal{H}$ and $H_t \leftarrow \text{TCR.Gen}(1^\lambda)$.
- Generate two obfuscated circuits $\tilde{P} \leftarrow i\mathcal{O}(P[C_{\text{mc}}, K])$ and $\tilde{P}^\perp \leftarrow i\mathcal{O}(P[C_{\text{mc}}^\perp, K^\perp])$, where P is the circuit described in Figure 11.
- Output $\text{msk} := (K, K^\perp, h, H_t)$ and $s\mathcal{K} := (|A_{s,t}\rangle, \tilde{P}, \tilde{P}^\perp, h, H_t)$.

$\text{CP-PRF.Eval}(\text{msk}, x)$:

- Parse $\text{msk} = (K, K^\perp, h, H_t)$.
- Compute $u \leftarrow h(x)$ and $v \leftarrow H_t(x)$.
- Output $Y \leftarrow F_K(u \| v) \oplus F_{K^\perp}(u \| v)$.

$\text{CP-PRF.QEval}(s\mathcal{K}, x)$:

- Parse $s\mathcal{K} = (\rho, \tilde{P}, \tilde{P}^\perp, h, H_t)$, where ρ is a quantum state, and $\tilde{P}$ and $\tilde{P}^\perp$ are interpreted as descriptions of circuits.
- Compute $u \leftarrow h(x)$ and $v \leftarrow H_t(x)$.
- Let E and $E^\perp$ be unitaries that respectively work as follows:

$$|z_1\rangle |z_2\rangle \mapsto |z_1\rangle \left| z_2 \oplus \tilde{P}(z_1, u, v) \right\rangle \quad \text{and} \quad |z_1\rangle |z_2\rangle \mapsto |z_1\rangle \left| z_2 \oplus \tilde{P}^\perp(z_1, u, v) \right\rangle$$

for any $z_1 \in \{0, 1\}^n$ and $z_2 \in \{0, 1\}^{\ell_{\text{out}}}$.

- Apply E to $\rho \otimes |0^{\ell_{\text{out}}}\rangle$ and measure the second register to obtain an outcome y . Then, trace out the second register and obtain the state ρ' .
- Apply $E^\perp$ to $H^{\otimes n} \rho' \otimes |0^{\ell_{\text{out}}}\rangle$ and measure the second register to obtain an outcome $y^\perp$.
- Output $Y := y \oplus y^\perp$.

$P[C, K](a, u, v)$
Hardwired constant: A description of a boolean circuit C and a string K .
Input: (a, u, v) .
1. If $C(a) = 0$, output $\perp$. Otherwise, go to the next step.
2. Output $F_K(u v)$.

Figure 11: The description of the circuit P .

Perfect correctness. We prove

$$\Pr [\text{CP-PRF.QEval}(s\kappa, x) = \text{CP-PRF.Eval}(\text{msk}, x) \mid (\text{msk}, s\kappa) \leftarrow \text{CP-PRF.KG}(1^\lambda)] = 1$$

for any $x \in \{0, 1\}^{\ell_{\text{inp}}}$. Suppose we generate $\text{msk} := (K, K^\perp, h, H_t)$ and $s\kappa := (|A_{s,t}\rangle, \tilde{P}, \tilde{P}^\perp, h, H_t)$ as prescribed in the description of CP-PRF.KG . Fix $x \in \{0, 1\}^{\ell_{\text{inp}}}$ and let $u \leftarrow h(m)$ and $v \leftarrow H_t(m)$. We have $\text{CP-PRF.Eval}(\text{msk}, x) = F_K(u||v) \oplus F_{K^\perp}(u||v)$. Letting $y \oplus y^\perp \leftarrow \text{CP-PRF.QEval}(s\kappa, x)$, we show that $y = F_K(u||v)$ and $y^\perp = F_{K^\perp}(u||v)$ holds with probability 1.

- We prove $y = F_K(u||v)$. For any $a \in A + s$, we have $\tilde{P}(a, u, v) = F_K(u||v)$ from the correctness of $i\mathcal{O}$. Thus, given that $|A_{s,t}\rangle = \frac{1}{\sqrt{|A|}} \sum_{a \in A} (-1)^{\langle a, t \rangle} |a + s\rangle$, if we apply the unitary E to $|A_{s,t}\rangle \otimes |0^{\ell_{\text{out}}}\rangle$, we obtain the state $|A_{s,t}\rangle \otimes |F_K(u||v)\rangle$. This means $y = F_K(u||v)$ holds, and the state obtained by tracing out the second register after measuring y is $|A_{s,t}\rangle$.
- We then prove $y^\perp = F_{K^\perp}(u||v)$. For any $a \in A^\perp + t$, we have $\tilde{P}_2(a, u, v) = F_{K^\perp}(u||v)$ from the correctness of $i\mathcal{O}$. Thus, given that $H^{\otimes n} |A_{s,t}\rangle = \frac{1}{\sqrt{|A^\perp|}} \sum_{a \in A^\perp} (-1)^{\langle a, s \rangle} |a + t\rangle$, if we apply the unitary $E^\perp$ to $H^{\otimes n} |A_{s,t}\rangle \otimes |0^{\ell_{\text{out}}}\rangle$, we obtain the state $H^{\otimes n} |A_{s,t}\rangle \otimes |F_{K^\perp}(u||v)\rangle$. This means $y^\perp = F_{K^\perp}(u||v)$ holds.

The above shows the perfect correctness of CP-PRF.

8.2 Proof of Security

In the security proof, we additionally use the following primitive, where $\ell_A := n^2/2$ is the number of bits needed to represent a subspace $A \subseteq \mathbb{F}_2^n$ of dimension $n/2$.

- A key-robust non-committing encryption scheme $\text{NCE} = (\text{NCE.Enc}, \text{NCE.Dec}, \text{NCE.Fake}, \text{NCE.Open})$ with the message space $\{0, 1\}^{\ell_A + n}$ and the key space $\{0, 1\}^{\ell_k}$.

We prove the following theorem.

Theorem 8.2. *Let $\mathcal{D}$ be a distribution over $\{0, 1\}^{\ell_{\text{inp}}} \times \{0, 1\}^{\ell_{\text{aux}}}$ with conditional min-entropy at least $\ell_k + \ell_{\text{tcr}} + \omega(\log \lambda)$. Then, assuming the existence of key-robust non-committing encryption NCE, CP-PRF satisfies $\mathcal{D}$ -IND⁺anti-piracy security.*

We may choose ℓ_k and ℓ_{tcr} to be arbitrarily small polynomials, while allowing ℓ_{inp} and ℓ_{out} to be any polynomials. Therefore, Theorem 8.1 follows.

Proof of Theorem 8.2. Let $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ be a QPT adversary against $\mathcal{D}$ -IND⁺anti-piracy security of CP-PRF. We consider the following sequence of hybrids.

Hyb₀: This is the original security experiment $\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{indplus-anti-piracy}}(\lambda)$ in Definition 6.5. More specifically, it works as follows.

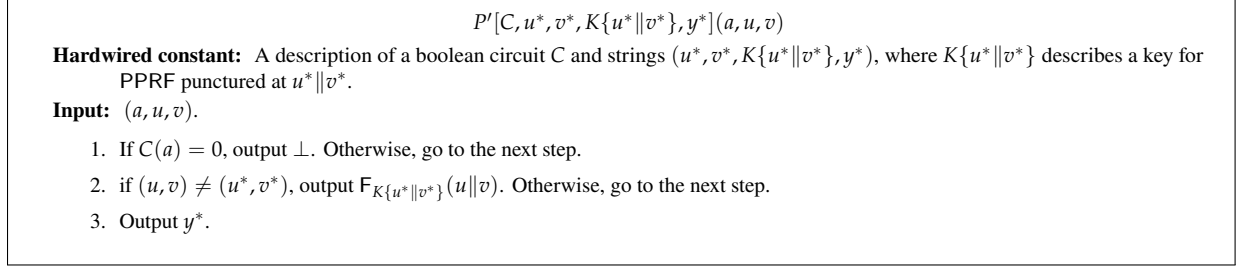


Figure 12: The description of the circuit P' .

1. The challenger generates $\text{msk} := (K, K^\perp, h, H_t)$, and $s_K := (|A_{s,t}\rangle, \tilde{P}, \tilde{P}^\perp, h, H_t)$ as prescribed in the description of CP-PRF.XG . The challenger also generates $(x_B, \text{aux}_B) \leftarrow \mathcal{D}(1^\lambda)$ and $(x_C, \text{aux}_C) \leftarrow \mathcal{D}(1^\lambda)$. The challenger sends s_K , aux_B , and aux_C to $\mathcal{A}$.
2. $\mathcal{A}$ can make arbitrarily many queries to the evaluation oracle, which on input x , returns $Y \leftarrow F_K(u||v) \oplus F_{K^\perp}(u||v)$ where $u \leftarrow h(x)$ and $v \leftarrow H_t(x)$.
3. $\mathcal{A}$ creates a bipartite state q over registers R_B and R_C . Then, $\mathcal{A}$ sends R_B to $\mathcal{B}$, and R_C to $\mathcal{C}$.
4. The challenger does the following.
 - Choose $\text{coin}_B \leftarrow \{0, 1\}$, and generate $Y_{B,0} \leftarrow F_K(u_B||v_B) \oplus F_{K^\perp}(u_B||v_B)$, and $Y_{B,1} \leftarrow \{0, 1\}^{\ell_{\text{out}}}$, where $u_B \leftarrow h(x_B)$ and $v_B \leftarrow H_t(x_B)$.
 - Choose $\text{coin}_C \leftarrow \{0, 1\}$, and generate $Y_{C,0} \leftarrow F_K(u_C||v_C) \oplus F_{K^\perp}(u_C||v_C)$, and $Y_{C,1} \leftarrow \{0, 1\}^{\ell_{\text{out}}}$, where $u_C \leftarrow h(x_C)$ and $v_C \leftarrow H_t(x_C)$.

The challenger sends $(x_B, Y_{B, \text{coin}_B})$ and $(x_C, Y_{C, \text{coin}_C})$ to $\mathcal{B}$ and $\mathcal{C}$, respectively.

5. $\mathcal{B}$ and $\mathcal{C}$ can make arbitrarily many queries to the evaluation oracle described in the second item, with the restriction that $\mathcal{B}$ and $\mathcal{C}$ are not allowed to query x_B and x_C , respectively.
6. $\mathcal{B}$ and $\mathcal{C}$ respectively output coin'_B and coin'_C . The challenger outputs 1 if $\text{coin}'_B \oplus \text{coin}'_C = \text{coin}_B \oplus \text{coin}_C$ otherwise outputs 0.

By the definition, we clearly have

$$\Pr[\text{Hyb}_0 = 1] = \Pr\left[\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{CP}}}^{\text{indplus-anti-piracy}}(\lambda) = 1\right].$$

Hyb₁: This is identical to Hyb_0 except for the following changes.

- It immediately outputs 0 if $v_B = v_C$.
- The signing oracle returns $\perp$ on input m if $H_t(x) = v_B$ or $H_t(x) = v_C$ holds.

By the min-entropy condition of $\mathcal{D}$, we have $x_B \neq x_C$ with overwhelming probability. Thus, by the security of TCR, $v_B = v_C$ occurs with negligible probability.

For the second item, note that the probability that the adversary queries m_B or m_C is negligible, since $\mathcal{B}$ and $\mathcal{C}$ are prohibited to query x_B and x_C , respectively, and they are statistically unpredictable from other part of the adversary. On the other hand, by the security of TCR, the probability that the adversary queries the signing oracle on x such that $(H_t(x) = v_B \wedge x \neq x_B)$ or $(H_t(x) = v_C \wedge x \neq x_C)$ is negligible. Therefore, we have

$$|\Pr[\text{Hyb}_0 = 1] - \Pr[\text{Hyb}_1 = 1]| = \text{negl}(\lambda).$$

We below use the circuit P' described in Figure 12.

Hyb₂: This is identical to Hyb_1 except for the following changes.

- $\tilde{P}$ is generated as $\tilde{P} \leftarrow i\mathcal{O}(P'[C_{mc}, u_B, v_B, K\{u_B\|v_B\}, y_{B,0}])$, where $K\{u_B\|v_B\} \leftarrow \text{Puncture}(K, u_B\|v_B)$ and $y_{B,0} \leftarrow F_K(u_B\|v_B)$.
- $\tilde{P}^\perp$ is generated as $\tilde{P}^\perp \leftarrow i\mathcal{O}(P'[C_{mc}^\perp, u_C, v_C, K^\perp\{u_C\|v_C\}, y_{C,0}^\perp])$, where $K^\perp\{u_C\|v_C\} \leftarrow \text{Puncture}(K^\perp, u_C\|v_C)$ and $y_{C,0}^\perp \leftarrow F_{K^\perp}(u_C\|v_C)$.

Moreover, we set $Y_{B,0} = y_{B,0} \oplus F_{K^\perp}(u_B\|v_B)$ and $Y_{C,0} = F_K(u_C\|v_C) \oplus y_{C,0}^\perp$.

The values of $Y_{B,0}$ and $Y_{C,0}$ are clearly unchanged from the previous hybrid. Moreover, it is easy to verify that each replacement of the obfuscated circuits described above involves functionally equivalent circuits. Thus, from the security of $i\mathcal{O}$, we have

$$|\Pr[\text{Hyb}_1 = 1] - \Pr[\text{Hyb}_2 = 1]| = \text{negl}(\lambda).$$

Hyb₃: This is identical to **Hyb₂** except that $y_{B,0}$ and $y_{C,0}^\perp$ are generated as uniformly random strings instead of $F_K(u_B\|v_B)$ and $F_{K^\perp}(u_C\|v_C)$, respectively.

Note that, due to the modifications made in **Hyb₁**, the simulation of the evaluation oracle and the computation of $F_{K^\perp}(u_B\|v_B)$ and $F_K(u_C\|v_C)$ —required for computing $Y_{B,0}$ and $Y_{C,0}$, respectively—can be performed using only $K\{u_B\|v_B\}$ and $K^\perp\{u_C\|v_C\}$. Thus, from the security of PPRF, we have

$$|\Pr[\text{Hyb}_2 = 1] - \Pr[\text{Hyb}_3 = 1]| = \text{negl}(\lambda).$$

Hyb₄: This is identical to **Hyb₃** except for the following changes.

- $Y_{B,1}$ is replaced with $y_{B,1} \oplus F_{K^\perp}(u_B\|v_B)$, where $y_{B,1} \leftarrow \{0,1\}^{\ell_{\text{out}}}$.
- $Y_{C,1}$ is replaced with $F_K(u_C\|v_C) \oplus y_{C,1}^\perp$, where $y_{C,1}^\perp \leftarrow \{0,1\}^{\ell_{\text{out}}}$.

The joint distribution of $(Y_{B,1}, Y_{C,1})$ is uniformly random in both **Hyb₃** and **Hyb₄**. Thus we have

$$\Pr[\text{Hyb}_3 = 1] = \Pr[\text{Hyb}_4 = 1].$$

Note that, in **Hyb₄**, $\mathcal{B}$ is given $(x_B, y_{B,\text{coin}_B} \oplus F_{K^\perp}(u_B\|v_B))$ as the challenge, and $\mathcal{C}$ is given $(x_C, F_K(u_C\|v_C) \oplus y_{C,\text{coin}_C}^\perp)$ as the challenge, where $y_{B,0}, y_{B,1}, y_{C,0}^\perp, y_{C,1}^\perp \leftarrow \{0,1\}^{\ell_{\text{out}}}$.

Hyb₅: This is identical to **Hyb₄** except for the following changes.

- $\mathcal{B}$ is given $(x_B, y_{B,0} \oplus F_{K^\perp}(u_B\|v_B))$ as the challenge regardless of the value of coin_B .
- $\mathcal{C}$ is given $(x_C, F_K(u_C\|v_C) \oplus y_{C,0}^\perp)$ as the challenge regardless of the value of coin_C .
- $\tilde{P}$ is generated as $\tilde{P} \leftarrow i\mathcal{O}(P'[C_{mc}, u_B, v_B, K\{u_B\|v_B\}, y_{B,\text{coin}_B}])$. That is, the hardwired value is switched from $y_{B,0}$ to y_{B,coin_B} .
- $\tilde{P}^\perp$ is generated as $\tilde{P}^\perp \leftarrow i\mathcal{O}(P'[C_{mc}^\perp, u_C, v_C, K^\perp\{u_C\|v_C\}, y_{C,\text{coin}_C}^\perp])$. That is, the hardwired values is switched from $y_{C,0}^\perp$ to $y_{C,\text{coin}_C}^\perp$.

Note that **Hyb₅** is obtained by simply renaming $(y_{B,0}, y_{B,1})$ to $(y_{B,\text{coin}_B}, y_{B,1-\text{coin}_B})$ and $(y_{C,0}, y_{C,1})$ to $(y_{C,\text{coin}_C}, y_{C,1-\text{coin}_C})$. This change is perfectly indistinguishable from the adversary's view, since $y_{B,0}$ and $y_{B,1}$, and $y_{C,0}$ and $y_{C,1}$ are each identically distributed. Thus, we have

$$\Pr[\text{Hyb}_4 = 1] = \Pr[\text{Hyb}_5 = 1].$$

Hyb₆: This is identical to **Hyb₅** except that coin_B and coin_C are replaced with $\text{coin}_B \oplus \langle r_B, s \rangle$ and $\text{coin}_C \oplus \langle r_C, t \rangle$, respectively, where $r_B, r_C \leftarrow \{0,1\}^n$. More specifically, the following changes are applied.

- $\tilde{P}$ is generated as $\tilde{P} \leftarrow i\mathcal{O}(P'[C_{mc}, u_B, v_B, K\{u_B\|v_B\}, y_{B,\text{coin}_B \oplus \langle r_B, s \rangle}])$.

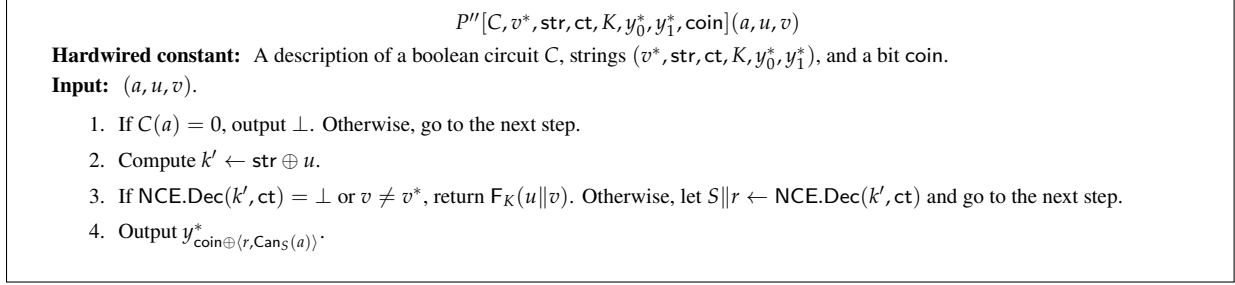


Figure 13: The description of the circuit P'' .

- $\tilde{P}^\perp$ is generated as $\tilde{P}^\perp \leftarrow i\mathcal{O}(P'[C_{\text{mc}}^\perp, u_C, v_C, K^\perp\{u_C||v_C\}, y_{C, \text{coin}_C \oplus \langle r_C, t \rangle}^\perp])$.
- The challenger outputs 1 if $\text{coin}'_B \oplus \text{coin}'_C = \text{coin}_B \oplus \text{coin}_C \oplus \langle r_B, s \rangle \oplus \langle r_C, t \rangle$ and otherwise outputs 0.

Since $(\text{coin}_B, \text{coin}_C)$ is distributed uniformly at random, so is $(\text{coin}_B \oplus \langle r_B, s \rangle, \text{coin}_C \oplus \langle r_C, t \rangle)$. Thus, we have

$$\Pr[\text{Hyb}_5 = 1] = \Pr[\text{Hyb}_6 = 1].$$

We below use the circuit P'' described in Figure 13.

Hyb₇: This is identical to Hyb₆ except the following changes.

- $\tilde{P}$ is generated as $\tilde{P} \leftarrow i\mathcal{O}(P''[C_{\text{mc}}, v_B, \text{str}_B, \text{ct}_B, K, y_{B,0}, y_{B,1}, \text{coin}_B])$, where $k_B \leftarrow \text{NCE.Gen}(1^\lambda)$, $\text{ct}_B \leftarrow \text{NCE.Enc}(k_B, A||r_B)$, and $\text{str}_B = k_B \oplus u_B$.
- $\tilde{P}^\perp$ is generated as $\tilde{P}^\perp \leftarrow i\mathcal{O}(P''[C_{\text{mc}}^\perp, v_C, \text{str}_C, \text{ct}_C, K^\perp, y_{C,0}^\perp, y_{C,1}^\perp, \text{coin}_C])$, where $k_C \leftarrow \text{NCE.Gen}(1^\lambda)$, $\text{ct}_C \leftarrow \text{NCE.Enc}(k_C, A^\perp||r_C)$, and $\text{str}_C = k_C \oplus u_C$.

Below, we assume that both k_B and k_C are good as per Definition 4.1, i.e., the perfect correctness and key-robustness hold under those keys, since this holds with overwhelming probability.

We prove that $P'[C_{\text{mc}}, u_B, v_B, K\{u_B||v_B\}, y_{B, \text{coin}_B \oplus \langle r_B, s \rangle}]$ and $P''[C_{\text{mc}}, v_B, \text{str}_B, \text{ct}_B, K, y_{B,0}, y_{B,1}, \text{coin}_B]$ are functionally equivalent. For an input (a, u, v) , we consider the following three cases.

- The case where $C_{\text{mc}}(a) = 0$ (i.e., $a \notin A + s$). In this case, both circuits output $\perp$ according to the first items of Figures 12 and 13, respectively.
- The case where $C_{\text{mc}}(x) = 1$ (i.e., $a \in A + s$), and $(u \neq u_B$ or $v \neq v_B)$.
 - If $u \neq u_B$, $P'[C_{\text{mc}}, u_B, v_B, K\{u_B||v_B\}, y_{B, \text{coin}_B \oplus \langle r_B, s \rangle}]$ outputs $F_{K\{u^*||v^*\}}(u||v) = F_K(u||v)$ according to the second item of Figure 12. We show that $P''[C_{\text{mc}}, v_B, \text{str}_B, \text{ct}_B, K, y_{B,0}, y_{B,1}, \text{coin}_B]$ also outputs $F_K(u||v)$. Let $k' \leftarrow \text{str}_B \oplus u$. Then $k' \neq \text{str}_B \oplus u_B = k_B$ since $u \neq u_B$. Thus, by the key-robustness of NCE and the fact that ct_B is a ciphertext under the key k_B , $\text{NCE.Dec}(k', \text{ct}_B) = \perp$. Hence, it also outputs $F_K(u||v)$ according to the third item of Figure 13.
 - If $v \neq v_B$, both circuits clearly output $F_K(u||v)$ according to the second item of Figure 12 and the third item of Figure 13, respectively.
- The case where $C_{\text{mc}}(a) = 1$ (i.e., $a \in A + s$), $u = u_B$, and $v = v_B$. In this case, $P^*[C_{\text{mc}}, u_B, v_B, K\{u_B||v_B\}, y_{B, \text{coin}_B \oplus \langle r_B, s \rangle}]$ outputs the hardwired value $y_{B, \text{coin}_B \oplus \langle r_B, s \rangle}$ according to the third item of Figure 12. We show that $P''[C_{\text{mc}}, v_B, \text{str}_B, \text{ct}_B, K, y_{B,0}, y_{B,1}, \text{coin}_B]$ also outputs $y_{B, \text{coin}_B \oplus \langle r_B, s \rangle}$ in this case. Let $k' \leftarrow \text{str}_B \oplus u$. Since $u = u_B$ and $\text{str}_B = k_B \oplus u_B$, we have $k' = k_B$. Thus, by the correctness of NCE, we have $\text{NCE.Dec}(k', \text{ct}_B) = A||r_B$, and combining it with the fact that $v = v_B$, the computation goes to the forth item of Figure 13. Since $a \in A + s$ holds, we have $\text{Can}_A(a) = s$. We then can conclude that it outputs $y_{B, \text{coin}_B \oplus \langle r_B, s \rangle}$.

This shows that the two circuits are functionally equivalent.

We can similarly prove that $P'[C_{mc}^\perp, u_C, v_C, K^\perp\{u_C\|v_C\}, y_{C, \text{coin}_C \oplus \langle r_C, t \rangle}^\perp]$ and $P''[C_{mc}^\perp, v_C, \text{str}_C, \text{ct}_C, K^\perp, y_{C,0}^\perp, y_{C,1}^\perp, \text{coin}_C]$ are functionally equivalent. Then, from the above discussion and the security of $i\mathcal{O}$, we have

$$|\Pr[\text{Hyb}_6 = 1] - \Pr[\text{Hyb}_7 = 1]| = \text{negl}(\lambda).$$

Hyb₈: This is identical to **Hyb₇** except for the following changes.

- (ct_B, k_B) are generated as $(\text{ct}_B, \text{st}_B) \leftarrow \text{NCE.Fake}(1^\lambda)$ and $k_B \leftarrow \text{NCE.Open}(\text{st}_B, A\|r_B)$.
- (ct_C, k_C) are generated as $(\text{ct}_C, \text{st}_C) \leftarrow \text{NCE.Fake}(1^\lambda)$ and $k_C \leftarrow \text{NCE.Open}(\text{st}_C, A^\perp\|r_C)$.

From the non-committing property of NCE, we have

$$|\Pr[\text{Hyb}_7 = 1] - \Pr[\text{Hyb}_8 = 1]| = \text{negl}(\lambda).$$

We below show that

$$\Pr[\text{Hyb}_8 = 1] \leq \frac{1}{2} + \text{negl}(\lambda)$$

by a reduction to the computational strong indistinguishability monogamy property of coset states with simulatable auxiliary input (Theorem 5.2) with respect to the following auxiliary input generator $\mathcal{Aux} = (\mathcal{AuxSetup}, \mathcal{AuxGen}_B, \mathcal{AuxGen}_C)$ and the corresponding simulator $\mathcal{Sim} = (\mathcal{Sim}_B, \mathcal{Sim}_C)$.

$\mathcal{AuxSetup}(1^\lambda)$:

- Generate $h \leftarrow \mathcal{H}$ and $H_t \leftarrow \text{TCR.KG}(1^\lambda)$.
- Output $\text{pp} = (h, H_t)$.

$\mathcal{AuxGen}_X(A, r_X)$ where $X \in \{B, C\}$:

- Parse $\text{pp} = (h, H_t)$.
- Sample $(x_X, \text{aux}_X) \leftarrow \mathcal{D}_X(1^\lambda)$ and compute $u_X \leftarrow h(x_X)$ and $v_X \leftarrow H_t(x_X)$.
- Generate $(\text{ct}_X, \text{st}_X) \leftarrow \text{NCE.Fake}(1^\lambda)$ and

$$k_X \leftarrow \begin{cases} \text{NCE.Open}(\text{st}_B, A\|r_B) & \text{if } X = B \\ \text{NCE.Open}(\text{st}_C, A^\perp\|r_C) & \text{if } X = C \end{cases}$$

- Set $\text{str}_X := k_X \oplus u_X$.
- Output $z_{1,X} := (\text{aux}_X, \text{str}_X, v_X, \text{ct}_X)$ and $z_2 := x$.

$\mathcal{Sim}_X(1^\lambda)$ where $X \in \{B, C\}$:

- Generate $(\text{aux}_X, v_X, \text{ct}_X)$ in the same way as in $\mathcal{AuxGen}_X$.
- Sample $\text{str}_X \leftarrow \{0, 1\}^{\ell_k + n}$.
- Output $z_{1,X} := (\text{aux}_X, \text{str}_X, v_X, \text{ct}_X)$.

First, we show that they satisfy the simulatability condition of Theorem 5.2.

Lemma 8.3. For $X \in \{B, C\}$ and any subspace $A \subseteq \mathbb{F}_2^n$ and $r_X \in \{0, 1\}^n$,

$$\left\{ (\text{pp}, z_{1,X}) : \begin{array}{l} \text{pp} \leftarrow \mathcal{AuxSetup}(1^\lambda) \\ (z_{1,X}, z_{2,X}) \leftarrow \mathcal{AuxGen}_X(\text{pp}, A, r_X) \end{array} \right\} \approx \left\{ (\text{pp}, z_1) : \begin{array}{l} \text{pp} \leftarrow \mathcal{AuxSetup}(1^\lambda) \\ z_{1,X} \leftarrow \mathcal{Sim}_X(\text{pp}) \end{array} \right\}.$$

Proof of Lemma 8.3. In the execution of $\mathcal{AuxGen}_X(\text{pp}, A, r_X)$, note that $\text{aux}_X, u_X = h(x_X)$, and $v_X = H_t(x_X)$ are the only variables that depend on x_X . By the assumption, x_X has conditional min-entropy at least $\ell_k + \ell_{\text{tr}} + \omega(\log \lambda)$ given aux_X . Since v_X is an ℓ_{tr} -bit string, by the chain rule, x_X has conditional min-entropy at least $\ell_k + \omega(\log \lambda)$ given both aux_X and v_X . Thus, by the leftover hash lemma (Lemma 3.3), $(h, H_t, \text{aux}_X, v_X, \text{ct}_X, k_X, h(x_X)) \approx (h, H_t, \text{aux}_X, v_X, \text{ct}_X, k_X, u'_X)$ where $u'_X \leftarrow \{0, 1\}^{\ell_k}$. This immediately implies Lemma 8.3. $\square$

We now construct the following adversary $\mathcal{A}_{\text{moe}} = (\hat{\mathcal{A}}, \hat{\mathcal{B}}, \hat{\mathcal{C}})$ for the experiment in Theorem 5.2 using $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$.

1. On input $|A_{s,t}\rangle$, $C_{\text{mc}} = i\mathcal{O}(A + s)$, $C_{\text{mc}}^\perp = i\mathcal{O}(A^\perp + t)$, $\text{pp} = (h, H_t)$, $z_{1,\mathcal{B}} = (\text{aux}_{\mathcal{B}}, \text{str}_{\mathcal{B}}, v_{\mathcal{B}}, \text{ct}_{\mathcal{B}})$, and $z_{1,\mathcal{C}} = (\text{aux}_{\mathcal{C}}, \text{str}_{\mathcal{C}}, v_{\mathcal{C}}, \text{ct}_{\mathcal{C}})$. $\hat{\mathcal{A}}$ generates $\text{msk} := (K, K^\perp, h, H_t)$ and $s\kappa := (|A_{s,t}\rangle, \tilde{P}, \tilde{P}^\perp, h, H_t)$ as in Hyb_8 . Note that $\hat{\mathcal{A}}$ can generate them with the given inputs and $(K, K^\perp, \text{coin}_{\mathcal{B}}, y_{\mathcal{B},0}, y_{\mathcal{B},1}, \text{coin}_{\mathcal{C}}, y_{\mathcal{C},0}^\perp, y_{\mathcal{C},1}^\perp)$ sampled by $\hat{\mathcal{A}}$ itself. $\hat{\mathcal{A}}$ then invokes $\mathcal{A}$ with $s\kappa$, $\text{aux}_{\mathcal{B}}$, and $\text{aux}_{\mathcal{C}}$. $\mathcal{A}_{\text{moe}}$ below simulates the evaluation oracle for $\mathcal{A}_{\text{cp}}$ as in Hyb_8 , which is possible with msk , $v_{\mathcal{B}}$, and $v_{\mathcal{C}}$.
2. When $\mathcal{A}$ outputs a bipartite state q over registers $R_{\mathcal{B}}$ and $R_{\mathcal{C}}$, $\hat{\mathcal{A}}$ sends $(\text{msk}, v_{\mathcal{B}}, v_{\mathcal{C}}, \text{coin}_{\mathcal{B}}, y_{\mathcal{B},0}, q[R_{\mathcal{B}}])$ to $\hat{\mathcal{B}}$ and $(\text{msk}, v_{\mathcal{B}}, v_{\mathcal{C}}, \text{coin}_{\mathcal{C}}, y_{\mathcal{C},1}^\perp, q[R_{\mathcal{C}}])$ to $\hat{\mathcal{C}}$.
3. $\hat{\mathcal{B}}$ and $\hat{\mathcal{C}}$ are respectively given $(A, r_{\mathcal{B}}, z_{2,\mathcal{B}} = x_{\mathcal{B}})$ and $(A, r_{\mathcal{C}}, z_{2,\mathcal{C}} = x_{\mathcal{C}})$ and behave as follows.
 - $\hat{\mathcal{B}}$ computes $u_{\mathcal{B}} \leftarrow h(x_{\mathcal{B}})$, invokes $\mathcal{B}$ with $q[R_{\mathcal{B}}]$ and $(x_{\mathcal{B}}, y_{\mathcal{B},0} \oplus F_{K^\perp}(u_{\mathcal{B}} \| v_{\mathcal{B}}))$, and obtains $\text{coin}'_{\mathcal{B}}$. $\hat{\mathcal{B}}$ outputs $\text{coin}'_{\mathcal{B}} \oplus \text{coin}_{\mathcal{B}}$.
 - $\hat{\mathcal{C}}$ computes $u_{\mathcal{C}} \leftarrow h(x_{\mathcal{C}})$, invokes $\mathcal{C}$ with $q[R_{\mathcal{C}}]$ and $(x_{\mathcal{C}}, F_K(u_{\mathcal{C}} \| v_{\mathcal{C}}) \oplus y_{\mathcal{C},0}^\perp)$, and obtains $\text{coin}'_{\mathcal{C}}$. $\hat{\mathcal{C}}$ outputs $\text{coin}'_{\mathcal{C}} \oplus \text{coin}_{\mathcal{C}}$.

$\mathcal{A}_{\text{moe}}$ perfectly simulates Hyb_8 for $\mathcal{A}_{\text{cp}}$. If $\mathcal{A}_{\text{cp}}$ wins the simulated experiment (i.e., Hyb_8 outputs 1), we have $\text{coin}'_{\mathcal{B}} \oplus \text{coin}'_{\mathcal{C}} = \text{coin}_{\mathcal{B}} \oplus \text{coin}_{\mathcal{C}} \oplus \langle r_{\mathcal{B}}, s \rangle \oplus \langle r_{\mathcal{C}}, t \rangle$, that is, $(\text{coin}'_{\mathcal{B}} \oplus \text{coin}_{\mathcal{B}}) \oplus (\text{coin}'_{\mathcal{C}} \oplus \text{coin}_{\mathcal{C}}) = \langle r_{\mathcal{B}}, s \rangle \oplus \langle r_{\mathcal{C}}, t \rangle$. Thus, by Theorem 5.2 and Lemma 8.3, we have

$$\Pr[\text{Hyb}_8 = 1] = \Pr\left[\text{Exp}_{i\mathcal{O}, \text{Aux}, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-ind-moe-aux}}(\lambda) = 1\right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

Combining the above, we have

$$\Pr\left[\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{indplus-anti-piracy}}(\lambda) = 1\right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

This completes the proof of Theorem 8.2. $\square$

9 Construction of UPO

We prove the following theorem.

Theorem 9.1. *Assuming the existence of polynomially secure $i\mathcal{O}$ and the LWE assumption, for any constant $c > 0$ and any polynomials ℓ_{inp} and ℓ_{out} , there exists a UPO for the circuit class $\text{Circ} = \{C : \{0, 1\}^{\ell_{\text{inp}}} \rightarrow \{0, 1\}^{\ell_{\text{out}}}\}$ that satisfies $\mathcal{D}\text{-UPO}^+$ anti-piracy security for any QPT-samplable distributions $\mathcal{D} = (\mathcal{D}_{\mathcal{B}}, \mathcal{D}_{\mathcal{C}})$, where $\mathcal{D}_{\mathcal{B}}$ and $\mathcal{D}_{\mathcal{C}}$ have conditional min-entropy at least λ^c .*

Moreover, if $\mathcal{D}_{\mathcal{B}}$ and $\mathcal{D}_{\mathcal{C}}$ are uniformly random, the existence of OWFs (in place of the LWE assumption) together with $i\mathcal{O}$ suffices.

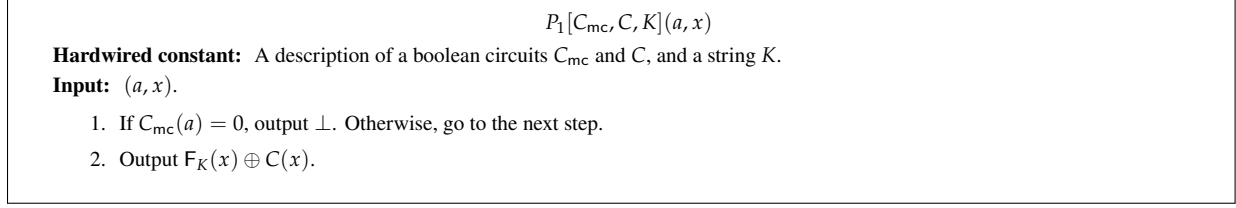


Figure 14: The description of the circuit P_1 .

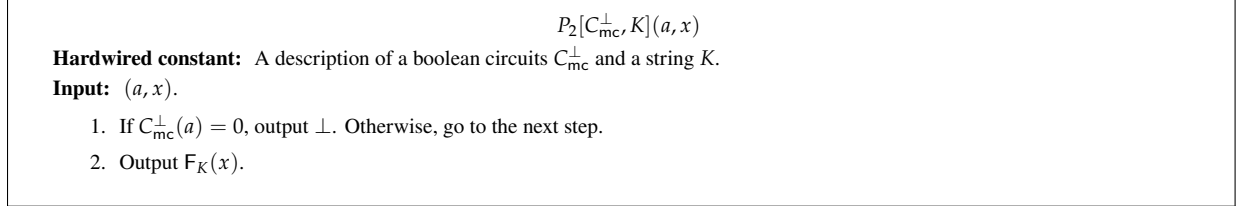


Figure 15: The description of the circuit P_2 .

9.1 Construction

Let n , ℓ_{inp} , and ℓ_{out} be polynomials in the security parameter λ .

We use the following building blocks.

- $i\mathcal{O}$ for polynomial size classical circuits $i\mathcal{O}$.
- PPRF $\text{PPRF} = (\text{F}, \text{Puncture})$ with the domain $\{0, 1\}^{\ell_{\text{inp}}}$ and the range $\{0, 1\}^{\ell_{\text{out}}}$.

We remark that additional cryptographic primitives are used in the security proof, as described in the relevant subsections.

We below construct a UPO scheme $\text{UPO} = \text{UPO}(\text{Obf}, \text{Eval})$ for the circuit class $\text{Circ} = \{C : \{0, 1\}^{\ell_{\text{inp}}} \rightarrow \{0, 1\}^{\ell_{\text{out}}}\}$.

$\text{UPO.Obf}(1^\lambda, C)$:

- Generate a uniformly random subspace $A \subseteq \mathbb{F}_2^n$ of dimension $\frac{n}{2}$, and two uniformly random elements $(s, t) \in \text{CS}(A) \times \text{CS}(A^\perp)$, where n is specified later.
- Define circuits $C_{mc} := i\mathcal{O}(A + s)$ and $C_{mc}^\perp := i\mathcal{O}(A^\perp + t)$.
- Generate a key $K \leftarrow \{0, 1\}^\lambda$ for PPRF.
- Generate two obfuscated circuits $\tilde{P}_1 \leftarrow i\mathcal{O}(P_1[C_{mc}, C, K])$ and $\tilde{P}_2 \leftarrow i\mathcal{O}(P_2[C_{mc}^\perp, K])$, where P_1 and P_2 are the circuits described in Figure 14 and Figure 15.
- Output $\rho_C := (|A_{s,t}\rangle, \tilde{P}_1, \tilde{P}_2)$.

$\text{UPO.Eval}(\rho_C, x)$:

- Parse $\rho_C = (\rho, \tilde{P}_1, \tilde{P}_2)$, where ρ is a quantum state, and $\tilde{P}_1$ and $\tilde{P}_2$ are interpreted as descriptions of circuits.
- Let E_1 and E_2 be unitaries that respectively work as follows:

$$|z_1\rangle |z_2\rangle \mapsto |z_1\rangle \left| z_2 \oplus \tilde{P}_1(z_1, x) \right\rangle \quad \text{and} \quad |z_1\rangle |z_2\rangle \mapsto |z_1\rangle \left| z_2 \oplus \tilde{P}_2(z_1, x) \right\rangle$$

for any $z_1 \in \{0, 1\}^n$ and $z_2 \in \{0, 1\}^{\ell_{\text{out}}}$.

- Apply E_1 to $\rho \otimes |0^{\ell_{\text{out}}}\rangle$ and measure the second register to obtain an outcome y_1 . Then, trace out the second register and obtain the state ρ' .
- Apply E_2 to $H^{\otimes n} \rho' \otimes |0^{\ell_{\text{out}}}\rangle$ and measure the second register to obtain an outcome y_2 .
- Output $Y := y_1 \oplus y_2$.

Perfect correctness. We prove

$$\Pr [\text{UPO.Eval}(\rho_C, x) = C(x) \mid \rho_C \leftarrow \text{UPO.Obf}(1^\lambda, C)] = 1$$

for any $C \in \text{Circ}$ and $x \in \{0, 1\}^{\ell_{\text{inp}}}$. Fix $C \in \text{Circ}$ and $x \in \{0, 1\}^{\ell_{\text{inp}}}$. Suppose we generate $\rho_C := (|A_{s,t}\rangle, \tilde{P}, \tilde{P}_2)$ as prescribed in the description of UPO.Obf . Letting y_1 and y_2 be the values respectively computed in the third and fourth item of $\text{UPO.Eval}(\rho_C, x)$, we prove the followings.

- We prove that $y_1 = F_K(x) \oplus C(x)$ holds with probability 1. For any $a \in A + s$, we have $\tilde{P}_1(a, x) = F_K(x) \oplus C(x)$ from the correctness of $i\mathcal{O}$. Thus, given that $|A_{s,t}\rangle = \frac{1}{\sqrt{|A|}} \sum_{a \in A} (-1)^{\langle a, t \rangle} |a + s\rangle$, if we apply the unitary E_1 to $|A_{s,t}\rangle \otimes |0^{\ell_{\text{out}}}\rangle$, we obtain the state $|A_{s,t}\rangle \otimes |F_K(x) \oplus C(x)\rangle$. This means $y = F_K(x) \oplus C(x)$ holds with probability 1, and the state obtained by tracing out the second register after measuring y is $|A_{s,t}\rangle$.
- We then prove $y_2 = F_K(x)$. For any $a \in A^\perp + t$, we have $\tilde{P}_2(a, x) = F_K(x)$ from the correctness of $i\mathcal{O}$. Thus, given that $H^{\otimes n} |A_{s,t}\rangle = \frac{1}{\sqrt{|A^\perp|}} \sum_{a \in A^\perp} (-1)^{\langle a, s \rangle} |a + t\rangle$, if we apply the unitary E_2 to $H^{\otimes n} |A_{s,t}\rangle \otimes |0^{\ell_{\text{out}}}\rangle$, we obtain the state $H^{\otimes n} |A_{s,t}\rangle \otimes |F_K(x)\rangle$. This means $y_2 = F_K(x)$ holds.

The above shows the perfect correctness of UPO.

9.2 Proof of Security

In the security proof, we additionally use the following primitives, where ℓ_{loss} is a polynomial in λ and $\ell_A := n^2/2$ is the number of bits needed to represent a subspace $A \subseteq \mathbb{F}_2^n$ of dimension $n/2$.

- Universal hash function family $\mathcal{H} : \{0, 1\}^{\ell_{\text{inp}}} \rightarrow \{0, 1\}^{\ell_A + n}$.
- Collection of $(\ell_{\text{inp}}, \ell_{\text{loss}})$ -lossy functions $\text{LF} = (\text{Gen}_{\text{inj}}, \text{Gen}_{\text{loss}})$.

We prove the following theorem.

Theorem 9.2. *Let $\mathcal{D}$ be a distribution over $\{0, 1\}^{\ell_{\text{inp}}} \times \{0, 1\}^{\ell_{\text{aux}}}$ with conditional min-entropy at least $\ell_A + n + (\ell_{\text{inp}} - \ell_{\text{loss}}) + \omega(\log \lambda)$. Then, assuming the existence of $(\ell_{\text{inp}}, \ell_{\text{loss}})$ -lossy functions LF, UPO satisfies $\mathcal{D}$ -UPO⁺ security.*

We may choose ℓ_A and n to be arbitrarily small polynomials. Assuming the LWE assumption, by Theorem 3.9, we may also make $(\ell_{\text{inp}} - \ell_{\text{loss}})$ an arbitrarily small polynomial while allowing ℓ_{inp} and ℓ_{out} to be any polynomials. Therefore, the former part of Theorem 9.1 follows.

Proof of Theorem 9.2. Let $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ be a QPT adversary against $\mathcal{D}$ -UPO⁺ security of UPO. We consider the following sequence of hybrids.

Hyb₀: This is the original security experiment $\text{Expt}_{\text{UPO}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{upoplus}}(\lambda)$ in Definition 6.9. More specifically, it works as follows.

1. On input 1^λ , $\mathcal{A}$ sends $C \in \text{Circ}$ together with two circuits $\mu_{\mathcal{B}} : \{0, 1\}^{\ell_{\text{inp}}} \rightarrow \{0, 1\}^{\ell_{\text{out}}}$ and $\mu_{\mathcal{C}} : \{0, 1\}^{\ell_{\text{inp}}} \rightarrow \{0, 1\}^{\ell_{\text{out}}}$.
2. The challenger does the following.
 - Choose $\text{coin}_{\mathcal{B}} \leftarrow \{0, 1\}$, generate $(x_{\mathcal{B}}, \text{aux}_{\mathcal{B}}) \leftarrow \mathcal{D}_{\mathcal{B}}(1^\lambda)$, $Y_{\mathcal{B},0} \leftarrow C(x_{\mathcal{B}})$, and $Y_{\mathcal{B},1} \leftarrow \mu_{\mathcal{B}}(x_{\mathcal{B}})$.
 - Choose $\text{coin}_{\mathcal{C}} \leftarrow \{0, 1\}$, generate $(x_{\mathcal{C}}, \text{aux}_{\mathcal{C}}) \leftarrow \mathcal{D}_{\mathcal{C}}(1^\lambda)$, $Y_{\mathcal{C},0} \leftarrow C(x_{\mathcal{C}})$, and $Y_{\mathcal{C},1} \leftarrow \mu_{\mathcal{C}}(x_{\mathcal{C}})$.

The challenger generates $\rho_C := (|A_{s,t}\rangle, \tilde{P}_1, \tilde{P}_2) \leftarrow \text{UPO.Obf}(1^\lambda, C^*[x_{\mathcal{B}}, x_{\mathcal{C}}, Y_{\mathcal{B}, \text{coin}_{\mathcal{B}}}, Y_{\mathcal{C}, \text{coin}_{\mathcal{C}}}])$ and sends $(\rho_C, \text{aux}_{\mathcal{B}}, \text{aux}_{\mathcal{C}})$ to $\mathcal{A}$.

3. $\mathcal{A}$ creates a bipartite state q over registers $R_{\mathcal{B}}$ and $R_{\mathcal{C}}$. Then, $\mathcal{A}$ sends register $R_{\mathcal{B}}$ to $\mathcal{B}$ and register $R_{\mathcal{C}}$ to $\mathcal{C}$.

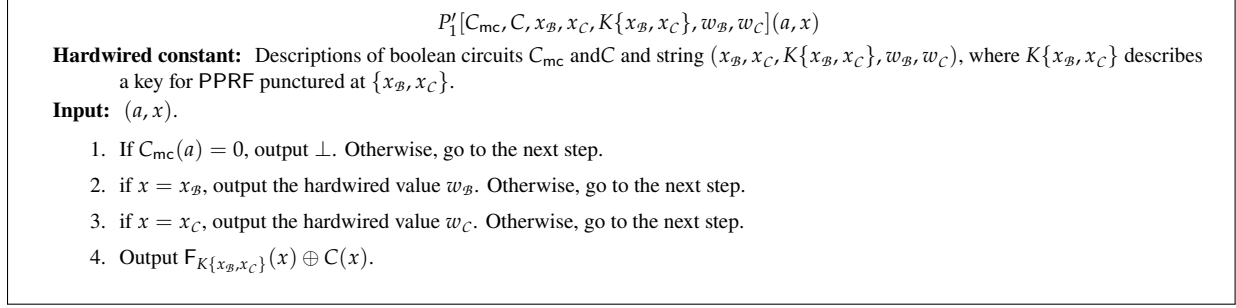


Figure 16: The description of the circuit P'_1 .

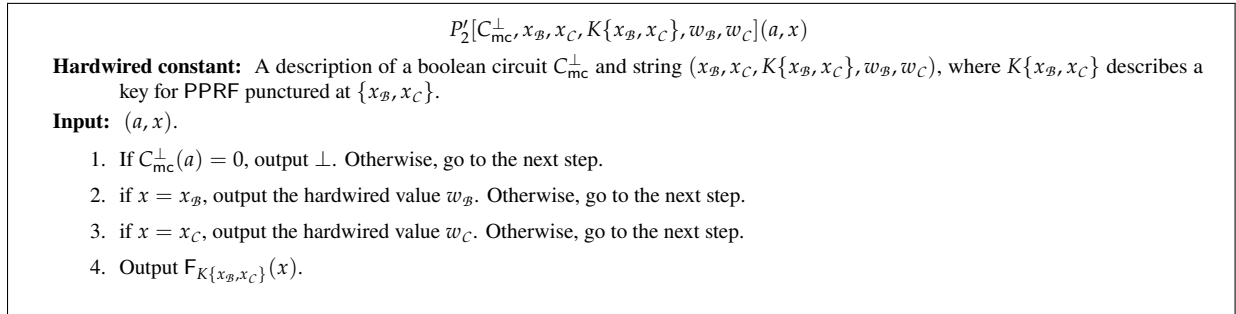


Figure 17: The description of the circuit P'_2 .

4. The challenger sends x_B and x_C to $\mathcal{B}$ and $\mathcal{C}$, respectively.
5. $\mathcal{B}$ and $\mathcal{C}$ respectively output coin'_B and coin'_C . The challenger outputs 1 if $\text{coin}'_B \oplus \text{coin}'_C = \text{coin}_B \oplus \text{coin}_C$ otherwise outputs 0.

By the definition, we clearly have

$$\Pr[\text{Hyb}_0 = 1] = \Pr[\text{Expt}_{\text{UPO}, \mathcal{D}, \mathcal{A}_{cp}}^{\text{upoplus}}(\lambda) = 1].$$

We below use the circuits P'_1 and P'_2 respectively described in Figure 16 and Figure 17.

Hyb₁: This is identical to **Hyb₀** except for the following changes, where $K\{x_B, x_C\} \leftarrow \text{Puncture}(K, \{x_B, x_C\})$, $y_B \leftarrow F_K(x_B)$, and $y_C \leftarrow F_K(x_C)$.

- $\tilde{P}_1$ is generated as $\tilde{P}_1 \leftarrow i\mathcal{O}(P'_1[C_{mc}, C, x_B, x_C, K\{x_B, x_C\}, y_B \oplus Y_{B, \text{coin}_B}, y_C \oplus Y_{C, \text{coin}_C}])$.
- $\tilde{P}_2$ is generated as $\tilde{P}_2 \leftarrow i\mathcal{O}(P'_2[C_{mc}^\perp, x_B, x_C, K\{x_B, x_C\}, y_B, y_C])$.

From the security of $i\mathcal{O}$, we have

$$|\Pr[\text{Hyb}_0 = 1] - \Pr[\text{Hyb}_1 = 1]| = \text{negl}(\lambda).$$

Hyb₂: This is identical to **Hyb₁** except that y_B and y_C are generated as uniformly random strings instead of $F_K(x_B)$ and $F_K(x_C)$, respectively. From the security of PPRF, we have

$$|\Pr[\text{Hyb}_1 = 1] - \Pr[\text{Hyb}_2 = 1]| = \text{negl}(\lambda).$$

Hereafter, we assume that $x_B \neq x_C$, which holds with overwhelming probability.

Hyb₃: This is identical to Hyb₂ except that we replace y_C with $y_C \oplus Y_{C,\text{coin}_C}$. More specifically, the following changes are applied.

- $\tilde{P}_1$ is generated as $\tilde{P}_1 \leftarrow i\mathcal{O}(P'_1[C_{\text{mc}}, C, x_B, x_C, K\{x_B, x_C\}, y_B \oplus Y_{B,\text{coin}_B}, y_C])$.
- $\tilde{P}_2$ is generated as $\tilde{P}_2 \leftarrow i\mathcal{O}(P'_2[C_{\text{mc}}^\perp, x_B, x_C, K\{x_B, x_C\}, y_B, y_C \oplus Y_{C,\text{coin}_C}])$.

Since y_C is uniformly at random, so is $y_C \oplus Y_{C,\text{coin}_C}$. Thus, we have

$$\Pr[\text{Hyb}_2 = 1] = \Pr[\text{Hyb}_3 = 1].$$

Hyb₄: This is identical to Hyb₃ except that coin_B and coin_C are replaced with $\text{coin}_B \oplus \langle r_B, s \rangle$ and $\text{coin}_C \oplus \langle r_C, t \rangle$, respectively, where $r_B, r_C \leftarrow \{0, 1\}^n$. More specifically, the following three changes are applied.

- $\tilde{P}_1$ is generated as $\tilde{P}_1 \leftarrow i\mathcal{O}(P'_1[C_{\text{mc}}, C, x_B, x_C, K\{x_B, x_C\}, y_B \oplus Y_{B,\text{coin}_B \oplus \langle r_B, s \rangle}, y_C])$.
- $\tilde{P}_2$ is generated as $\tilde{P}_2 \leftarrow i\mathcal{O}(P'_2[C_{\text{mc}}^\perp, x_B, x_C, K\{x_B, x_C\}, y_B, y_C \oplus Y_{C,\text{coin}_C \oplus \langle r_C, t \rangle}])$.
- The challenger outputs 1 if $\text{coin}'_B \oplus \text{coin}'_C = \text{coin}_B \oplus \text{coin}_C \oplus \langle r_B, s \rangle \oplus \langle r_C, t \rangle$ and otherwise outputs 0.

Since $(\text{coin}_B, \text{coin}_C)$ is distributed uniformly at random, so is $(\text{coin}_B \oplus \langle r_B, s \rangle, \text{coin}_C \oplus \langle r_C, t \rangle)$. Thus, we have

$$\Pr[\text{Hyb}_3 = 1] = \Pr[\text{Hyb}_4 = 1].$$

We below use the circuits Q_1 and Q_2 respectively described in Figure 18 and Figure 19.

Hyb₅: This is identical to Hyb₄ except for the following changes, where $h \leftarrow \mathcal{H}$, $F_{\text{f}} \leftarrow \text{Gen}_{\text{inj}}(1^\lambda)$, $\eta_B = F_{\text{f}}(x_B)$, and $\eta_C = F_{\text{f}}(x_C)$.

- $\tilde{P}_1$ is generated as $\tilde{P}_1 \leftarrow i\mathcal{O}(Q_1[C_{\text{mc}}, C, \mu_B, h, F_{\text{f}}, \text{str}_B, \eta_B, \eta_C, K, y_B, y_C, \text{coin}_B])$, where $\text{str}_B = (A \| r_B) \oplus h(x_B)$.
- $\tilde{P}_2$ is generated as $\tilde{P}_2 \leftarrow i\mathcal{O}(Q_2[C_{\text{mc}}^\perp, C, \mu_C, h, F_{\text{f}}, \text{str}_C, \eta_B, \eta_C, K, y_B, y_C, \text{coin}_C])$, where $\text{str}_C = (A^\perp \| r_C) \oplus h(x_C)$.

We prove that $P'_1[C_{\text{mc}}, C, x_B, x_C, K\{x_B, x_C\}, y_B \oplus Y_{B,\text{coin}_B \oplus \langle r_B, s \rangle}, y_C]$ and $Q_1[C_{\text{mc}}, C, \mu_B, h, F_{\text{f}}, \text{str}_B, \eta_B, \eta_C, K, y_B, y_C, \text{coin}_B]$ are functionally equivalent. For an input (a, x) , we consider the following cases.

- The case where $C_{\text{mc}}(a) = 0$ (i.e., $a \notin A + s$). In this case, both circuits output $\perp$ according to the first items of Figures 16 and 18, respectively.
- The case where $C_{\text{mc}}(a) = 1$ (i.e., $a \in A + s$), and $x = x_B$. In this case, $P'_1[C_{\text{mc}}, C, x_B, x_C, K\{x_B, x_C\}, y_B \oplus Y_{B,\text{coin}_B \oplus \langle r_B, s \rangle}, y_C]$ clearly outputs the hardwired value $y_B \oplus Y_{B,\text{coin}_B \oplus \langle r_B, s \rangle}$ according to the second item of Figure 16. We show that $Q_1[C_{\text{mc}}, C, \mu_B, h, F_{\text{f}}, \text{str}_B, \eta_B, \eta_C, K, y_B, y_C, \text{coin}_B]$ also outputs $y_B \oplus Y_{B,\text{coin}_B \oplus \langle r_B, s \rangle}$ in this case. $x = x_B$ implies $F_{\text{f}}(x) = \eta_B$, and thus the computation goes to the second item of Figure 18. Since $x = x_B$, we have $\text{str}_B \oplus h(x) = A \| r_B$. Since $a \in A + s$, we also have $\text{Can}_A(a) = s$. Thus, we can conclude that it outputs $y_B \oplus Y_{B,\text{coin}_B \oplus \langle r_B, s \rangle}$, where we recall that $Y_{B,0} = C(x_B)$ and $Y_{B,1} = \mu_B(x_B)$.
- The case where $C_{\text{mc}}(a) = 1$ (i.e., $a \in A + s$), and $x = x_C$. Since F_{f} is injective and we assume $x_B \neq x_C$, the latter condition implies $F_{\text{f}}(x) \neq \eta_B$ and $F_{\text{f}}(x) = \eta_C$. Then, we can see that both circuits output y_C in this case according to the third items of Figures 16 and 18, respectively.
- The case where $C_{\text{mc}}(x) = 1$ (i.e., $a \in A + s$), and $x \notin \{x_B, x_C\}$. Since F_{f} is injective, the latter condition implies $F_{\text{f}}(x) \notin \{\eta_B, \eta_C\}$. Then, we can see that both circuits output $F_K(x) \oplus C(x)$ according to the fourth items of Figures 16 and 18, respectively.

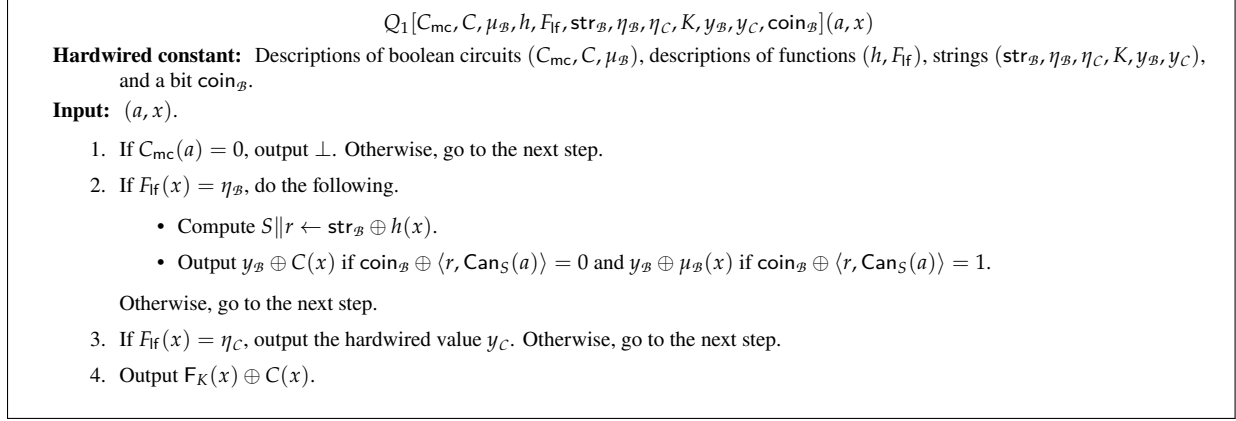


Figure 18: The description of the circuit Q_1 .

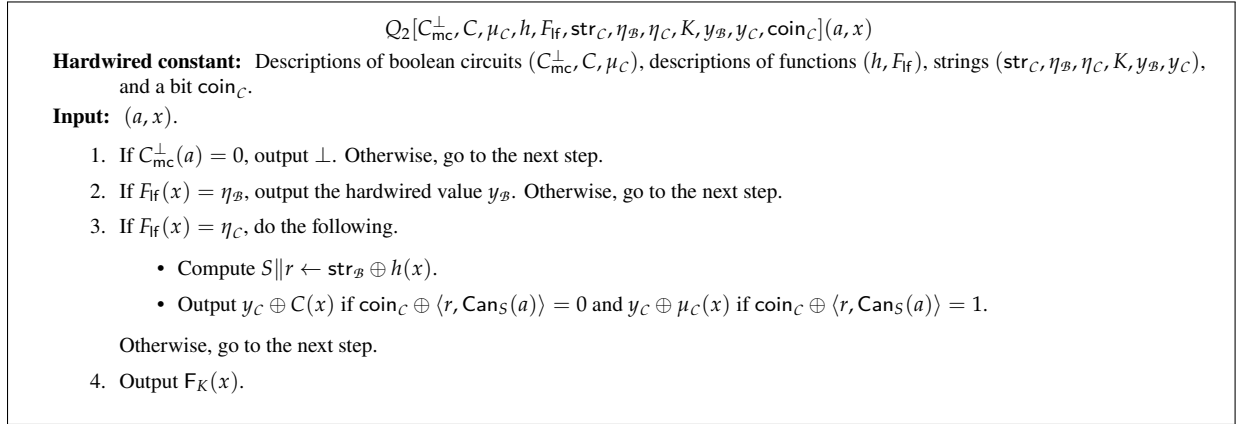


Figure 19: The description of the circuit Q_2 .

This shows that the two circuits are functionally equivalent.

We can similarly prove that $P'_2[C_{mc}^\perp, x_B, x_C, K\{x_B, x_C\}, y_B, y_C \oplus Y_{C, \text{coin}_C \oplus \langle r_C, t \rangle}]$ and $Q_2[C_{mc}^\perp, C, \mu_C, h, F_{ff}, \text{str}_C, \eta_B, \eta_C, K, y_B, y_C, \text{coin}_C]$ are functionally equivalent. Then, from the above discussion and the security of $i\mathcal{O}$, we have

$$|\Pr[\text{Hyb}_4 = 1] - \Pr[\text{Hyb}_5 = 1]| = \text{negl}(\lambda).$$

Hyb₆: This is identical to Hyb₅ except F_{ff} is generated in lossy mode, that is, $F_{ff} \leftarrow \text{Gen}_{\text{loss}}(1^\lambda)$. From the mode indistinguishability of LF, we have

$$|\Pr[\text{Hyb}_5 = 1] - \Pr[\text{Hyb}_6 = 1]| = \text{negl}(\lambda).$$

We below show that

$$\Pr[\text{Hyb}_6 = 1] \leq \frac{1}{2} + \text{negl}(\lambda)$$

by a reduction to the computational strong indistinguishability monogamy property of coset states with simulatable auxiliary input (Theorem 5.2) with respect to the following auxiliary input generator $\mathcal{Aux} = (\mathcal{AuxSetup}, \mathcal{AuxGen}_B, \mathcal{AuxGen}_C)$ and the corresponding simulator $\text{Sim} = (\text{Sim}_B, \text{Sim}_C)$.

$\mathcal{AuxSetup}(1^\lambda)$:

- Generate $h \leftarrow \mathcal{H}$ and $F_{\text{ff}} \leftarrow \text{Gen}_{\text{loss}}(1^\lambda)$.
- Output $\text{pp} = (h, F_{\text{ff}})$.

$\mathcal{AuxGen}_X(\text{pp}, A, r_X)$ where $X \in \{\mathcal{B}, \mathcal{C}\}$:

- Parse $\text{pp} = (h, F_{\text{ff}})$.
- Sample $(x_X, \text{aux}_X) \leftarrow \mathcal{D}_X(1^\lambda)$ and compute $\eta_X := F_{\text{ff}}(x_X)$.
- Set

$$\text{str}_X := \begin{cases} (A \| r_{\mathcal{B}}) \oplus h(x_{\mathcal{B}}) & \text{if } X = \mathcal{B} \\ (A^\perp \| r_{\mathcal{C}}) \oplus h(x_{\mathcal{C}}) & \text{if } X = \mathcal{C} \end{cases}$$

- Output $z_{1,X} := (\text{aux}_X, \eta_X, \text{str}_X)$ and $z_{2,X} := x_X$.

$\text{Sim}_X(1^\lambda)$ where $X \in \{\mathcal{B}, \mathcal{C}\}$:

- Generate aux_X and η_X in the same way as in $\mathcal{AuxGen}_X$.
- Sample $\text{str}_X \leftarrow \{0, 1\}^{\ell_A + n}$.
- Output $z_{1,X} := (\text{aux}_X, \eta_X, \text{str}_X)$.

First, we show that they satisfy the simulatability condition of Theorem 5.2.

Lemma 9.3. For $X \in \{\mathcal{B}, \mathcal{C}\}$, any subspace $A \subseteq \mathbb{F}_2^n$, and any $r_X \in \{0, 1\}^n$,

$$\left\{ (\text{pp}, z_{1,X}) : \begin{array}{l} \text{pp} \leftarrow \mathcal{AuxSetup}(1^\lambda) \\ (z_{1,X}, z_{2,X}) \leftarrow \mathcal{AuxGen}_X(\text{pp}, A, r_X) \end{array} \right\} \approx \left\{ (\text{pp}, z_{1,X}) : \begin{array}{l} \text{pp} \leftarrow \mathcal{AuxSetup}(1^\lambda) \\ z_{1,X} \leftarrow \text{Sim}_X(\text{pp}) \end{array} \right\}.$$

Proof of Lemma 8.3. In the execution of $\mathcal{AuxGen}_X(\text{pp}, A, r_X)$, note that aux_X and $\eta_X := F_{\text{ff}}(x_X)$ are the only variables that depend on x_X . By the assumption, x_X has conditional min-entropy at least $\ell_A + n + (\ell_{\text{inp}} - \ell_{\text{loss}}) + \omega(\log \lambda)$ given aux_X . Since the image size of F_{ff} is at most $2^{\ell_{\text{inp}} - \ell_{\text{loss}}}$, by the chain rule, x_X has conditional min-entropy at least $\ell_A + n + \omega(\log \lambda)$ given both aux_X and η_X . Thus, by the leftover hash lemma (Lemma 3.3), $(h, F_{\text{ff}}, \text{aux}_X, \eta_X, h(x_X)) \approx (h, F_{\text{ff}}, \text{aux}_X, \eta_X, u'_X)$ where $u'_X \leftarrow \{0, 1\}^{\ell_k}$. This immediately implies Lemma 9.3. $\square$

We now construct the following adversary $\mathcal{A}_{\text{moe}} = (\hat{\mathcal{A}}, \hat{\mathcal{B}}, \hat{\mathcal{C}})$ for the experiment in Theorem 5.2 using $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$.

1. $\hat{\mathcal{A}}$ is given $|A_{s,t}\rangle$, $C_{\text{mc}} = i\mathcal{O}(A + s)$, $C_{\text{mc}}^\perp = i\mathcal{O}(A^\perp + t)$, $\text{pp} = (h, F_{\text{ff}})$, $z_{1,\mathcal{B}} = (\text{aux}_{\mathcal{B}}, \eta_{\mathcal{B}}, \text{str}_{\mathcal{B}})$, and $z_{1,\mathcal{C}} = (\text{aux}_{\mathcal{C}}, \eta_{\mathcal{C}}, \text{str}_{\mathcal{C}})$. $\hat{\mathcal{A}}$ invokes $\mathcal{A}$ with 1^λ and obtains the circuits $(C, \mu_{\mathcal{B}}, \mu_{\mathcal{C}})$. $\hat{\mathcal{A}}$ generates $\rho_{\mathcal{C}} := (|A_{s,t}\rangle, \tilde{P}, \tilde{P}^\perp)$ as in Hyb_6 . $\hat{\mathcal{A}}$ can generate it with the given inputs, circuits $(C, \mu_{\mathcal{B}}, \mu_{\mathcal{C}})$ declared by $\mathcal{A}$, and $(K, \text{coin}_{\mathcal{B}}, y_{\mathcal{B}}, \text{coin}_{\mathcal{C}}, y_{\mathcal{C}})$ sampled by $\hat{\mathcal{A}}$ itself. $\hat{\mathcal{A}}$ then sends $\rho_{\mathcal{C}}$, $\text{aux}_{\mathcal{B}}$, and $\text{aux}_{\mathcal{C}}$ to $\mathcal{A}$.
2. When $\mathcal{A}$ outputs a bipartite state q over registers $R_{\mathcal{B}}$ and $R_{\mathcal{C}}$, $\hat{\mathcal{A}}$ sends $(\text{coin}_{\mathcal{B}}, q[R_{\mathcal{B}}])$ to $\mathcal{B}$ and $(\text{coin}_{\mathcal{C}}, q[R_{\mathcal{C}}])$ to $\mathcal{C}$.
3. $\hat{\mathcal{B}}$ and $\hat{\mathcal{C}}$ are respectively given $(A, r_{\mathcal{B}}, z_{2,\mathcal{B}} = x_{\mathcal{B}})$ and $(A, r_{\mathcal{C}}, z_{2,\mathcal{C}} = x_{\mathcal{C}})$ and behave as follows.
 - $\hat{\mathcal{B}}$ invokes $\mathcal{B}$ with $q[R_{\mathcal{B}}]$ and $x_{\mathcal{B}}$, and obtains $\text{coin}'_{\mathcal{B}}$. $\hat{\mathcal{B}}$ outputs $\text{coin}'_{\mathcal{B}} \oplus \text{coin}_{\mathcal{B}}$.
 - $\hat{\mathcal{C}}$ invokes $\mathcal{C}$ with $q[R_{\mathcal{C}}]$ and $x_{\mathcal{C}}$, and obtains $\text{coin}'_{\mathcal{C}}$. $\hat{\mathcal{C}}$ outputs $\text{coin}'_{\mathcal{C}} \oplus \text{coin}_{\mathcal{C}}$.

$\mathcal{A}_{\text{moe}}$ perfectly simulates Hyb_6 for $\mathcal{A}_{\text{cp}}$. If $\mathcal{A}_{\text{cp}}$ wins the simulated experiment (i.e., Hyb_6 outputs 1), we have $\text{coin}'_{\mathcal{B}} \oplus \text{coin}'_{\mathcal{C}} = \text{coin}_{\mathcal{B}} \oplus \text{coin}_{\mathcal{C}} \oplus \langle r_{\mathcal{B}}, s \rangle \oplus \langle r_{\mathcal{C}}, t \rangle$, that is, $(\text{coin}'_{\mathcal{B}} \oplus \text{coin}_{\mathcal{B}}) \oplus (\text{coin}'_{\mathcal{C}} \oplus \text{coin}_{\mathcal{C}}) = \langle r_{\mathcal{B}}, s \rangle \oplus \langle r_{\mathcal{C}}, t \rangle$. Thus, by Theorem 5.2 and Lemma 9.3, we have

$$\Pr[\text{Hyb}_6 = 1] = \Pr\left[\text{Exp}_{i\mathcal{O}, \mathcal{Aux}, \mathcal{A}_{\text{moe}}}^{\text{comp-strong-ind-moe-aux}}(\lambda) = 1\right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

Combining the above, we have

$$\Pr\left[\text{Exp}_{\text{UPO}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{upoplus}}(\lambda) = 1\right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

This completes the proof of Theorem 9.2. $\square$

9.3 Alternative Proof of Security for Uniform Inputs

Here, we give an alternative security proof for the special case of the uniform input distribution. An advantage of this alternative proof is that it does not rely on the additional assumption of lossy functions. The security proof is very similar to that for copy-protected PRF in Section 8.

In the security proof, instead of universal hash functions and lossy functions, we rely on the following primitive, where ℓ_k is a polynomial in λ and $\ell_A := n^2/2$ is the number of bits needed to represent a subspace $A \subseteq \mathbb{F}_2^n$ of dimension $n/2$.

- A key-robust non-committing encryption scheme $\text{NCE} = (\text{NCE.Enc}, \text{NCE.Dec}, \text{NCE.Fake}, \text{NCE.Open})$ with the message space $\{0, 1\}^{\ell_A + n}$ and the key space $\{0, 1\}^{\ell_k}$.

We prove the following theorem.

Theorem 9.4. *Let $\mathcal{U}$ be the uniformly random distribution over $\{0, 1\}^{\ell_{\text{inp}}}$ where $\ell_{\text{inp}} \geq \ell_k$. Then, UPO satisfies $\mathcal{U}$ -UPO⁺ security.*

For any polynomial ℓ_{inp} , we may choose sufficiently small polynomial ℓ_k such that $\ell_{\text{inp}} \geq \ell_k$. Also, we may choose ℓ_{out} to be any polynomial. Therefore, the latter part of Theorem 9.1 follows.

Proof of Theorem 9.4. For simplicity, we give the proof assuming $\ell_{\text{inp}} = \ell_k$ while the extension to the general case is straightforward.

Let $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ be a QPT adversary against $\mathcal{D}$ -UPO⁺ security of UPO. We consider the following sequence of hybrids.

Hyb₀-Hyb₄: They are identical to those in the security proof in Section 9.2 except that $x_{\mathcal{B}}$ and $x_{\mathcal{C}}$ are chosen uniformly randomly instead of from the distributions $\mathcal{D}_{\mathcal{B}}$ and $\mathcal{D}_{\mathcal{C}}$, respectively.²⁶

We below use the circuits Q'_1 and Q'_2 respectively described in Figure 20 and Figure 21.

Hyb₅: This is identical to Hyb₄ except for the following changes, where $k_{\mathcal{B}} \leftarrow \text{NCE.Gen}(1^\lambda)$, $k_{\mathcal{C}} \leftarrow \text{NCE.Gen}(1^\lambda)$, $\text{ct}_{\mathcal{B}} \leftarrow \text{NCE.Enc}(k_{\mathcal{B}}, A \| r_{\mathcal{B}})$, and $\text{ct}_{\mathcal{C}} \leftarrow \text{NCE.Enc}(k_{\mathcal{C}}, A \| r_{\mathcal{C}})$.

- $\tilde{P}_1$ is generated as $\tilde{P}_1 \leftarrow i\mathcal{O}(Q'_1[C_{\text{mc}}, C, \mu_{\mathcal{B}}, \text{str}_{\mathcal{B}}, \text{str}_{\mathcal{C}}, \text{ct}_{\mathcal{B}}, \text{ct}_{\mathcal{C}}, K, y_{\mathcal{B}}, y_{\mathcal{C}}, \text{coin}_{\mathcal{B}}])$, where $\text{str}_{\mathcal{B}} = k_{\mathcal{B}} \oplus x_{\mathcal{B}}$.
- $\tilde{P}_2$ is generated as $\tilde{P}_2 \leftarrow i\mathcal{O}(Q'_2[C_{\text{mc}}^\perp, C, \mu_{\mathcal{C}}, \text{str}_{\mathcal{B}}, \text{str}_{\mathcal{C}}, \text{ct}_{\mathcal{B}}, \text{ct}_{\mathcal{C}}, K, y_{\mathcal{B}}, y_{\mathcal{C}}, \text{coin}_{\mathcal{C}}])$, where $\text{str}_{\mathcal{C}} = k_{\mathcal{C}} \oplus x_{\mathcal{C}}$.

Below, we assume that both $k_{\mathcal{B}}$ and $k_{\mathcal{C}}$ are good as per Definition 4.1, i.e., the perfect correctness and key-robustness hold under those keys, since this holds with overwhelming probability.

We prove that $P'_1[C_{\text{mc}}, C, x_{\mathcal{B}}, x_{\mathcal{C}}, K\{x_{\mathcal{B}}, x_{\mathcal{C}}\}, y_{\mathcal{B}} \oplus Y_{\mathcal{B}, \text{coin}_{\mathcal{B}} \oplus \langle r_{\mathcal{B}}, s \rangle}, y_{\mathcal{C}}]$ and $Q'_1[C_{\text{mc}}, C, \mu_{\mathcal{B}}, \text{str}_{\mathcal{B}}, \text{ct}_{\mathcal{B}}, \text{ct}_{\mathcal{C}}, K, y_{\mathcal{B}}, y_{\mathcal{C}}, \text{coin}_{\mathcal{B}}]$ are functionally equivalent. For an input (a, x) , we consider the following cases.

- The case where $C_{\text{mc}}(a) = 0$ (i.e., $a \notin A + s$). In this case, both circuits output $\perp$ according to the first items of Figures 16 and 20, respectively.
- The case where $C_{\text{mc}}(a) = 1$ (i.e., $a \in A + s$), and $x = x_{\mathcal{B}}$. In this case, $P'_1[C_{\text{mc}}, C, x_{\mathcal{B}}, x_{\mathcal{C}}, K\{x_{\mathcal{B}}, x_{\mathcal{C}}\}, y_{\mathcal{B}} \oplus Y_{\mathcal{B}, \text{coin}_{\mathcal{B}} \oplus \langle r_{\mathcal{B}}, s \rangle}, y_{\mathcal{C}}]$ clearly outputs the hardwired value $y_{\mathcal{B}} \oplus Y_{\mathcal{B}, \text{coin}_{\mathcal{B}} \oplus \langle r_{\mathcal{B}}, s \rangle}$ according to the second item of Figure 16. We show that $Q'_1[C_{\text{mc}}, C, \mu_{\mathcal{B}}, \text{str}_{\mathcal{B}}, \text{ct}_{\mathcal{B}}, \text{ct}_{\mathcal{C}}, K, y_{\mathcal{B}}, y_{\mathcal{C}}, \text{coin}_{\mathcal{B}}]$ also outputs $y_{\mathcal{B}} \oplus Y_{\mathcal{B}, \text{coin}_{\mathcal{B}} \oplus \langle r_{\mathcal{B}}, s \rangle}$ in this case. $x = x_{\mathcal{B}}$ implies $k'_{\mathcal{B}} = \text{str}_{\mathcal{B}} \oplus x = k_{\mathcal{B}}$, and thus $\text{NCE.Dec}(k'_{\mathcal{B}}, \text{ct}_{\mathcal{B}}) = A \| r_{\mathcal{B}} \neq \perp$ by the correctness of NCE. Therefore, the computation goes to the second item of Figure 20. Since $a \in A + s$, we also have $\text{Can}_A(a) = s$. Thus, we can conclude that it outputs $y_{\mathcal{B}} \oplus Y_{\mathcal{B}, \text{coin}_{\mathcal{B}} \oplus \langle r_{\mathcal{B}}, s \rangle}$, where we recall that $Y_{\mathcal{B}, 0} = C(x_{\mathcal{B}})$ and $Y_{\mathcal{B}, 1} = \mu_{\mathcal{B}}(x_{\mathcal{B}})$.
- The case where $C_{\text{mc}}(a) = 1$ (i.e., $a \in A + s$), and $x = x_{\mathcal{C}}$. Since we assume $x_{\mathcal{B}} \neq x_{\mathcal{C}}$, we have $k'_{\mathcal{B}} = \text{str}_{\mathcal{B}} \oplus x \neq k_{\mathcal{B}}$ and $k'_{\mathcal{C}} = \text{str}_{\mathcal{B}} \oplus x = k_{\mathcal{C}}$. Thus, by the key-robustness and correctness of NCE, $\text{NCE.Dec}(k'_{\mathcal{B}}, \text{ct}_{\mathcal{B}}) = \perp$ and $\text{NCE.Dec}(k'_{\mathcal{C}}, \text{ct}_{\mathcal{C}}) = A \| r_{\mathcal{C}} \neq \perp$. Then, we can see that both circuits output $y_{\mathcal{C}}$ in this case according to the third items of Figures 16 and 20, respectively.

²⁶Note that the universal hash functions and lossy functions are not used at or before Hyb₄.

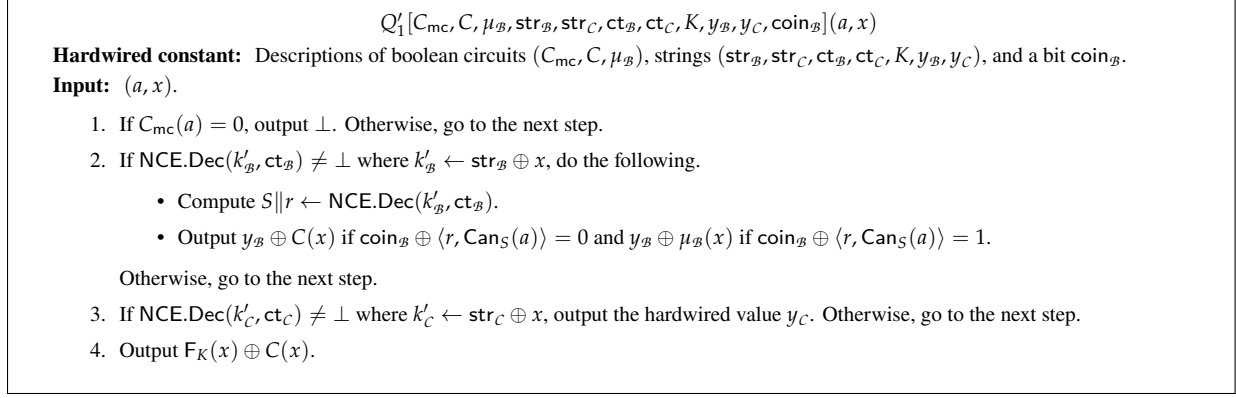


Figure 20: The description of the circuit Q'_1 .

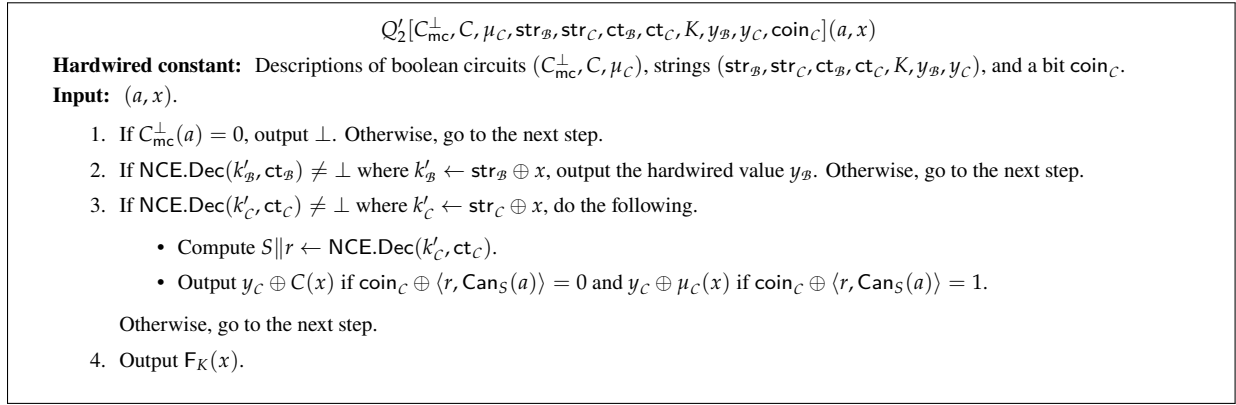


Figure 21: The description of the circuit Q'_2 .

- The case where $C_{mc}(x) = 1$ (i.e., $a \in A + s$), and $x \notin \{x_B, x_C\}$. Then we have $k'_B = \text{str}_B \oplus x \neq k_B$ and $k'_C = \text{str}_B \oplus x \neq k_C$. By the key-robustness of NCE, $\text{NCE.Dec}(k'_B, \text{ct}_B) = \perp$ and $\text{NCE.Dec}(k'_C, \text{ct}_C) = \perp$. Then, we can see that both circuits output $F_K(x) \oplus C(x)$ according to the fourth items of Figures 16 and 20, respectively.

This shows that the two circuits are functionally equivalent.

We can similarly prove that $P'_2[C_{mc}^\perp, x_B, x_C, K\{x_B, x_C\}, y_B, y_C \oplus Y_{C, \text{coin}_C \oplus \langle r_C, t \rangle}]$ and $Q'_2[C_{mc}^\perp, C, \mu_C, \text{str}_B, \text{str}_C, \text{ct}_B, \text{ct}_C, K, y_B, y_C, \text{coin}_C]$ are functionally equivalent. Then, from the above discussion and the security of $i\mathcal{O}$, we have

$$|\Pr[\text{Hyb}_4 = 1] - \Pr[\text{Hyb}_5 = 1]| = \text{negl}(\lambda).$$

Hyb₆: This is identical to Hyb₅ except for the following changes.

- (ct_B, k_B) are generated as $(\text{ct}_B, \text{st}_B) \leftarrow \text{NCE.Fake}(1^\lambda)$ and $k_B \leftarrow \text{NCE.Open}(\text{st}_B, A \| r_B)$.
- (ct_C, k_C) are generated as $(\text{ct}_C, \text{st}_C) \leftarrow \text{NCE.Fake}(1^\lambda)$ and $k_C \leftarrow \text{NCE.Open}(\text{st}_C, A^\perp \| r_C)$.

From the non-committing property of NCE, we have

$$|\Pr[\text{Hyb}_5 = 1] - \Pr[\text{Hyb}_6 = 1]| = \text{negl}(\lambda).$$

Hyb₇: This is identical to Hyb₆ except that str_B and str_C are chosen as uniformly random strings and then we set $x_B := \text{str}_B \oplus k_B$ and $x_C := \text{str}_C \oplus k_C$.

Since the challenge inputs x_B, x_C are uniformly random, the strings $\text{str}_B = k_B \oplus x_B, \text{str}_C = k_C \oplus x_C$ are also uniformly random. Thus, this hybrid is identical to the previous one from the view of the adversary, and thus,

$$\Pr[\text{Hyb}_6 = 1] = \Pr[\text{Hyb}_7 = 1]$$

We below show that

$$\Pr[\text{Hyb}_7 = 1] \leq \frac{1}{2} + \text{negl}(\lambda)$$

by a reduction to the computational strong indistinguishability monogamy property of coset states with no auxiliary information, i.e., the special case of Theorem 5.2 where $\mathcal{Aux} = (\mathcal{AuxSetup}, \mathcal{AuxGen})$ outputs nothing, and thus the simulatability condition is trivially satisfied.²⁷

We now construct the following adversary $\mathcal{A}_{\text{moe}} = (\hat{\mathcal{A}}, \hat{\mathcal{B}}, \hat{\mathcal{C}})$ for the experiment in Theorem 5.2 with no auxiliary input using $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$.

1. $\hat{\mathcal{A}}$ is given $|A_{s,t}\rangle$, and $C_{\text{mc}} = i\mathcal{O}(A + s)$, $C_{\text{mc}}^\perp = i\mathcal{O}(A^\perp + t)$. $\hat{\mathcal{A}}$ invokes $\mathcal{A}$ with 1^λ and obtains the circuits (C, μ_B, μ_C) . $\hat{\mathcal{A}}$ generates $\rho_C := (|A_{s,t}\rangle, \tilde{P}, \tilde{P}^\perp)$ as in Hyb₇. $\hat{\mathcal{A}}$ can generate it with the given inputs, circuits (C, μ_B, μ_C) declared by $\mathcal{A}$, and $(K, \text{coin}_B, y_B, \text{str}_B, \text{ct}_B, \text{st}_B, \text{coin}_C, y_C, \text{str}_C, \text{ct}_C, \text{st}_C)$ sampled by $\hat{\mathcal{A}}$ itself. $\hat{\mathcal{A}}$ then sends ρ_C , aux_B , and aux_C to $\mathcal{A}$.
2. When $\mathcal{A}$ outputs a bipartite state q over registers R_B and R_C , $\hat{\mathcal{A}}$ sends $(\text{coin}_B, q[R_B])$ to $\mathcal{B}$ and $(\text{coin}_C, q[R_C])$ to $\mathcal{C}$.
3. $\hat{\mathcal{B}}$ and $\hat{\mathcal{C}}$ are respectively given (A, r_B) and (A, r_C) and behave as follows.
 - $\hat{\mathcal{B}}$ computes $k_B \leftarrow \text{NCE.Open}(\text{st}_B, A \| r_B)$ and $x_B := \text{str}_B \oplus k_B$, invokes $\mathcal{B}$ with $q[R_B]$ and x_B , and obtains coin'_B . $\hat{\mathcal{B}}$ outputs $\text{coin}'_B \oplus \text{coin}_B$.
 - $\hat{\mathcal{C}}$ computes $k_C \leftarrow \text{NCE.Open}(\text{st}_C, A^\perp \| r_B)$ and $x_C := \text{str}_C \oplus k_C$, invokes $\mathcal{C}$ with $q[R_C]$ and x_C , and obtains coin'_C . $\hat{\mathcal{C}}$ outputs $\text{coin}'_C \oplus \text{coin}_C$.

$\mathcal{A}_{\text{moe}}$ perfectly simulates Hyb₇ for $\mathcal{A}_{\text{cp}}$. If $\mathcal{A}_{\text{cp}}$ wins the simulated experiment (i.e., Hyb₇ outputs 1), we have $\text{coin}'_B \oplus \text{coin}'_C = \text{coin}_B \oplus \text{coin}_C \oplus \langle r_B, s \rangle \oplus \langle r_C, t \rangle$, that is, $(\text{coin}'_B \oplus \text{coin}_B) \oplus (\text{coin}'_C \oplus \text{coin}_C) = \langle r_B, s \rangle \oplus \langle r_C, t \rangle$. Thus, by Theorem 5.2, we have

$$\Pr[\text{Hyb}_7 = 1] \leq \frac{1}{2} + \text{negl}(\lambda).$$

Combining the above, we have

$$\Pr[\text{Expt}_{\text{UPO}, \mathcal{U}, \mathcal{A}_{\text{cp}}}^{\text{upoplus}}(\lambda) = 1] \leq \frac{1}{2} + \text{negl}(\lambda).$$

This completes the proof of Theorem 9.4. □

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²⁷For the special case of Theorem 5.2 with no auxiliary information, the proof given in Section 5.1 is redundant. Simply applying the simultaneous quantum Goldreich-Levin lemma (Lemma 3.10) to the computational strong monogamy property of coset states [CLLZ21] suffices, as shown in [KY25].

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A Relationship among Security Definitions for Copy-Protected PRF

We present direct extensions of the two security notions for copy-protected PRFs considered in [CLLZ21], called anti-piracy security and indistinguishability anti-piracy security, to the non-uniform challenge setting. We show that $\mathcal{D}\text{-IND}^+$ anti-piracy security implies both.

Search anti-piracy security. This is an extension of anti-piracy security of [CLLZ21] to the non-uniform challenge setting.

Definition A.1 ($\mathcal{D}$ -Search Anti-Piracy Security). Let $\text{CP-PRF} = (\mathcal{KG}, \text{Eval}, Q\text{Eval})$ be a copy protected PRF scheme. Let $\mathcal{D} = (\mathcal{D}_B, \mathcal{D}_C)$ be a pair of distributions over $D_{\text{prf}} \times \{0, 1\}^{\ell_{\text{aux}}}$. We consider the $\mathcal{D}$ -search anti-piracy game $\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{search-anti-piracy}}(\lambda)$ between the challenger and an adversary $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ below.

1. The challenger generates $(\text{msk}, s_k) \leftarrow \mathcal{KG}(1^\lambda)$, $(x_B, \text{aux}_B) \leftarrow \mathcal{D}_B(1^\lambda)$, and $(x_C, \text{aux}_C) \leftarrow \mathcal{D}_C(1^\lambda)$. The challenger sends s_k , aux_B , and aux_C to $\mathcal{A}$.
2. $\mathcal{A}$ may make an arbitrary number of queries to the evaluation oracle, which, on input $x \in D_{\text{prf}}$, returns $\text{Eval}(\text{msk}, x)$.
3. $\mathcal{A}$ creates a bipartite state q over registers R_B and R_C . Then, $\mathcal{A}$ sends register R_B to $\mathcal{B}$ and register R_C to $\mathcal{C}$.
4. The challenger sends x_B and x_C to $\mathcal{B}$ and $\mathcal{C}$, respectively.
5. $\mathcal{B}$ and $\mathcal{C}$ may make an arbitrary number of queries to the evaluation oracle described in the second item, with the restriction that $\mathcal{B}$ and $\mathcal{C}$ are not allowed to query x_B and x_C , respectively.
6. $\mathcal{B}$ and $\mathcal{C}$ respectively output y'_B and y'_C . The challenger outputs 1 if $y'_B = \text{Eval}(\text{msk}, x_B)$ and $y'_C = \text{Eval}(\text{msk}, x_C)$. Otherwise the challenger outputs 0.

We say that CP-PRF is $\mathcal{D}$ -search anti-piracy secure if for any QPT adversary $\mathcal{A}_{\text{cp}}$, it holds that

$$\Pr \left[\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{search-anti-piracy}}(\lambda) = 1 \right] \leq \text{negl}(\lambda).$$

Indistinguishability anti-piracy security. This is an extension of indistinguishability anti-piracy security of [CLLZ21] to the non-uniform challenge setting.

Definition A.2 ($\mathcal{D}$ -IND Anti-Piracy Security). Let $\text{CP-PRF} = (\mathcal{KG}, \text{Eval}, Q\text{Eval})$ be a copy protected PRF scheme. Let $\mathcal{D} = (\mathcal{D}_B, \mathcal{D}_C)$ be a pair of distributions over $D_{\text{prf}} \times \{0, 1\}^{\ell_{\text{aux}}}$. We define the $\mathcal{D}$ -IND anti-piracy game $\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{ind-anti-piracy}}(\lambda)$ between the challenger and an adversary $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$, in the same way as $\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{indplus-anti-piracy}}(\lambda)$ in Definition 6.5 with the difference that the challenger outputs 1 if and only if $\text{coin}'_B = \text{coin}_B$ and $\text{coin}'_C = \text{coin}_C$.

We say that CP-PRF is $\mathcal{D}$ -IND anti-piracy secure if for any QPT adversary $\mathcal{A}_{\text{cp}}$, it holds that

$$\Pr \left[\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{ind-anti-piracy}}(\lambda) = 1 \right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

It is obvious from the definition that $\mathcal{D}\text{-IND}^+$ anti-piracy security implies $\mathcal{D}$ -IND anti-piracy security. Below, we show that $\mathcal{D}\text{-IND}^+$ anti-piracy security also implies $\mathcal{D}$ -search anti-piracy security.

A.1 $\mathcal{D}$ -IND⁺ Anti-Piracy Implies $\mathcal{D}$ -Search Anti-Piracy

Theorem A.3. Let $\text{CP-PRF} = (\text{KG}, \text{Eval}, \text{QEval})$ be a copy-protected PRF scheme with the domain D_{prf} and the range R_{prf} such that $|R_{\text{prf}}|^{-1} = \text{negl}(\lambda)$. Let $\mathcal{D}$ be a distribution over $D_{\text{prf}} \times \{0,1\}^{\ell_{\text{aux}}}$. If CP-PRF satisfies $\mathcal{D}$ -IND⁺ anti-piracy security, then it also satisfies $\mathcal{D}$ -search anti-piracy security.

Proof. Assume toward contradiction that there is an adversary $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ that breaks $\mathcal{D}$ -search anti-piracy security of CP-PRF with non-negligible probability ϵ . Then we construct a QPT adversary $\tilde{\mathcal{A}}_{\text{cp}} = (\tilde{\mathcal{A}}, \tilde{\mathcal{B}}, \tilde{\mathcal{C}})$ that breaks $\mathcal{D}$ -IND⁺ anti-piracy security of CP-PRF as follows.

$\tilde{\mathcal{A}}$: Given $s\mathcal{K}$, $\text{aux}_{\mathcal{B}}$, and $\text{aux}_{\mathcal{C}}$, $\tilde{\mathcal{A}}$ executes $\mathcal{A}$ with them to obtain a bipartite state q over registers $R_{\mathcal{B}}$ and $R_{\mathcal{C}}$. Throughout the reduction, $\tilde{\mathcal{A}}_{\text{cp}}$ simulates the evaluation oracle of $\mathcal{A}_{\text{cp}}$ by its evaluation oracle. $\tilde{\mathcal{A}}$ sends the register $R_{\mathcal{B}}$ to $\tilde{\mathcal{B}}$, and the register $R_{\mathcal{C}}$ to $\tilde{\mathcal{C}}$.

$\tilde{\mathcal{B}}$: Given register $R_{\mathcal{B}}$ and the challenge $(x_{\tilde{\mathcal{B}}}, y_{\tilde{\mathcal{B}}})$, $\tilde{\mathcal{B}}$ runs $\mathcal{B}$ on input $R_{\mathcal{B}}$ and $x_{\tilde{\mathcal{B}}}$ to obtain $y'_{\tilde{\mathcal{B}}}$. $\tilde{\mathcal{B}}$ outputs $\text{coin}_{\tilde{\mathcal{B}}} := 0$ if $y'_{\tilde{\mathcal{B}}} = y_{\tilde{\mathcal{B}}}$. Otherwise, $\tilde{\mathcal{B}}$ outputs $\text{coin}_{\tilde{\mathcal{B}}} \leftarrow \{0,1\}$.

$\tilde{\mathcal{C}}$: Given register $R_{\mathcal{C}}$ and the challenge $(x_{\tilde{\mathcal{C}}}, y_{\tilde{\mathcal{C}}})$, $\tilde{\mathcal{C}}$ runs $\mathcal{C}$ on input $R_{\mathcal{C}}$ and $x_{\tilde{\mathcal{C}}}$ to obtain $y'_{\tilde{\mathcal{C}}}$. $\tilde{\mathcal{C}}$ outputs $\text{coin}'_{\tilde{\mathcal{C}}} := 0$ if $y'_{\tilde{\mathcal{C}}} = y_{\tilde{\mathcal{C}}}$. Otherwise, $\tilde{\mathcal{C}}$ outputs $\text{coin}'_{\tilde{\mathcal{C}}} \leftarrow \{0,1\}$.

In an execution of $\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \tilde{\mathcal{A}}_{\text{cp}}}^{\text{prplus-anti-piracy}}(\lambda)$, let $\text{coin}_{\tilde{\mathcal{B}}}$ and $\text{coin}_{\tilde{\mathcal{C}}}$ be the challenge bits chosen by the challenger for $\tilde{\mathcal{B}}$ and $\tilde{\mathcal{C}}$, respectively. Regardless of the values of $\text{coin}_{\tilde{\mathcal{B}}}$ and $\text{coin}_{\tilde{\mathcal{C}}}$, $\tilde{\mathcal{A}}_{\text{cp}}$ perfectly simulates $\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{search-anti-piracy}}(\lambda)$. We consider the following two cases separately.

- The case where either $\text{coin}_{\tilde{\mathcal{B}}} = 1$ or $\text{coin}_{\tilde{\mathcal{C}}} = 1$. Suppose $\text{coin}_{\tilde{\mathcal{B}}} = 1$. Then, $y_{\tilde{\mathcal{B}}}$ is uniformly at random and independent of $x_{\tilde{\mathcal{B}}}$. Thus, $y'_{\tilde{\mathcal{B}}} = y_{\tilde{\mathcal{B}}}$ occurs with only negligible probability, and $\tilde{\mathcal{B}}$ outputs uniformly at random $\text{coin}'_{\tilde{\mathcal{B}}}$ with overwhelming probability. Similarly, if $\text{coin}_{\tilde{\mathcal{C}}} = 1$, $\tilde{\mathcal{C}}$ outputs uniformly at random $\text{coin}'_{\tilde{\mathcal{C}}}$ with overwhelming probability. Therefore, we have $\Pr[\text{coin}'_{\tilde{\mathcal{B}}} \oplus \text{coin}'_{\tilde{\mathcal{C}}} = \text{coin}_{\tilde{\mathcal{B}}} \oplus \text{coin}_{\tilde{\mathcal{C}}}] = \frac{1}{2} - \text{negl}(\lambda)$.
- The case where $\text{coin}_{\tilde{\mathcal{B}}} = 0$ and $\text{coin}_{\tilde{\mathcal{C}}} = 0$. We first consider the sub-case where $y_{\mathcal{B}} = \text{Eval}(\text{msk}, x_{\tilde{\mathcal{B}}})$ and $y_{\mathcal{C}} = \text{Eval}(\text{msk}, x_{\tilde{\mathcal{C}}})$, which occurs with probability ϵ . In this case, $\tilde{\mathcal{B}}$ and $\tilde{\mathcal{C}}$ simultaneously output 0 and thus $\text{coin}'_{\tilde{\mathcal{B}}} \oplus \text{coin}'_{\tilde{\mathcal{C}}} = \text{coin}_{\tilde{\mathcal{B}}} \oplus \text{coin}_{\tilde{\mathcal{C}}}$ holds with probability 1. We then consider the other sub-case where $y_{\mathcal{B}} \neq \text{Eval}(\text{msk}, x_{\tilde{\mathcal{B}}})$ and $y_{\mathcal{C}} \neq \text{Eval}(\text{msk}, x_{\tilde{\mathcal{C}}})$, which occurs with probability $1 - \epsilon$. In this case, at least either one of $\tilde{\mathcal{B}}$ or $\tilde{\mathcal{C}}$ outputs a uniformly random bit, and thus $\text{coin}'_{\tilde{\mathcal{B}}} \oplus \text{coin}'_{\tilde{\mathcal{C}}} = \text{coin}_{\tilde{\mathcal{B}}} \oplus \text{coin}_{\tilde{\mathcal{C}}}$ holds with probability $\frac{1}{2}$.

Since the first case occurs with probability $\frac{3}{4}$ and the second case occurs with probability $\frac{1}{4}$, $\tilde{\mathcal{A}}_{\text{cp}}$ has total advantage

$$\frac{3}{4} \cdot \left(\frac{1}{2} - \text{negl}(\lambda) \right) + \frac{1}{4} \cdot \left(\epsilon \cdot 1 + (1 - \epsilon) \cdot \frac{1}{2} \right) = \frac{1}{2} + \frac{\epsilon}{8} - \text{negl}(\lambda).$$

This contradicts the $\mathcal{D}$ -IND⁺ anti-piracy security of CP-PRF. This completes the proof of Theorem A.3. $\square$

B Relationship among Security Notions for UPO

We present a direct extension of the original security definition of (generalized) UPO from [AB24] to the non-uniform challenge input setting, focusing on the independent challenge input case. In this definition, *challenge bits* are identical while *challenge inputs* are independent.

Definition B.1 ($\mathcal{D}$ -UPO Security). Let $\text{UPO} = (\text{Obf}, \text{Eval})$ be a UPO scheme for the circuit class $\text{Circ} = \{C : \{0,1\}^{\ell_{\text{cin}}} \rightarrow \{0,1\}^{\ell_{\text{cout}}}\}$. Let $\mathcal{D} = (\mathcal{D}_{\mathcal{B}}, \mathcal{D}_{\mathcal{C}})$ be a pair of distributions over $\{0,1\}^{\ell_{\text{cin}}} \times \{0,1\}^{\ell_{\text{aux}}}$. We consider the $\mathcal{D}$ -UPO anti-piracy game $\text{Exp}_{\text{UPO}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{upo}}(\lambda)$ between the challenger and an adversary $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ below.

1. On input 1^λ , $\mathcal{A}$ sends $C \in \text{Circ}$ together with two circuits $\mu_B : \{0,1\}^{\ell_{\text{cin}}} \rightarrow \{0,1\}^{\ell_{\text{cout}}}$ and $\mu_C : \{0,1\}^{\ell_{\text{cin}}} \rightarrow \{0,1\}^{\ell_{\text{cout}}}$.
2. The challenger generates $\text{coin} \leftarrow \{0,1\}$ and does the following.
 - If $\text{coin} = 0$, generate $\rho_C \leftarrow \text{Obf}(1^\lambda, C)$.
 - If $\text{coin} = 1$, generate $\rho_C \leftarrow \text{Obf}(1^\lambda, C^*[x_B, x_C, \mu_B(x_B), \mu_C(x_C)])$.

The challenger sends $(\rho_C, \text{aux}_B, \text{aux}_C)$ to $\mathcal{A}$
3. $\mathcal{A}$ creates a bipartite state q over registers R_B and R_C . Then, $\mathcal{A}$ sends register R_B to $\mathcal{B}$ and register R_C to $\mathcal{C}$.
4. The challenger sends x_B and x_C to $\mathcal{B}$ and $\mathcal{C}$, respectively.
5. $\mathcal{B}$ and $\mathcal{C}$ respectively output coin'_B and coin'_C . The challenger outputs 1 if $\text{coin}'_B = \text{coin}'_C = \text{coin}$ otherwise outputs 0.

We say that UPO is $\mathcal{D}$ -UPO secure if for any QPT adversary $\mathcal{A}_{\text{cp}}$, it holds that

$$\Pr[\text{Expt}_{\text{UPO}, \mathcal{A}_{\text{cp}}}^{\text{upo}}(\lambda) = 1] \leq \frac{1}{2} + \text{negl}(\lambda).$$

For clarity, we define what it means for a UPO to satisfy iO security.

Definition B.2 (iO Security). For any PPT Sampler and QPT adversary $\mathcal{A}$, the following holds:

If $\Pr[\forall x C_0(x) = C_1(x) \wedge |C_0| = |C_1| \mid (C_0, C_1, \text{aux}) \leftarrow \text{Sampler}(1^\lambda)] > 1 - \text{negl}(\lambda)$, then we have

$$\begin{aligned} \text{Adv}_{\text{iO}, \mathcal{D}}^{\text{iO}}(\lambda) := & \left| \Pr[\mathcal{A}(\text{Obf}(1^\lambda, C_0), \text{aux}) = 1 \mid (C_0, C_1, \text{aux}) \leftarrow \text{Sampler}(1^\lambda)] \right. \\ & \left. - \Pr[\mathcal{A}(\text{Obf}(1^\lambda, C_1), \text{aux}) = 1 \mid (C_0, C_1, \text{aux}) \leftarrow \text{Sampler}(1^\lambda)] \right| \leq \text{negl}(\lambda). \end{aligned}$$

We then prove that $\mathcal{D}$ -UPO⁺ security implies $\mathcal{D}$ -UPO security, under the assumption that the UPO satisfies iO security. We note that the proof is essentially identical to the implication from CPA⁺ anti-piracy security to $\mathcal{D}_{\text{idn-bit, ind-msg}}$ -CPA anti-piracy security for SDE given in [KY25].

Theorem B.3. Let $\text{UPO} = (\text{Obf}, \text{Eval})$ be a UPO scheme for the circuit class $\text{Circ} = \{C : \{0,1\}^{\ell_{\text{cin}}} \rightarrow \{0,1\}^{\ell_{\text{cout}}}\}$. Let $\mathcal{D} = (\mathcal{D}_B, \mathcal{D}_C)$ be a pair of distributions over $\{0,1\}^{\ell_{\text{inp}}} \times \{0,1\}^{\ell_{\text{aux}}}$. If CP-PRF satisfies iO security and $\mathcal{D}$ -UPO⁺ security, then it also satisfies $\mathcal{D}$ -UPO security.

Proof. Let $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ be any adversary for the $\mathcal{D}$ -UPO security of UPO. Consider the following hybrid experiments.

Hyb₀: This is $\text{Expt}_{\text{UPO}, \mathcal{A}_{\text{cp}}}^{\text{upo}}(\lambda)$.

We have

$$\Pr[\text{Hyb}_0 = 1] = \Pr[\text{Expt}_{\text{UPO}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{upo}}(\lambda) = 1].$$

Hyb₁: Same as Hyb₀ except that if $\text{coin} = 0$, ρ_C is generated as $\rho_C \leftarrow \text{Obf}(1^\lambda, C^*[x_B, x_C, C(x_B), C(x_C)])$.

From the iO-security of UPO, we have

$$|\Pr[\text{Hyb}_0 = 1] - \Pr[\text{Hyb}_1 = 1]| = \text{negl}(\lambda).$$

We see that Hyb₁ is the $\mathcal{D}$ -UPO⁺ security experiment $\text{Expt}_{\text{UPO}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{upoplus}}(\lambda)$ where the challenge bits are always sampled with the constraint that $\text{coin}_B = \text{coin}_C$. We below prove that

$$\Pr[\text{Hyb}_1 = 1] \leq \Pr[\text{Expt}_{\text{UPO}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{upoplus}}(\lambda) = 1].$$

In an execution of $\text{Expt}_{\text{UPO}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{upoplus}}(\lambda)$, let $\text{coin}_{\mathcal{B}}$ and $\text{coin}_{\mathcal{C}}$ be the challenge bits chosen by the challenger for $\mathcal{B}$ and $\mathcal{C}$ and $\text{coin}'_{\mathcal{B}}$ and $\text{coin}'_{\mathcal{C}}$ be the outputs of $\mathcal{B}$ and $\mathcal{C}$, respectively. Let $E_{\mathcal{B}}$ be the event that $\text{coin}'_{\mathcal{B}} = \text{coin}_{\mathcal{B}}$, $E_{\mathcal{C}}$ be the event that $\text{coin}'_{\mathcal{C}} = \text{coin}_{\mathcal{C}}$, and $E_{\text{idn-bit}}$ be the event that $\text{coin}_{\mathcal{B}} = \text{coin}_{\mathcal{C}}$. Clearly, we have

$$\Pr[\text{Expt}_{\text{UPO}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{upoplus}}(\lambda) = 1] = \Pr[E_{\mathcal{B}} \wedge E_{\mathcal{C}}] + \Pr[\neg E_{\mathcal{B}} \wedge \neg E_{\mathcal{C}}]$$

and

$$\Pr[\text{Hyb}_1 = 1] = \Pr[E_{\mathcal{B}} \wedge E_{\mathcal{C}} \mid E_{\text{idn-bit}}].$$

Our goal is to demonstrate that the former is greater than the latter. To prove this, we borrow the following claim that is implicitly shown in [AKY25].²⁸

Claim B.4. Let E_0, E_1, E_2 be events. If $\Pr[E_1 \mid E_0] = \Pr[E_1 \mid \neg E_0]$ and $\Pr[E_2 \mid E_0] = \Pr[E_2 \mid \neg E_0]$, then it holds that

$$\Pr[E_1 \wedge E_2] + \Pr[\neg E_1 \wedge \neg E_2] = \Pr[E_1 \wedge E_2 \mid E_0] + \Pr[\neg E_1 \wedge \neg E_2 \mid \neg E_0].$$

Note that the view of $\mathcal{B}$ is independent of $E_{\text{idn-bit}}$. Thus, by the no-signalling principle, we have

$$\Pr[E_{\mathcal{B}} \mid E_{\text{idn-bit}}] = \Pr[E_{\mathcal{B}} \mid \neg E_{\text{idn-bit}}].$$

Similarly, we have

$$\Pr[E_{\mathcal{C}} \mid E_{\text{idn-bit}}] = \Pr[E_{\mathcal{C}} \mid \neg E_{\text{idn-bit}}].$$

Thus, we can apply Claim B.4 with $E_0 = E_{\text{idn-bit}}$, $E_1 = E_{\mathcal{B}}$, and $E_2 = E_{\mathcal{C}}$ to obtain

$$\Pr[E_{\mathcal{B}} \wedge E_{\mathcal{C}}] + \Pr[\neg E_{\mathcal{B}} \wedge \neg E_{\mathcal{C}}] = \Pr[E_{\mathcal{B}} \wedge E_{\mathcal{C}} \mid E_{\text{idn-bit}}] + \Pr[\neg E_{\mathcal{B}} \wedge \neg E_{\mathcal{C}} \mid \neg E_{\text{idn-bit}}].$$

Since $\Pr[\neg E_{\mathcal{B}} \wedge \neg E_{\mathcal{C}} \mid \neg E_{\text{idn-bit}}] \geq 0$, we have

$$\Pr[E_{\mathcal{B}} \wedge E_{\mathcal{C}}] + \Pr[\neg E_{\mathcal{B}} \wedge \neg E_{\mathcal{C}}] \geq \Pr[E_{\mathcal{B}} \wedge E_{\mathcal{C}} \mid E_{\text{idn-bit}}].$$

By combining the above argument with the $\mathcal{D}$ -UPO⁺ security of UPO, we conclude that UPO satisfies $\mathcal{D}$ -UPO security. $\square$

C Applications of UPO

We present applications of UPO with $\mathcal{D}$ -UPO⁺ security. As shown in Appendix B, $\mathcal{D}$ -UPO⁺ security, when combined with iO security, implies $\mathcal{D}$ -UPO security. This is sufficient for the search-type applications discussed in [AB24], such as copy-protection of puncturable signatures. In what follows, we demonstrate that UPO with $\mathcal{D}$ -UPO⁺ and iO security also enables decision-type applications, including copy-protection of puncturable PRFs and puncturable PKE.

C.1 Copy-Protecting Puncturable PRFs via UPO

We show that UPO with $\mathcal{D}$ -UPO⁺ security and iO security enables us to copy-protect a secret key of any puncturable PRF.

Let $\text{PPRF} = (\text{F}, \text{Puncture})$ be a PPRF with the domain $\{0, 1\}^{\ell_{\text{inp}}}$ and the range $\{0, 1\}^{\ell_{\text{out}}}$, and $\text{UPO} = (\text{Obf}, \text{Eval})$ be a UPO scheme for a circuit class that contains a circuit computing F_K for any key K . Then we construct a copy-protected PRF scheme $\text{CP-PRF} = (\mathcal{K}\mathcal{G}, \text{Eval}, \mathcal{Q}\text{Eval})$ as follows:

$\mathcal{K}\mathcal{G}(1^\lambda)$:

- Generate a key $K \leftarrow \{0, 1\}^\lambda$ for PPRF.
- Generate $\rho_F \leftarrow \text{Obf}(1^\lambda, \text{F}_K)$.

²⁸The claim is implicitly proven in the proof of Lemma 18 of the arXiv version [AKY25]: <https://arxiv.org/abs/2405.10274v1>

- Output $\text{msk} := K$ and $s\kappa := \rho_F$.

$\text{Eval}(\text{msk}, x)$:

- Parse $\text{msk} = K$.
- Output $F_K(x)$.

$Q\text{Eval}(s\kappa, x)$:

- Parse $s\kappa = \rho_F$.
- Output $\text{Eval}(\rho_F, x)$.

The correctness of the above scheme clearly follows from the evaluation correctness of UPO. Below, we prove security.

Theorem C.1. *For any pair of QPT-samplable distributions $\mathcal{D} = (\mathcal{D}_B, \mathcal{D}_C)$ over $\{0, 1\}^{\ell_{\text{inp}}} \times \{0, 1\}^{\ell_{\text{aux}}}$, if UPO satisfies $\mathcal{D}$ -UPO⁺ security and iO security, then CP-PRF satisfies $\mathcal{D}$ -IND⁺ anti-piracy security.*

Proof. We first remark that both $\mathcal{D}_B$ and $\mathcal{D}_C$ must have super-logarithmic conditional min-entropy if a UPO scheme satisfying $\mathcal{D}$ -UPO⁺ security exists. We therefore assume this condition in what follows.

Let $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ be a QPT adversary against $\mathcal{D}$ -IND⁺ anti-piracy security of CP-PRF. We consider the following sequence of hybrids.

Hyb₀: This is the original security experiment $\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{indplus-anti-piracy}}(\lambda)$ in Definition 6.5. More specifically, it works as follows.

1. The challenger generates $K \leftarrow \{0, 1\}^\lambda$ and $\rho_F \leftarrow \text{Obf}(1^\lambda, F_K)$. The challenger also generates $(x_B, \text{aux}_B) \leftarrow \mathcal{D}(1^\lambda)$ and $(x_C, \text{aux}_C) \leftarrow \mathcal{D}(1^\lambda)$. The challenger sends $s\kappa := \rho_F$, aux_B , and aux_C to $\mathcal{A}$.
2. $\mathcal{A}$ can make arbitrarily many queries to the evaluation oracle, which on input x , returns $y \leftarrow F_K(x)$.
3. $\mathcal{A}$ creates a bipartite state q over registers R_B and R_C . Then, $\mathcal{A}$ sends R_B to $\mathcal{B}$, and R_C to $\mathcal{C}$.
4. The challenger does the following.
 - Choose $\text{coin}_B \leftarrow \{0, 1\}$, and generate $y_{B,0} \leftarrow F_K(x_B)$, and $y_{B,1} \leftarrow \{0, 1\}^{\ell_{\text{out}}}$.
 - Choose $\text{coin}_C \leftarrow \{0, 1\}$, and generate $y_{C,0} \leftarrow F_K(x_C)$ and $y_{C,1} \leftarrow \{0, 1\}^{\ell_{\text{out}}}$.

The challenger sends $(x_B, y_{B, \text{coin}_B})$ and $(x_C, y_{C, \text{coin}_C})$ to $\mathcal{B}$ and $\mathcal{C}$, respectively.

5. $\mathcal{B}$ and $\mathcal{C}$ can make arbitrarily many queries to the evaluation oracle described in the second item, with the restriction that $\mathcal{B}$ and $\mathcal{C}$ are not allowed to query x_B and x_C , respectively.
6. $\mathcal{B}$ and $\mathcal{C}$ respectively output coin'_B and coin'_C . The challenger outputs 1 if $\text{coin}'_B \oplus \text{coin}'_C = \text{coin}_B \oplus \text{coin}_C$ otherwise outputs 0.

By the definition, we clearly have

$$\Pr[\text{Hyb}_0 = 1] = \Pr[\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{indplus-anti-piracy}}(\lambda) = 1].$$

Hyb₁: This is identical to Hyb₀ except that $y_{B,0} \leftarrow F_K(x_B)$ and $y_{C,0} \leftarrow F_K(x_C)$ are generated in the first item, the challenger generates $K\{x_B, x_C\} \leftarrow \text{Puncture}(K, \{x_B, x_C\})$, and ρ_F is generated as $\rho_F \leftarrow \text{Obf}(1^\lambda, F_{K\{x_B, x_C\}}^*[x_B, x_C, y_{B,0}, y_{C,0}])$ where $F_{K\{x_B, x_C\}}^*[x_B, x_C, y_{B,0}, y_{C,0}]$ is a punctured circuit as defined in Definition 6.8.

It is clear that F_K and $F_{K\{x_B, x_C\}}^*[x_B, x_C, y_{B,0}, y_{C,0}]$ are functionally equivalent. Therefore, by the iO security of UPO, we have

$$|\Pr[\text{Hyb}_0 = 1] - \Pr[\text{Hyb}_1 = 1]| = \text{negl}(\lambda).$$

Hyb₂: This is identical to Hyb₁ except that $y_{B,0}$ and $y_{C,0}$ are generated as uniformly random strings instead of set as $F_K(x_B)$ and $F_K(x_C)$, respectively.

Note that the hybrid still uses $F_K(x_B)$ and $F_K(x_C)$ if the adversary queries x_B and x_C , respectively. Since B and C are prohibited from querying x_B and x_C , respectively, this can only occur if: (i) $\mathcal{A}$ queries x_B or x_C , (ii) B queries x_C , or (iii) C queries x_B . However, each of these events occurs only with negligible probability due to the super-logarithmic min-entropy of x_B and x_C . Thus, the only places where we use $F_K(x_B)$ and $F_K(x_C)$ are to define $y_{B,0}$ and $y_{C,0}$. Hence, by the security of PPRF,

$$|\Pr[\text{Hyb}_1 = 1] - \Pr[\text{Hyb}_2 = 1]| = \text{negl}(\lambda).$$

Hyb₃: This is identical to Hyb₂ except for the following changes:

- ρ_F is generated as $\rho_F \leftarrow \text{Obf}(1^\lambda, F_{K\{x_B, x_C\}}^*[x_B, x_C, y_{B, \text{coin}_B}, y_{C, \text{coin}_C}])$
- B and C always receive $(x_B, y_{B,1})$ and $(x_C, y_{C,1})$, respectively, regardless of the values of coin_B and coin_C .

Note that Hyb₅ is obtained by simply renaming $(y_{B,0}, y_{B,1})$ to $(y_{B,1-\text{coin}_B}, y_{B, \text{coin}_B})$ and $(y_{C,0}, y_{C,1})$ to $(y_{C,1-\text{coin}_C}, y_{C, \text{coin}_C})$, and then overriding coin_B and coin_C by $1 - \text{coin}_B$ and $1 - \text{coin}_C$, respectively.²⁹ This change is perfectly indistinguishable from the adversary's view, since $y_{B,0}$ and $y_{B,1}$, and $y_{C,0}$ and $y_{C,1}$ are each identically distributed. Thus, we have

$$\Pr[\text{Hyb}_3 = 1] = \Pr[\text{Hyb}_2 = 1].$$

Hyb₄: This is identical to Hyb₃ except that $y_{B,0}$ and $y_{C,0}$ are set as $y_{B,0} \leftarrow F_K(x_B)$ and $y_{C,0} \leftarrow F_K(x_C)$ instead of being chosen uniformly randomly.

Based on a similar argument to the hop from Hyb₁ to Hyb₂, by the security of PPRF,

$$|\Pr[\text{Hyb}_4 = 1] - \Pr[\text{Hyb}_3 = 1]| = \text{negl}(\lambda).$$

Hyb₅: This is identical to Hyb₄ except that ρ_F is generated as $\rho_F \leftarrow \text{Obf}(1^\lambda, F_K^*[x_B, x_C, y_{B, \text{coin}_B}, y_{C, \text{coin}_C}])$, that is, the unpunctured key K is used instead of the punctured key $K\{x_B, x_C\}$.

It is clear that $F_K^*[x_B, x_C, y_{B, \text{coin}_B}, y_{C, \text{coin}_C}]$ and $F_{K\{x_B, x_C\}}^*[x_B, x_C, y_{B, \text{coin}_B}, y_{C, \text{coin}_C}]$ are functionally equivalent. Therefore, by the iO security of UPO, we have

$$|\Pr[\text{Hyb}_5 = 1] - \Pr[\text{Hyb}_4 = 1]| = \text{negl}(\lambda).$$

Below, we show

$$|\Pr[\text{Hyb}_5 = 1]| \leq \frac{1}{2} + \text{negl}(\lambda)$$

by the $\mathcal{D}$ -UPO⁺ security of UPO.

To show this, we consider the following adversary $\hat{\mathcal{A}}_{\text{cp}} = (\hat{\mathcal{A}}, \hat{\mathcal{B}}, \hat{\mathcal{C}})$ against the $\mathcal{D}$ -UPO⁺ security of UPO using $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$.

1. On input 1^λ , $\hat{\mathcal{A}}$ generates $K \leftarrow \{0, 1\}^\lambda$ and chooses $y_{B,1}, y_{C,1} \leftarrow \{0, 1\}^{\ell_{\text{out}}}$. Let $\mu_B : \{0, 1\}^{\ell_{\text{inp}}} \rightarrow \{0, 1\}^{\ell_{\text{out}}}$ and $\mu_C : \{0, 1\}^{\ell_{\text{inp}}} \rightarrow \{0, 1\}^{\ell_{\text{out}}}$ be circuits that compute constant functions that output $y_{B,1}$ and $y_{C,1}$ on all inputs, respectively. It sends F_K, μ_B , and μ_C to the challenger.
2. Upon receiving $(\rho_F, \text{aux}_B, \text{aux}_C)$ from the challenger, $\hat{\mathcal{A}}$ invokes $\mathcal{A}$ on input $(\rho_F, \text{aux}_B, \text{aux}_C)$ to obtain a bipartite state q over registers R_B and R_C . It sends $(y_{B,1}, R_B)$ to B and $(y_{C,1}, R_C)$ to C . $\hat{\mathcal{A}}_{\text{cp}}$ below simulates the evaluation oracle for $\mathcal{A}_{\text{cp}}$ by using K .

²⁹This change does not affect the adversary's winning condition (i.e., the condition for outputting 1), since $\text{coin}_B \oplus \text{coin}_C = (1 - \text{coin}_B) \oplus (1 - \text{coin}_C)$.

3. Upon receiving R_B from $\hat{\mathcal{A}}$ and x_B from the challenger, $\hat{\mathcal{B}}$ runs $\mathcal{B}$ on R_B and $(x_B, y_{B,1})$ to obtain coin_B , and outputs coin_B .
4. Upon receiving R_C from $\hat{\mathcal{A}}$ and x_C from the challenger, $\hat{\mathcal{C}}$ runs $\mathcal{C}$ on R_C and $(x_C, y_{C,1})$ to obtain coin_C , and outputs coin_C .

$\hat{\mathcal{A}}_{\text{cp}}$ perfectly simulates Hyb_5 for $\mathcal{A}_{\text{cp}}$. Thus,

$$\Pr[\text{Hyb}_5 = 1] = \Pr\left[\text{Expt}_{\text{UPO}, \mathcal{D}, \hat{\mathcal{A}}_{\text{cp}}}^{\text{upoplus}}(\lambda) = 1\right] \leq \frac{1}{2} + \text{negl}(\lambda)$$

where the latter inequality follows from the $\mathcal{D}$ -UPO⁺ security of UPO.

Combining the above, we obtain

$$\Pr[\text{Hyb}_0 = 1] = \Pr\left[\text{Exp}_{\text{CP-PRF}, \mathcal{D}, \mathcal{A}_{\text{cp}}}^{\text{indplus-anti-piracy}}(\lambda) = 1\right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

This completes the proof of Theorem C.1. □

C.2 Copy-Protecting Puncturable PKE via UPO

We show that UPO with $\mathcal{D}$ -UPO⁺ security and iO security enables us to copy-protect a decryption key of any puncturable PKE scheme.

Definition of puncturable PKE. We define puncturable PKE.

Definition C.2 (Puncturable PKE). A puncturable public key encryption (PPKE) scheme PPKE with the message space $\mathcal{M}$ and the ciphertext space $\mathcal{CT}$ is a tuple of PPT algorithms $(\text{Gen}, \text{Puncture}, \text{Enc}, \text{Dec})$.

$\text{Gen}(1^\lambda) \rightarrow (\text{pk}, \text{sk})$: The key generation algorithm takes as input the security parameter 1^λ and outputs a public key pk and a secret key sk .

$\text{Puncture}(\text{sk}, (\text{ct}_1^*, \text{ct}_2^*)) \rightarrow \text{sk}\{\text{ct}_1^*, \text{ct}_2^*\}$: The puncturing algorithm takes as input a secret key sk and two ciphertexts $\text{ct}_1^*, \text{ct}_2^* \in \mathcal{CT}$, and outputs a punctured secret key $\text{sk}\{\text{ct}_1^*, \text{ct}_2^*\}$.

$\text{Enc}(\text{pk}, m) \rightarrow \text{ct}$: The encryption algorithm takes as input a public key pk and a message $m \in \mathcal{M}$, and outputs a ciphertext $\text{ct} \in \mathcal{CT}$.

$\text{Dec}(\text{sk}', \text{ct}') \rightarrow m' \text{ or } \perp$: The decryption algorithm is a deterministic algorithm that takes as input a (possibly punctured) secret key sk' and a ciphertext $\text{ct}' \in \mathcal{CT}$, and outputs either a message $m' \in \mathcal{M}$ or the special symbol $\perp$.

Correctness. For all messages $m \in \mathcal{M}$ and for all key pairs $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$, it holds that:

$$\text{Dec}(\text{pk}, \text{Enc}(\text{sk}, m)) = m.$$

Punctured correctness. For all key pairs $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$, all pairs of ciphertexts $\text{ct}_1^*, \text{ct}_2^* \in \mathcal{CT}$, all punctured secret keys $\text{sk}\{\text{ct}_1^*, \text{ct}_2^*\} \leftarrow \text{Puncture}(\text{sk}, (\text{ct}_1^*, \text{ct}_2^*))$, and all ciphertexts $\text{ct} \in \mathcal{CT} \setminus \{\text{ct}_1^*, \text{ct}_2^*\}$, it holds that:

$$\text{Dec}(\text{sk}, \text{ct}) = \text{Dec}(\text{sk}\{\text{ct}_1^*, \text{ct}_2^*\}, \text{ct}).$$

Indistinguishability with punctured keys. We consider the following experiment $\text{Exp}_{\text{PPKE}, \mathcal{A}}^{\text{ppke}}(\lambda)$ between the challenger and an adversary $\mathcal{A}$.

1. The adversary $\mathcal{A}$ receives 1^λ and sends two pairs of messages $(m_{1,0}^*, m_{1,1}^*) \in \mathcal{M}^2$ and $(m_{2,0}^*, m_{2,1}^*) \in \mathcal{M}^2$ such that $|m_{1,0}^*| = |m_{1,1}^*|$ and $|m_{2,0}^*| = |m_{2,1}^*|$ to the challenger.
2. The challenger proceeds as follows:
 - Generate $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$.
 - Choose a uniform bit $\text{coin} \leftarrow \{0, 1\}$
 - Compute the ciphertext $\text{ct}_1^* \leftarrow \text{Enc}(\text{pk}, m_{1,\text{coin}}^*)$ and $\text{ct}_2^* \leftarrow \text{Enc}(\text{pk}, m_{2,\text{coin}}^*)$.
 - Generate a punctured key $\text{sk}\{\text{ct}_1^*, \text{ct}_2^*\} \leftarrow \text{Puncture}(\text{sk}, (\text{ct}_1^*, \text{ct}_2^*))$.
 - Send $(\text{ct}_1^*, \text{ct}_2^*, \text{pk}, \text{sk}\{\text{ct}_1^*, \text{ct}_2^*\})$ to $\mathcal{A}$.
3. The adversary $\mathcal{A}$ outputs a guess coin' . The experiment outputs 1 if $\text{coin}' = \text{coin}$, and 0 otherwise.

For any QPT adversary $\mathcal{A}$, we have

$$\left| \Pr \left[\text{Exp}_{\text{PPKE}, \mathcal{A}}^{\text{ppke}}(\lambda) = 1 \right] - \frac{1}{2} \right| = \text{negl}(\lambda).$$

While several works have defined puncturable PKE (e.g., [GM15, CHN⁺18, SSS⁺20, SDLP20, KN22b]), our definition differs from all of them. The closest to ours is the one in [KN22b], but their definition only allows for puncturing on one ciphertext rather than two. Thus, in Appendix D, we present a construction that satisfies our definition, based on polynomially secure iO and OWFs.

Definition of SDE. We define SDE and its security following [CLLZ21]. Specifically, we adopt the CPA⁺ anti-piracy security introduced in [CLLZ21]. It is shown in [CLLZ21] that CPA⁺ anti-piracy security implies most existing security notions for SDE; see [CLLZ21] for details.

Definition C.3 (Single-Decryptor Encryption [CLLZ21]). A single-decryptor encryption (SDE) scheme SDE is a tuple of four algorithms $(\text{Setup}, \mathcal{KG}, \text{Enc}, \text{Dec})$. Below, let $\mathcal{M}$ be the message space of SDE.

$\text{Setup}(1^\lambda) \rightarrow (\text{pk}, \text{sk})$: The setup algorithm takes a security parameter 1^λ , and outputs a public key pk and a secret key sk .

$\mathcal{KG}(\text{sk}) \rightarrow s\mathcal{k}$: The key generation algorithm $\mathcal{KG}$ takes a secret key sk , and outputs a quantum secret key $s\mathcal{k}$.

$\text{Enc}(\text{pk}, m) \rightarrow \text{ct}$: The encryption algorithm takes a public key pk and a message $m \in \mathcal{M}$, and outputs a ciphertext ct .

$\text{Dec}(s\mathcal{k}, \text{ct}) \rightarrow \tilde{m}$: The decryption algorithm takes a quantum secret key $s\mathcal{k}$ and a ciphertext ct , and outputs a message $\tilde{m} \in \{\perp\} \cup \mathcal{M}$.

Correctness: For every $m \in \mathcal{M}$, we require

$$\Pr \left[\text{Dec}(s\mathcal{k}, \text{ct}) = m \mid \begin{array}{l} (\text{pk}, \text{sk}) \leftarrow \text{Setup}(1^\lambda) \\ s\mathcal{k} \leftarrow \mathcal{KG}(\text{sk}) \\ \text{ct} \leftarrow \text{Enc}(\text{pk}, m) \end{array} \right] = 1 - \text{negl}(\lambda).$$

Definition C.4 (CPA⁺ Anti-Piracy Security). Let $\text{SDE} = (\text{Setup}, \mathcal{KG}, \text{Enc}, \text{Dec})$ be an SDE scheme. We consider the $\mathcal{D}$ -CPA⁺ anti-piracy game $\text{Exp}_{\text{SDE}, \mathcal{A}_{\text{sde}}}^{\text{cpaplus-anti-piracy}}(\lambda)$ between the challenger and an adversary $\mathcal{A}_{\text{sde}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ below.

1. The challenger generates $(\text{pk}, \text{sk}) \leftarrow \text{Setup}(1^\lambda)$.
2. The challenger generates $s\mathcal{k} \leftarrow \mathcal{KG}(\text{sk})$ and sends $(1^\lambda, \text{pk}, s\mathcal{k})$ to $\mathcal{A}$.

3. $\mathcal{A}$ chooses two pairs of messages $(m_{\mathcal{B},0}, m_{\mathcal{B},1}) \in \mathcal{M}^2$ and $(m_{\mathcal{C},0}, m_{\mathcal{C},1}) \in \mathcal{M}^2$ such that $|m_{\mathcal{B},0}| = |m_{\mathcal{B},1}|$ and $|m_{\mathcal{C},0}| = |m_{\mathcal{C},1}|$, and creates a bipartite state q over registers $\mathcal{R}_{\mathcal{B}}$ and $\mathcal{R}_{\mathcal{C}}$. Then, $\mathcal{A}$ sends $(m_{\mathcal{B},0}, m_{\mathcal{B},1}, m_{\mathcal{C},0}, m_{\mathcal{C},1})$ to the challenger, register $\mathcal{R}_{\mathcal{B}}$ to $\mathcal{B}$, and register $\mathcal{R}_{\mathcal{C}}$ to $\mathcal{C}$.
4. The challenger chooses $\text{coin}_{\mathcal{B}} \leftarrow \{0,1\}$ and $\text{coin}_{\mathcal{C}} \leftarrow \{0,1\}$, generates $\text{ct}_{\mathcal{B}} \leftarrow \text{Enc}(\text{pk}, m_{\mathcal{B}, \text{coin}_{\mathcal{B}}})$ and $\text{ct}_{\mathcal{C}} \leftarrow \text{Enc}(\text{pk}, m_{\mathcal{C}, \text{coin}_{\mathcal{C}}})$, and sends $\text{ct}_{\mathcal{B}}$ and $\text{ct}_{\mathcal{C}}$ to $\mathcal{B}$ and $\mathcal{C}$, respectively.
5. $\mathcal{B}$ and $\mathcal{C}$ respectively output $\text{coin}'_{\mathcal{B}}$ and $\text{coin}'_{\mathcal{C}}$. The challenger outputs 1 if $\text{coin}'_{\mathcal{B}} \oplus \text{coin}'_{\mathcal{C}} = \text{coin}_{\mathcal{B}} \oplus \text{coin}_{\mathcal{C}}$ otherwise outputs 0.

We say that SDE is $\mathcal{D}\text{-CPA}^+$ anti-piracy secure if for any QPT adversary $\mathcal{A}_{\text{sde}}$, it holds that

$$\Pr \left[\text{Exp}_{\text{SDE}, \mathcal{A}_{\text{sde}}}^{\text{cpaplus-anti-piracy}}(\lambda) = 1 \right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

We say that SDE is selectively $\mathcal{D}\text{-CPA}^+$ anti-piracy secure if the above holds for all QPT adversaries that declare $(m_{\mathcal{B},0}, m_{\mathcal{B},1})$ and $(m_{\mathcal{C},0}, m_{\mathcal{C},1})$ before receiving pk and sk .

Remark C.5. For schemes with single-bit messages, selective $\mathcal{D}\text{-CPA}^+$ anti-piracy security is equivalent to full $\mathcal{D}\text{-CPA}^+$ anti-piracy security. [KY25] demonstrated that the message space can be expanded through bit-by-bit encryption while preserving $\mathcal{D}\text{-CPA}^+$ anti-piracy security. Therefore, in terms of feasibility, selective $\mathcal{D}\text{-CPA}^+$ anti-piracy secure SDE and full $\mathcal{D}\text{-CPA}^+$ anti-piracy secure SDE are equivalent.

Construction We describe how to copy-protect a decryption key of any puncturable PKE scheme that has sufficiently large min-entropy in ciphertexts.

Let $\text{PPKE} = (\text{PPKE.Gen}, \text{PPKE.Puncture}, \text{PPKE.Enc}, \text{PPKE.Dec})$ be a PPKE scheme with the message space $\mathcal{M}$ and the ciphertext space $\mathcal{CT}$, and $\text{UPO} = (\text{Obf}, \text{Eval})$ be a UPO scheme for a circuit class that contains a circuit computing $\text{SDE.Dec}(\text{sk}, \cdot)$ for all sk generated by $\text{PPKE.Gen}(1^\lambda)$. Then we construct a SDE scheme $\text{SDE} = (\text{SDE.Setup}, \text{SDE.XG}, \text{SDE.Enc}, \text{SDE.Dec})$ as follows:

$\text{SDE.Setup}(1^\lambda)$:

- Generate a key $(\text{pk}, \text{sk}) \leftarrow \text{PPKE.Gen}(1^\lambda)$.
- Output pk and sk .

$\text{SDE.XG}(\text{sk})$:

- Generate $\rho_{\text{Dec}} \leftarrow \text{Obf}(1^\lambda, \text{SDE.Dec}(\text{sk}, \cdot))$.
- Output $\text{sk} := \rho_{\text{Dec}}$.

$\text{SDE.Enc}(\text{pk}, m)$:

- Generate $\text{ct} \leftarrow \text{PPKE.Enc}(\text{pk}, m)$.
- Output ct .

$\text{SDE.Dec}(\text{sk}, \text{ct})$:

- Parse $\text{sk} = \rho_{\text{Dec}}$.
- Output $\text{Eval}(\rho_{\text{Dec}}, \text{ct})$.

The correctness of the above scheme clearly follows from the evaluation correctness of UPO.

Below, we prove security. For a public key pk and a message $m \in \mathcal{M}$, let $\mathcal{D}_{\text{pk}, m}$ be the distribution of $\text{ct} \leftarrow \text{PPKE.Enc}(\text{pk}, m)$. In light of Definition 3.1, $\mathcal{D}_{\text{pk}, m}$ is regarded as a distribution without auxiliary information.

Theorem C.6. Suppose that UPO satisfies iO security and $\mathcal{D}\text{-UPO}^+$ security where $\mathcal{D} = (\mathcal{D}_{\mathcal{B}}, \mathcal{D}_{\mathcal{C}}) = (\mathcal{D}_{\text{pk}, m_{\mathcal{B}}}, \mathcal{D}_{\text{pk}, m_{\mathcal{C}}})$ for all pk that is generated by SDE.Setup , $m_{\mathcal{B}} \in \mathcal{M}$, and $m_{\mathcal{C}} \in \mathcal{M}$. Then SDE satisfies selective CPA^+ anti-piracy security.

Remark C.7. In general, PKE schemes, including the PPKE scheme given in Appendix D, usually have a polynomial amount of min-entropy in the ciphertext for any fixed public key and message. Thus, we can instantiate the construction using UPO that supports high min-entropy challenge distributions as given in Section 9.

Proof of Theorem C.6. Let $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ be a QPT adversary against CPA^+ anti-piracy security of SDE. We consider the following sequence of hybrids.

Hyb₀: This is the original security experiment $\text{Exp}_{\text{SDE}, \mathcal{A}_{\text{sde}}}^{\text{cpaplus-anti-piracy}}(\lambda)$ in Definition C.4. More specifically, it works as follows.

1. $\mathcal{A}$ receives 1^λ and sends two pairs of messages $(m_{\mathcal{B},0}, m_{\mathcal{B},1}) \in \mathcal{M}^2$ and $(m_{\mathcal{C},0}, m_{\mathcal{C},1}) \in \mathcal{M}^2$ such that $|m_{\mathcal{B},0}| = |m_{\mathcal{B},1}|$ and $|m_{\mathcal{C},0}| = |m_{\mathcal{C},1}|$ to the challenger.
2. The challenger generates $(\text{pk}, \text{sk}) \leftarrow \text{PPKE.Gen}(1^\lambda)$ and $\rho_{\text{Dec}} \leftarrow \text{Obf}(1^\lambda, \text{SDE.Dec}(\text{sk}, \cdot))$. The challenger sends pk and $s\mathcal{K} := \rho_{\text{Dec}}$ to $\mathcal{A}$.
3. $\mathcal{A}$ creates a bipartite state q over registers $R_{\mathcal{B}}$ and $R_{\mathcal{C}}$. Then, $\mathcal{A}$ sends $R_{\mathcal{B}}$ to $\mathcal{B}$, and $R_{\mathcal{C}}$ to $\mathcal{C}$.
4. The challenger chooses $\text{coin}_{\mathcal{B}} \leftarrow \{0, 1\}$ and $\text{coin}_{\mathcal{C}} \leftarrow \{0, 1\}$, generates $\text{ct}_{\mathcal{B}} \leftarrow \text{PPKE.Enc}(\text{pk}, m_{\mathcal{B}, \text{coin}_{\mathcal{B}}})$ and $\text{ct}_{\mathcal{C}} \leftarrow \text{PPKE.Enc}(\text{pk}, m_{\mathcal{C}, \text{coin}_{\mathcal{C}}})$, and sends $\text{ct}_{\mathcal{B}}$ and $\text{ct}_{\mathcal{C}}$ to $\mathcal{B}$ and $\mathcal{C}$, respectively.
5. $\mathcal{B}$ and $\mathcal{C}$ respectively output $\text{coin}'_{\mathcal{B}}$ and $\text{coin}'_{\mathcal{C}}$. The challenger outputs 1 if $\text{coin}'_{\mathcal{B}} \oplus \text{coin}'_{\mathcal{C}} = \text{coin}_{\mathcal{B}} \oplus \text{coin}_{\mathcal{C}}$ otherwise outputs 0.

By the definition, we clearly have

$$\Pr[\text{Hyb}_0 = 1] = \Pr[\text{Exp}_{\text{SDE}, \mathcal{A}_{\text{sde}}}^{\text{cpaplus-anti-piracy}}(\lambda) = 1].$$

Hyb₁: This is identical to Hyb₀ except that $\text{coin}_{\mathcal{B}} \leftarrow \{0, 1\}$, $\text{coin}_{\mathcal{C}} \leftarrow \{0, 1\}$, $\text{ct}_{\mathcal{B}} \leftarrow \text{PPKE.Enc}(\text{pk}, m_{\mathcal{B}, \text{coin}_{\mathcal{B}}})$, and $\text{ct}_{\mathcal{C}} \leftarrow \text{PPKE.Enc}(\text{pk}, m_{\mathcal{C}, \text{coin}_{\mathcal{C}}})$ are generated in the first item, the challenger generates $\text{sk}\{\text{ct}_{\mathcal{B}}, \text{ct}_{\mathcal{C}}\} \leftarrow \text{Puncture}(\text{sk}, (\text{ct}_{\mathcal{B}}, \text{ct}_{\mathcal{C}}))$, and ρ_{Dec} is generated as $\rho_{\text{Dec}} \leftarrow \text{Obf}(1^\lambda, \text{PPKE.Dec}(\text{sk}\{\text{ct}_{\mathcal{B}}, \text{ct}_{\mathcal{C}}\}, \cdot)^*[\text{ct}_{\mathcal{B}}, \text{ct}_{\mathcal{C}}, m_{\mathcal{B}, \text{coin}_{\mathcal{B}}}, m_{\mathcal{C}, \text{coin}_{\mathcal{C}}}]])$ where $\text{PPKE.Dec}(\text{sk}\{\text{ct}_{\mathcal{B}}, \text{ct}_{\mathcal{C}}\}, \cdot)^*[\text{ct}_{\mathcal{B}}, \text{ct}_{\mathcal{C}}, m_{\mathcal{B}, \text{coin}_{\mathcal{B}}}, m_{\mathcal{C}, \text{coin}_{\mathcal{C}}}]])$ is a punctured circuit as defined in Definition 6.8.

It is clear that $\text{PPKE.Dec}(\text{sk}, \cdot)$ and $\text{PPKE.Dec}(\text{sk}\{\text{ct}_{\mathcal{B}}, \text{ct}_{\mathcal{C}}\}, \cdot)^*[\text{ct}_{\mathcal{B}}, \text{ct}_{\mathcal{C}}, m_{\mathcal{B}, \text{coin}_{\mathcal{B}}}, m_{\mathcal{C}, \text{coin}_{\mathcal{C}}}]])$ are functionally equivalent by the correctness and punctured correctness of PPKE. Therefore, by the iO security of UPO, we have

$$|\Pr[\text{Hyb}_0 = 1] - \Pr[\text{Hyb}_1 = 1]| = \text{negl}(\lambda).$$

Hyb₂: This is identical to Hyb₁ except that $\text{ct}_{\mathcal{B}}$ and $\text{ct}_{\mathcal{C}}$ are generated as $\text{ct}_{\mathcal{B}} \leftarrow \text{PPKE.Enc}(\text{pk}, m_{\mathcal{B},0})$ and $\text{ct}_{\mathcal{C}} \leftarrow \text{PPKE.Enc}(\text{pk}, m_{\mathcal{C},0})$ regardless of the values of $\text{coin}_{\mathcal{B}}$ and $\text{coin}_{\mathcal{C}}$.

By a straightforward reduction to the security of PPKE, where we set the challenge messages for PPKE as $(m_{1,0}, m_{1,1}) = (m_{\mathcal{B},0}, m_{\mathcal{B}, \text{coin}_{\mathcal{B}}})$ and $(m_{2,0}, m_{2,1}) = (m_{\mathcal{C},0}, m_{\mathcal{C}, \text{coin}_{\mathcal{C}}})$, we have

$$|\Pr[\text{Hyb}_1 = 1] - \Pr[\text{Hyb}_2 = 1]| = \text{negl}(\lambda).$$

Below, we show

$$|\Pr[\text{Hyb}_2 = 1]| \leq \frac{1}{2} + \text{negl}(\lambda)$$

by the $\mathcal{D}$ -UPO⁺ security of UPO.

To show this, we consider the following adversary $\hat{\mathcal{A}}_{\text{cp}} = (\hat{\mathcal{A}}, \hat{\mathcal{B}}, \hat{\mathcal{C}})$ against the $\mathcal{D}$ -UPO⁺ security of UPO using $\mathcal{A}_{\text{cp}} = (\mathcal{A}, \mathcal{B}, \mathcal{C})$ where $\mathcal{D} = (\mathcal{D}_{\mathcal{B}}, \mathcal{D}_{\mathcal{C}}) = (\mathcal{D}_{\text{pk}, m_{\mathcal{B},0}}, \mathcal{D}_{\text{pk}, m_{\mathcal{C},0}})$ for fixed (pk, sk) , $(m_{\mathcal{B},0}, m_{\mathcal{B},1}) \in \mathcal{M}^2$, and $(m_{\mathcal{C},0}, m_{\mathcal{C},1}) \in \mathcal{M}^2$. Let $\rho_{\mathcal{A}}$ be the internal state of $\mathcal{A}$ at the point of sending $(m_{\mathcal{B},0}, m_{\mathcal{B},1})$ and $(m_{\mathcal{C},0}, m_{\mathcal{C},1})$.³⁰

³⁰ $\hat{\mathcal{A}}_{\text{cp}}$ is a non-uniform QPT algorithm in which (pk, sk) , $m_{\mathcal{B},1}$, $m_{\mathcal{B},1}$, and $\rho_{\mathcal{A}}$ are hardwired.

1. $\hat{\mathcal{A}}$ lets $\mu_{\mathcal{B}} : \mathcal{CT} \rightarrow \mathcal{M}$ and $\mu_{\mathcal{C}} : \mathcal{CT} \rightarrow \mathcal{M}$ be circuits that compute constant functions that output $m_{\mathcal{B},1}$ and $m_{\mathcal{C},1}$ on all inputs, respectively. It sends $\text{SDE.Dec}(\text{sk}, \cdot)$, $\mu_{\mathcal{B}}$, and $\mu_{\mathcal{C}}$ to the challenger.
2. Upon receiving ρ_{Dec} from the challenger, $\hat{\mathcal{A}}$ invokes $\mathcal{A}$ with the internal state $\rho_{\mathcal{A}}$ on input $(\text{pk}, \rho_{\text{Dec}})$ to obtain a bipartite state q over registers $R_{\mathcal{B}}$ and $R_{\mathcal{C}}$. It sends $R_{\mathcal{B}}$ to $\mathcal{B}$ and $R_{\mathcal{C}}$ to $\mathcal{C}$.
3. Upon receiving $R_{\mathcal{B}}$ from $\hat{\mathcal{A}}$ and $\text{ct}_{\mathcal{B}}$ from the challenger, $\hat{\mathcal{B}}$ runs $\mathcal{B}$ on $R_{\mathcal{B}}$ and $\text{ct}_{\mathcal{B}}$ to obtain $\text{coin}_{\mathcal{B}}$, and outputs $\text{coin}_{\mathcal{B}}$.
4. Upon receiving $R_{\mathcal{C}}$ from $\hat{\mathcal{A}}$ and $\text{ct}_{\mathcal{C}}$ from the challenger, $\hat{\mathcal{C}}$ runs $\mathcal{C}$ on $R_{\mathcal{C}}$ and $\text{ct}_{\mathcal{C}}$ to obtain $\text{coin}_{\mathcal{C}}$, and outputs $\text{coin}_{\mathcal{C}}$.

$\hat{\mathcal{A}}_{\text{cp}}$ perfectly simulates Hyb_2 for $\mathcal{A}_{\text{cp}}$ conditioned on the fixed choice of (pk, sk) , $(m_{\mathcal{B},0}, m_{\mathcal{B},1})$ and $(m_{\mathcal{C},0}, m_{\mathcal{C},1})$. By the $\mathcal{D}$ - UPO^+ security of UPO , the probability that $\text{coin}'_{\mathcal{B}} \oplus \text{coin}'_{\mathcal{C}} = \text{coin}_{\mathcal{B}} \oplus \text{coin}_{\mathcal{C}}$ is at most $\frac{1}{2} + \text{negl}(\lambda)$. Thus, by taking average over (pk, sk) , $(m_{\mathcal{B},0}, m_{\mathcal{B},1})$, and $(m_{\mathcal{C},0}, m_{\mathcal{C},1})$,

$$\Pr[\text{Hyb}_2 = 1] \leq \frac{1}{2} + \text{negl}(\lambda).$$

Combining the above, we obtain

$$\Pr[\text{Hyb}_0 = 1] = \Pr\left[\text{Exp}_{\text{SDE}, \mathcal{A}_{\text{sde}}}^{\text{cpaplus-anti-piracy}}(\lambda)\right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

This completes the proof of Theorem C.1. □

D Construction of Puncturable Public Key Encryption

We give a construction of a PPKE scheme as defined in Definition C.2. The construction is very similar to the construction of functional encryption given in [GGH⁺16].

D.1 Definitions

Public key encryption. We define building blocks used in our construction.

The first is just a standard IND-CPA secure PKE.

Definition D.1 (PKE). A public key encryption (PKE) scheme PKE with the message space $\mathcal{M}$ and the ciphertext space $\mathcal{CT}$ is a tuple of PPT algorithms $(\text{Gen}, \text{Enc}, \text{Dec})$.

$\text{Gen}(1^\lambda) \rightarrow (\text{pk}, \text{sk})$: The key generation algorithm takes as input the security parameter 1^λ and outputs a public key pk and a secret key sk .

$\text{Enc}(\text{pk}, m) \rightarrow \text{ct}$: The encryption algorithm takes as input a public key pk and a message $m \in \mathcal{M}$, and outputs a ciphertext $\text{ct} \in \mathcal{CT}$.

$\text{Dec}(\text{sk}', \text{ct}') \rightarrow m' \text{ or } \perp$: The decryption algorithm is a deterministic algorithm that takes as input a secret key sk' and a ciphertext $\text{ct}' \in \mathcal{CT}$, and outputs either a message $m' \in \mathcal{M}$ or the special symbol $\perp$.

Correctness. For all messages $m \in \mathcal{M}$ and for all key pairs $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$, it holds that:

$$\text{Dec}(\text{pk}, \text{Enc}(\text{sk}, m)) = m.$$

IND-CPA security. We consider the following experiment $\text{Exp}_{\text{PKE}, \mathcal{A}}^{\text{pke}}(\lambda)$ between the challenger and an adversary $\mathcal{A}$.

1. The challenger generates $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$ and sends pk to $\mathcal{A}$.
2. The adversary $\mathcal{A}$ receives 1^λ and sends a pair of messages $(m_0^*, m_1^*) \in \mathcal{M}^2$ such that $|m_0^*| = |m_1^*|$.
3. The challenger chooses a uniform bit $\text{coin} \leftarrow \{0, 1\}$, compute the ciphertext $\text{ct}^* \leftarrow \text{Enc}(\text{pk}, m_{\text{coin}}^*)$, and sends ct^* to $\mathcal{A}$.
4. The adversary $\mathcal{A}$ outputs a guess coin' . The experiment outputs 1 if $\text{coin}' = \text{coin}$, and 0 otherwise.

For any QPT adversary $\mathcal{A}$, we have

$$\left| \Pr \left[\text{Exp}_{\text{PKE}, \mathcal{A}}^{\text{pke}}(\lambda) \right] - \frac{1}{2} \right| = \text{negl}(\lambda).$$

Note that IND-CPA secure PKE scheme exists if polynomially secure iO and OWFs exist [SW21].

Two-theorem statistically simulation sound non-interactive zero-knowledge proof. Next, we introduce a variant of the statistically simulation sound non-interactive zero-knowledge proofs (SSS-NIZK), where we consider simulation of proofs on two statements rather than one.

Definition D.2 (SSS-NIZK). A two-theorem statistically simulation sound non-interactive zero-knowledge proof (SSS-NIZK) scheme NIZK for an NP language L and the corresponding relation R_L is a tuple of PPT algorithms (Setup, Prove, Vrfy).

Setup(1^λ) $\rightarrow$ crs: The setup algorithm takes as input the security parameter 1^λ and outputs a common reference string crs.

Prove(crs, x, w) $\rightarrow$ ct: The prove algorithm takes as input a common reference string crs, a statement x and its witness w and outputs a proof π .

Vrfy(crs, x, π) $\rightarrow \top$ or $\perp$: The verification algorithm is a deterministic algorithm that takes as input a common reference string crs, a statement x , and a proof π , and outputs $\top$ (indicating acceptance) or $\perp$ (indicating rejection).

Completeness. For all crs $\leftarrow$ Setup(1^λ) and $(x, w) \in R_L$, it holds that:

$$\text{Vrfy}(\text{crs}, x, \text{Prove}(\text{crs}, x, \pi)) = \top.$$

Statistical soundness. With overwhelming probability over the choice of crs $\leftarrow$ Setup(1^λ), there are no $x \notin L$ and π such that $\text{Vrfy}(\text{crs}, x, \text{Prove}(\text{crs}, x, \pi)) = \top$.

Two-theorem non-adaptive computational zero-knowledge. There exists a PPT algorithm Sim such that for any $(x_1, w_1), (x_2, w_2) \in R_L$, the following computational indistinguishability against QPT adversaries holds:

$$\left\{ \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (\text{crs}, x_1, x_2, \pi_1, \pi_2) : \pi_1 \leftarrow \text{Prove}(\text{crs}, x_1, w_1) \\ \pi_2 \leftarrow \text{Prove}(\text{crs}, x_2, w_2) \end{array} \right\} \stackrel{c}{\approx} \left\{ (\text{crs}, x_1, x_2, \pi_1, \pi_2) : (\text{crs}, \pi_1, \pi_2) \leftarrow \text{Sim}(1^\lambda, x_1, x_2) \right\}.$$

Two-theorem statistical simulation soundness. Let Sim be as in the above computational zero-knowledge property. For any statements x_1, x_2 , with overwhelming probability over the choice of $(\text{crs}, \pi_1, \pi_2) \leftarrow \text{Sim}(1^\lambda, x_1, x_2)$, there are no $x \notin L \cup \{x_1, x_2\}$ and π such that $\text{Vrfy}(\text{crs}, x, \text{Prove}(\text{crs}, x, \pi)) = \top$.

[GGH⁺16] constructed SSS-NIZK in the single-theorem setting based on any statistically sound NIZK and one-way functions. Their construction directly extends to the two-theorem setting as above. We note that statistically sound NIZK exists assuming polynomially secure iO and OWFs [BP15, BPW16].

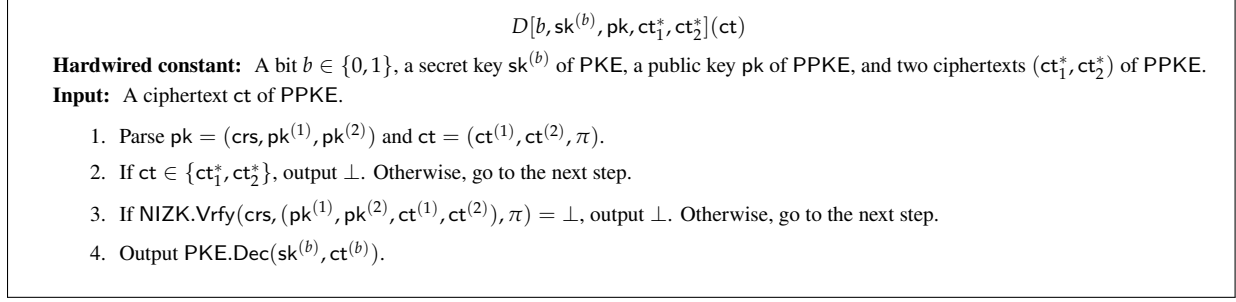


Figure 22: The description of the circuit D .

D.2 Construction

Let $i\mathcal{O}$ be $i\mathcal{O}$. Let $PKE = (PKE.Gen, PKE.Enc, PKE.Dec)$ be an IND-CPA secure PKE scheme and $NIZK = (NIZK.Setup, NIZK.Prove, NIZK.Vrfy)$ be a two-theorem SSS-NIZK scheme for the NP relation R_L defined below:

$$((pk^{(1)}, pk^{(2)}, ct^{(1)}, ct^{(2)}), (m, r^{(1)}, r^{(2)})) \in R_L \iff ct^{(1)} = PKE.Enc(pk^{(1)}, m; r^{(1)}) \wedge ct^{(2)} = PKE.Enc(pk^{(2)}, m; r^{(2)}).$$

Then we construct a PPKE scheme $PPKE = (PPKE.Gen, PPKE.Puncture, PPKE.Enc, PPKE.Dec)$ as follows.

$Gen(1^\lambda)$:

- Generate $crs \leftarrow NIZK.Setup(1^\lambda)$, $(pk^{(1)}, sk^{(1)}) \leftarrow PKE.Gen(1^\lambda)$, and $(pk^{(2)}, sk^{(2)}) \leftarrow PKE.Gen(1^\lambda)$.
- Output $pk = (crs, pk^{(1)}, pk^{(2)})$ and $sk = (pk, sk^{(1)})$.

$Puncture(sk, (ct_1^*, ct_2^*))$:

- Parse $sk = (pk, sk^{(1)})$.
- Generate $sk\{ct_1^*, ct_2^*\} \leftarrow i\mathcal{O}(D[1, sk^{(1)}, pk, ct_1^*, ct_2^*])$ where D is a circuit described in Figure 22.
- Output $sk\{ct_1^*, ct_2^*\}$.

$Enc(pk, m)$:

- Parse $pk = (crs, pk^{(1)}, pk^{(2)})$.
- Generate $ct^{(1)} = PKE.Enc(pk, m; r^{(1)})$ and $ct^{(2)} = PKE.Enc(pk, m; r^{(2)})$ where $r^{(1)}, r^{(2)}$ are uniform randomness.
- Generate $\pi \leftarrow NIZK.Prove(crs, (pk^{(1)}, pk^{(2)}, ct^{(1)}, ct^{(2)}), (m, r^{(1)}, r^{(2)}))$
- Output $ct = (ct^{(1)}, ct^{(2)}, \pi)$.

$Dec(sk', ct)$:

- If sk' is an unpunctured key, do the following:
 - Parse $sk' = (pk = (crs, pk^{(1)}, pk^{(2)}), sk^{(1)})$ and $ct = (ct^{(1)}, ct^{(2)}, \pi)$.
 - If $NIZK.Vrfy(crs, (pk^{(1)}, pk^{(2)}, ct^{(1)}, ct^{(2)}), \pi) = \perp$ output $\perp$.
 - Otherwise, output $m' \leftarrow PKE.Dec(sk^{(1)}, ct_1)$.
- If sk' is a punctured key, compute $m' \leftarrow sk'(ct)$, where sk' is viewed as an obfuscated program, and outputs m' .

The correctness and punctured correctness of PPKE follows from the correctness of PKE and $i\mathcal{O}$ and completeness of NIZK.

Below, we prove indistinguishability with punctured keys.

Let $\mathcal{A}$ be a QPT adversary against the indistinguishability with punctured keys of PPKE. We consider the following sequence of hybrids.

Hyb₀: This is the original security experiment $\text{Exp}_{\text{PPKE}, \mathcal{A}}^{\text{ppke}}(\lambda)$ in Definition C.2. More specifically, it proceeds as follows:

1. The adversary $\mathcal{A}$ receives 1^λ and sends two pairs of messages

$$(m_{1,0}, m_{1,1}) \in \mathcal{M}^2 \quad \text{and} \quad (m_{2,0}, m_{2,1}) \in \mathcal{M}^2$$

to the challenger, such that $|m_{1,0}| = |m_{1,1}|$ and $|m_{2,0}| = |m_{2,1}|$.

2. The challenger proceeds as follows:

- Generate $\text{crs} \leftarrow \text{NIZK.Setup}(1^\lambda)$, $(\text{pk}^{(1)}, \text{sk}^{(1)}) \leftarrow \text{PKE.Gen}(1^\lambda)$, and $(\text{pk}^{(2)}, \text{sk}^{(2)}) \leftarrow \text{PKE.Gen}(1^\lambda)$. and set $\text{pk} = (\text{crs}, \text{pk}^{(1)}, \text{pk}^{(2)})$.
- Choose a uniform bit coin $\leftarrow \{0, 1\}$.
- Compute the ciphertexts:

$$\begin{aligned} \text{ct}_1^{(1)} &= \text{PKE.Enc}(\text{pk}^{(1)}, m_{1,\text{coin}}; r_1^{(1)}), \\ \text{ct}_1^{(2)} &= \text{PKE.Enc}(\text{pk}^{(2)}, m_{1,\text{coin}}; r_1^{(2)}), \\ \text{ct}_2^{(1)} &= \text{PKE.Enc}(\text{pk}^{(1)}, m_{2,\text{coin}}; r_2^{(1)}), \\ \text{ct}_2^{(2)} &= \text{PKE.Enc}(\text{pk}^{(2)}, m_{2,\text{coin}}; r_2^{(2)}), \end{aligned}$$

where $r_1^{(1)}, r_1^{(2)}, r_2^{(1)}, r_2^{(2)}$ are uniformly random.

- Generate

$$\begin{aligned} \pi_1 &\leftarrow \text{NIZK.Prove}(\text{crs}, (\text{pk}^{(1)}, \text{pk}^{(2)}, \text{ct}_1^{(1)}, \text{ct}_1^{(2)}), (m_{1,\text{coin}}, r_1^{(1)}, r_1^{(2)})), \\ \pi_2 &\leftarrow \text{NIZK.Prove}(\text{crs}, (\text{pk}^{(1)}, \text{pk}^{(2)}, \text{ct}_2^{(1)}, \text{ct}_2^{(2)}), (m_{2,\text{coin}}, r_2^{(1)}, r_2^{(2)})). \end{aligned}$$

- Set

$$\text{ct}_1^* = (\text{ct}_1^{(1)}, \text{ct}_1^{(2)}, \pi_1), \quad \text{ct}_2^* = (\text{ct}_2^{(1)}, \text{ct}_2^{(2)}, \pi_2).$$

- Generate a punctured key:

$$\text{sk}\{\text{ct}_1^*, \text{ct}_2^*\} \leftarrow i\mathcal{O}(D[1, \text{sk}^{(1)}, \text{pk}, \text{ct}_1^*, \text{ct}_2^*]).$$

- Send

$$(\text{ct}_1^*, \text{ct}_2^*, \text{pk}, \text{sk}\{\text{ct}_1^*, \text{ct}_2^*\})$$

to $\mathcal{A}$.

3. The adversary $\mathcal{A}$ outputs a guess coin'. The hybrid outputs 1 if coin' = coin, and 0 otherwise.

Clearly, we have

$$\Pr[\text{Hyb}_0 = 1] = \Pr[\text{Exp}_{\text{PPKE}, \mathcal{A}}^{\text{ppke}}(\lambda) = 1].$$

Hyb₁: This is identical to Hyb₀ except that crs, π_1 , and π_2 are generated by the simulator. That is, they are generated as follows:

$$(\text{crs}, \pi_1, \pi_2) \leftarrow \text{Sim}(1^\lambda, (\text{pk}^{(1)}, \text{pk}^{(2)}, \text{ct}_1^{(1)}, \text{ct}_1^{(2)}), (\text{pk}^{(1)}, \text{pk}^{(2)}, \text{ct}_2^{(1)}, \text{ct}_2^{(2)})).$$

By the two-theorem non-adaptive computational zero-knowledge property of NIZK, we have

$$|\Pr[\text{Hyb}_0 = 1] - \Pr[\text{Hyb}_1 = 1]| = \text{negl}(\lambda).$$

Hyb₂: This is identical to Hyb₁ except that $ct_1^{(2)}$ $ct_2^{(2)}$ are generated as follows, regardless of the value of coin:

$$\begin{aligned} ct_1^{(2)} &\leftarrow \text{PKE.Enc}(\text{pk}^{(2)}, m_{1,0}), \\ ct_2^{(2)} &\leftarrow \text{PKE.Enc}(\text{pk}^{(2)}, m_{2,0}). \end{aligned}$$

Since $sk^{(2)}$ and the encryption randomness for generating $ct_1^{(2)}$ and $ct_2^{(2)}$ are not used throughout the experiment due to the modification made in Hyb₁, by the IND-CPA security of PKE, we have

$$|\Pr[\text{Hyb}_1 = 1] - \Pr[\text{Hyb}_2 = 1]| = \text{negl}(\lambda).$$

Hyb₃: This is identical to Hyb₂ except that $sk\{ct_1^*, ct_2^*\}$ is generated as follows.

$$sk\{ct_1^*, ct_2^*\} \leftarrow i\mathcal{O}(D[2, sk^{(2)}, \text{pk}, ct_1^*, ct_2^*]).$$

By the two-theorem statistical simulation soundness of NIZK, we can see that $D[1, sk^{(1)}, \text{pk}, ct_1^*, ct_2^*]$ and $D[2, sk^{(2)}, \text{pk}, ct_1^*, ct_2^*]$ are functionally equivalent with an overwhelming probability. Thus, by security of $i\mathcal{O}$, we have

$$|\Pr[\text{Hyb}_2 = 1] - \Pr[\text{Hyb}_3 = 1]| = \text{negl}(\lambda).$$

Hyb₄: This is identical to Hyb₃ except that $ct_1^{(1)}$ $ct_2^{(1)}$ are generated as follows, regardless of the value of coin:

$$\begin{aligned} ct_1^{(1)} &\leftarrow \text{PKE.Enc}(\text{pk}^{(1)}, m_{1,0}), \\ ct_2^{(1)} &\leftarrow \text{PKE.Enc}(\text{pk}^{(1)}, m_{2,0}). \end{aligned}$$

Since $sk^{(1)}$ and the encryption randomness for generating $ct_1^{(1)}$ and $ct_2^{(1)}$ are not used throughout the experiment due to the modifications made in Hyb₁ and Hyb₃, by the IND-CPA security of PKE, we have

$$|\Pr[\text{Hyb}_3 = 1] - \Pr[\text{Hyb}_4 = 1]| = \text{negl}(\lambda).$$

In Hyb₄, no information of coin is given to the adversary. Thus, we have

$$\Pr[\text{Hyb}_4 = 1] = \frac{1}{2}.$$

Combining the above, we have

$$\left| \Pr \left[\text{Exp}_{\text{PKE}, \mathcal{A}}^{\text{ppke}}(\lambda) = 1 \right] - \frac{1}{2} \right| = \text{negl}(\lambda).$$

This completes the proof of the indistinguishability with punctured keys.